The Useful Life of a Bubble

A forensic reading of the 2026 AI infrastructure buildout, from the filings rather than the narrative

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"I don't believe in the efficient market hypothesis.
I believe in the diligence hypothesis."

This report follows the method rather than the man: read the primary documents that almost nobody reads — the 10-K footnotes, the 8-K exhibits, the XBRL tags, the covenant amendments — and let the disclosures argue with the press releases.

Four companion artefacts accompany this report:

`AI_Depreciation_Sensitivity_Model.xlsx` — a working, formula-driven model of every figure in Section 2
`circular_financing_map.html` / `.svg` — the visual map and full edge register behind Section 3
`monitoring_dashboard.html` — Section 10 as a dated, checkable tracker

Sourcing standard: primary sources only. Where a figure could not be traced to a filing, a rating action, or an official announcement, this report says so rather than estimating it. Section 12 is a consolidated register of everything that could not be verified.

A note on method, and on what this document is

Michael Burry's edge in 2005 was not insight. It was that he read the prospectuses. The mortgage pools that later detonated were described, accurately and in detail, in documents that ran to several hundred pages and that essentially nobody outside the issuing banks' legal departments opened. The information was not hidden. It was merely tedious to find, and the consensus narrative was more comfortable than the footnotes.

This report applies that method to the AI infrastructure buildout as it stands in July 2026. Nine parallel research passes were run against SEC EDGAR — 10-Ks, 10-Qs, 8-Ks and their exhibits, 424B prospectuses, 13F-HR and 13G filings, Form 3s, SEC staff comment letters, and the XBRL rendered data behind each filing — plus rating agency actions, official government documents, the Bank for International Settlements, the Bank of England, the IMF, the Federal Reserve, the Financial Stability Board, the Office of Financial Research, and the companies' own investor relations material.

Three rules governed what made it into this document.

First, primary sources only. Every number below carries a filing, a date and a URL. Where the research could not reach a primary source, the claim is either absent or explicitly tagged. Press reporting appears only where it is the sole record of an event, and it is always labelled as such.

Second, verified fact and analyst inference are marked apart, everywhere. Four tags are used:

Tag	Meaning
[V]	Verified in a primary document that was actually retrieved and read
[R]	Reported in credible press or third-party research; not confirmed in a filing
[C]	Calculated by this report from verified inputs; the arithmetic is shown
[G]	Gap — looked for, not found. What was searched is stated

A **[G]** tag is not an admission of failure. Several of the most important findings in this report *are* gaps: the largest number in the entire AI financing complex appears in no SEC filing anywhere, and establishing that took more work than confirming it would have.

Third, this is not a forecast of collapse. It is the most rigorous documented case for concern that the available record supports, presented alongside the strongest honest case against it. Section 8 argues that the physical constraints on this buildout make "overbuild" the wrong frame entirely. Section 9 argues the bull case as forcefully as the bear case is argued in Sections 2 through 7, and it wins several of the arguments outright. Section 10 states in advance what would prove this document wrong.

The reader is meant to weigh it. That is the whole point.

One caveat on completeness, stated up front. The web research budget available to this session was exhausted partway through, which shaped where the evidence is thickest. Coverage of SEC filings, Federal Reserve and FRED data, BIS, BoE, IMF, FSB, OFR publications, and company investor-relations material is deep. Coverage of subscription rating-agency reports, state public-utility-commission dockets, and defence contract award identifiers is materially thinner, and Section 5 is explicit about which of its parts rest on retrieved documents and which do not. The consolidated gap register in Section 12 lists every one of these, by name.

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1. Executive summary

1.1 The thesis in one paragraph

The AI infrastructure buildout is not one trade. It is two, and the market is pricing them as though they were the same thing. The hyperscalers — Microsoft, Alphabet, Amazon, Meta — are spending enormous sums out of enormous cash flows against contracted revenue that is real, audited and accelerating, and four of the six largest buyers of AI compute are still in net cash. That trade is defensible. The periphery — Oracle, CoreWeave, the private-credit-funded developers, the special-purpose vehicles, and the labs whose finances are not disclosed at all — is doing something structurally different: financing a decade-length asset with four-year money, against backlog rather than revenue, using accounting elections that keep roughly $863 billion of signed obligations off six balance sheets, in a web of arrangements where the same institutions are simultaneously investor, supplier and customer to one another. **The risk is not that the AI buildout fails. The risk is that the second trade is being financed at the credit quality of the first, and that when it separates, it will separate at the vehicle level, not the parent level, where almost nobody is looking.**

1.2 What the filings actually say

1.2.1 1. The depreciation question is real, but it is not the scandal it is described as

Five of the six large operators extended server useful lives between 2020 and 2025, and they disclosed the effect. Meta's January 2025 extension to 5.5 years added **$2,590 million to FY2025 net income and $1.00 to diluted EPS** on its own arithmetic [V]. Microsoft's move from four to six years added **$1.1 billion of operating income in a single quarter** [V]. Alphabet's move to six years added **$770 million of net income in Q1 2023** [V].

But the fact that undermines the crude version of the bear case is that **Amazon went the other way.** Effective 1 January 2025 it *shortened* a subset of servers and networking equipment from six years to five, citing "the increased pace of technology development, particularly in the area of artificial intelligence and machine learning," at a cost of **$1,400 million of additional depreciation and $1,000 million of net income in FY2025** [V]. The operator with the largest disclosed server fleet moved toward the bears, publicly, and ate the earnings hit.

The real finding is narrower and harder to dismiss. It is a **disclosure asymmetry** — one extension in this period was never quantified at all, and it is the largest asset base of the lot:

Alphabet extended data-centre and office building lives from a 7-to-25-year range to a **7-to-40-year** range, first appearing in the FY2024 10-K, and **never quantified it as a change in accounting estimate.** The phrase "change in accounting estimate" appears in no Alphabet filing after the FY2023 10-K of 31 January 2024 [V]. Against technical infrastructure of $203.7 billion at year-end 2025, a fifteen-year extension on the shell portion is not immaterial.

A correction to this report's own working premise, made after an adversarial re-check: Oracle's five-to-six-year server extension **was** *quantified. Its FY2025 10-K states the change "decreased our total operating expenses by $733 million and increased our net income by $573 million, or $0.21 per basic and $0.20 per diluted share, during fiscal 2025"* [V]*. The disclosure it does not make is the share of its $638 billion backlog attributable to any single customer — which Section 3 takes up.*

The companion model quantifies the sensitivity. Reverting the whole disclosed server and IT asset base of all six operators to a five-year life reduces annual pre-tax earnings by roughly **$16.5 billion**; to four years, **$52.3 billion**; to three years — the pre-2020 industry norm — **$111.9 billion** [C]. Restricting the calculation to the three companies that disclose a genuinely separable server line (Meta, Amazon, CoreWeave) gives **$2.8 billion**, **$18.1 billion** and **$43.5 billion** respectively [C]. The model reproduces Meta's own disclosed $2,920 million FY2025 benefit to within 1.4%, which is the test of whether any of this arithmetic can be trusted.

1.2.2 2. The circular-financing web is real, but the money and the headlines do not match

The largest number in the entire complex — NVIDIA's "up to $100 billion" investment in OpenAI, announced 22 September 2025 — **has never been described in any NVIDIA SEC filing.** EDGAR full-text search returns exactly one occurrence of the phrase "up to $100 billion" in NVIDIA's entire filing history, in a 2020 8-K exhibit referring to the SoftBank Vision Fund, and **zero** hits for "first tranche." The Q3 FY2026 10-Q says only that NVIDIA intends "to invest in OpenAI," unquantified. The FY2026 10-K downgrades this to an unquantified "agreement with OpenAI" inside a new counterparty-risk factor. **The Q1 FY2027 10-Q, filed 20 May 2026, does not mention OpenAI at all [V]**.

Meanwhile the money that actually moved went somewhere else, and it is fully disclosed. Amazon's Q1 2026 10-Q states in plain language that it **invested $15.0 billion in OpenAI Series C preferred stock** and entered a **$35.0 billion binding equity commitment letter**, with mandatory drawdown on the earlier of specified milestones or an OpenAI public listing — against an **expansion of OpenAI's existing $38 billion AWS commitment by a further $100 billion over eight years [V]**. That is the same structure the NVIDIA deal was criticised for, at comparable scale, and it is funded, filed and quantified.

The one loop that closes completely in the documents is **NVIDIA and CoreWeave**. NVIDIA is simultaneously CoreWeave's sole GPU supplier, an **11.5% shareholder and Section 16 insider** since 26 January 2026, and a contractual **buyer of CoreWeave's unsold capacity through 13 April 2032** under a $6.3 billion order form that CoreWeave's own 8-K describes as "no longer immaterial" **[V]**. NVIDIA's non-marketable equity securities went from **$3,387 million to $42,336 million in five quarters**, including **$17,899 million of net additions in the single quarter to 26 April 2026**, with **$27 billion of further investment commitments** disclosed **[V]**.

Two widely repeated claims did not survive checking, and both are corrected here: the "$1.3 billion NVIDIA–CoreWeave backstop" does not exist — the only 1.3 in the prospectus is 1.3 *gigawatts* of contracted power — and Broadcom's $10 billion XPU order, attributed to OpenAI for months, was **Anthropic**, as Broadcom itself disclosed on 11 December 2025 **[V]**.

1.2.3 3. The off-balance-sheet channel is not the SPV. It is the lease footnote

This is the finding with the largest number attached to it, and it is the one that gets the least attention. Under ASC 842, a lease creates no asset and no liability until it *commences*. A sixteen-year data-centre lease signed today for 2029 delivery lives in a single footnote sentence for three years. Across six issuers at Q1 2026:

Entity	Leases not yet commenced	Recognised lease liabilities	Ratio
Oracle	$261.0bn (28 Feb 26); $260bn (31 May 26)	$27.6bn	**9.5x**
Microsoft	$196.6bn	$85.2bn	2.3x
Meta	$182.9bn	$28.0bn	6.5x
Amazon	$106.3bn	$126.6bn undiscounted	0.8x
Alphabet	$75.6bn	$18.4bn	4.1x
CoreWeave	$40.7bn	$10.3bn	4.0x
Total	**~$863bn**		

All figures **[V]** from the filings themselves. **Roughly $863 billion of contractually committed lease obligations sit outside these six balance sheets.** Moody's, in a sector report published around 22 July 2026, put the six-company lease commitment total at **$1.2 trillion, of which more than $820 billion has not yet commenced**, and called them "debt-equivalent liabilities" **[R]**.

The named SPVs are the supply side of that same trade. Meta's Louisiana venture is the template, and Meta's own Q1 2026 10-Q describes it with unusual candour: a 20% membership interest, approximately $27 billion of committed development costs, an aggregate initial lease commitment of **$12.31 billion** commencing in 2029, **residual value guarantees with an aggregate threshold of approximately $28 billion**, and — the number that matters — **maximum exposure to loss of $45.99 billion against a carrying value of $2.37 billion [V]**. Meta does not consolidate, on the stated basis that decisions about *remarketing* the campus most significantly affect the venture's economics and Meta does not control them. Meta is the sole tenant, the construction manager, the property manager and the guarantor.

The rating agencies disagree with each other about this in public. S&P rated the vehicle's bonds **A+**, one notch below Meta itself, and stated it makes "no debt adjustment for the RVG due to significant headroom between the RVG and a third-party appraisal value." Moody's replied that "the accounting at this time under the current rules is not in line with the expected economics" and that it foresees "a material increase in adjusted debt and lease-related cash outflows for these companies in the coming years" **[R, both quoted in trade press]**. **Meta names neither Blue Owl nor the issuing entity in any SEC filing. "Beignet" returns zero hits across every Meta filing ever made. [V]**

1.2.4 4. Oracle is where the leverage actually is

Oracle's FY2026, ended 31 May 2026: revenue **$67.4 billion**, capital expenditure **$55.7 billion**, operating cash flow **$32.0 billion**, and therefore **free cash flow of negative $23.7 billion [V]**. Capex to operating cash flow of **1.74x**, against 1.02x a year earlier and 0.37x two years earlier **[C]**. Gross debt **$129.5 billion**, up $37.0 billion in twelve months. Gross leverage rose from **3.88x to 4.33x [C]**, through the 4x level that IFR identifies as S&P's downgrade trigger. Depreciation rose 97% in a single year, and $40.0 billion of construction in progress — 32.6% of gross PP&E — is not yet depreciating at all **[V]**.

RPO is **$638 billion**, of which Oracle expects to recognise **approximately 12% over the next twelve months [V]**. That is 9.5x total FY2026 revenue and 35x OCI revenue **[C]**. Oracle's FY2026 10-K states that no single customer accounted for 10% or more of *revenue* — and discloses **no customer's share of RPO anywhere**. The concentration is measured against the denominator where it cannot yet appear.

New risk-factor language appeared for the first time in June 2026: "In certain OCI offerings, we are more concentrated among a number of large customers, which could increase these risks," and "if any of our key customers are unable to pay or otherwise perform under their contracts with us, we could be locked into multi-year commitments for excess data center space and related capital expenditures" **[V]**. The February 2026 forward prospectus introduced "ability to fund its commitments" — one hit in all of Oracle's filing history **[V]**.

The market has repriced this without waiting for the accounting. Oracle's thirty-year coupon moved from 5.950% in September 2025 to **6.700%** in February 2026, and the forty-year from 6.100% to **6.850%** — 75 basis points in four months, verified from the two prospectuses themselves **[V]**. S&P cut Oracle to **BBB−** on 9 July 2026 **[R]**, and its five-year CDS reached **203 basis points on 20 July 2026, the widest in the history of the series [R]**, against an investment-grade index at 79 basis points **[V]**.

1.2.5 5. CoreWeave's covenants have already been amended once, and the amendment is the tell

CoreWeave's FY2025 operating income was **negative $46 million** against **$1,229 million** of interest expense: **EBIT covers interest 0.0 times [C]**. Two-year cumulative free cash flow burn is **negative $17.9 billion**, funded almost entirely with debt **[C]**. Total debt principal went from $8.0 billion to $25.1 billion in fifteen months **[V]**.

On 2 January 2026 CoreWeave filed an 8-K amending its DDTL 3.0 facility. It reduced the minimum liquidity requirement to $100 million for a window in early 2026, deferred the first debt-service coverage test **from April 2027 to 31 October 2027** while leaving the 1.40x level intact, and permitted **"an unlimited number of equity cures" for any failure of the debt-service coverage ratio or the contract realisation ratio prior to 28 October 2026 [V]**. The stated reason was "the timing of deliveries described by the Parent on its earnings call." Lenders do not grant unlimited equity cures to borrowers who do not need them.

The company's cost of capital is bifurcating violently on identical collateral: **SOFR+225bp** for the investment-grade-rated DDTL 4.0 in March 2026, **SOFR+450bp** for DDTL 5.0 two months later, and **9.625% unsecured** in June 2026 **[V]**.

1.2.6 6. On the government: the story that dominates coverage has no federal money in it

Stargate — $500 billion, four equity funders, a White House podium — has, on the record retrieved here, **no identified federal money, no loan guarantee, and no dedicated tax expenditure.** Further, **none of the four announced equity funders has ever disclosed a Stargate equity contribution figure in any primary document**, and no operating agreement, ownership table or board roster for the vehicle exists in any public record eighteen months after announcement [**V, negative finding**]. Executive Order 14318 (28 July 2025) creates the *authority* for federal loans and guarantees to data centres above 100MW or $500 million, but no evidence was found that the Commerce financing initiative it directs was ever stood up [**V/G**].

The two largest federal financial positions in this sector have nothing to do with Stargate. The Intel stake — **433,323,000 shares for $8,869,800,000**, of which 274,583,000 were issued at closing against $5.695 billion actually disbursed — was executed through a private securities purchase agreement, with **no Federal Register instrument and no Schedule 13D ever filed on a 9.9% position** [**V**]. At Intel's 24 July 2026 close the shares actually held are worth roughly **$26.8 billion against a $6.0 billion basis** [**C**]. The taxpayer is up about $21 billion, unrealised, on a position that cannot be sold until 27 August 2026 at the earliest. The genuinely risky federal exposure is the one nobody discusses: the MP Materials agreement carries a **$110/kg NdPr price floor for ten years and a quarterly payment obligation against a $140 million EBITDA threshold** — uncapped contingent liabilities, in a securities purchase agreement, on the Department of Defense's books [**V**].

The real public subsidy to AI infrastructure is state and local. Doña Ana County, New Mexico authorised **up to $165 billion in industrial revenue bonds** with a thirty-year property tax abatement for a single project on 19 September 2025 [**V**].

1.2.7 7. The official sector has arrived, and it is not unanimous

The **BIS Quarterly Review of 16 March 2026** contains the single most important sentence written about this subject by a public institution: these arrangements "amount to 'shadow borrowing': obligations that are economically akin to debt but largely reside outside corporate balance sheets" [**V**]. The **Bank of England's July 2026 Financial Stability Report** records that the five hyperscalers were "just 3% of the stock of outstanding US IG debt at the end of 2025, but as of early May accounted for over 15% of year-to-date issuance" [**V**]. The **Federal Reserve's May 2026 Financial Stability Report** records that artificial intelligence was named as a financial-stability risk by **50% of surveyed market contacts, against 0% two quarters earlier** — and then does not address AI capex or data-centre lending anywhere in its four-pillar vulnerability framework [**V**]. The **IMF's April 2026 GFSR** notes the reliance on circular financing and adds, pointedly, that "the impact on financial stability appears modest currently" [**V**].

1.3 What cuts the other way, and cuts hard

The physical constraint is real and it is binding today. GE Vernova disclosed **116 GW** of gas equipment backlog plus slot reservations at Q2 2026 against **20 GW per year** of output — roughly **5.8 years sold out** — and raised 2026 free-cash-flow guidance from $6.5–7.5 billion to **$11.5–12.5 billion** while raising revenue guidance by only about $1 billion, which is customer cash arriving ahead of delivery [**V**]. Amy Hood told Microsoft's April 2026 call: **"We expect to remain constrained at least through 2026"** [**V**]. Across seven announced Stargate sites, **more than 9 GW is planned and roughly 0.3 GW is operational** [**R**]. If the binding constraint is turbines and transformers rather than demand, "overbuild" is the wrong word for what is happening.

The bull case wins several arguments outright. Cloud revenue growth is *accelerating* at the fourth year of the buildout — Azure +40%, Google Cloud +82%, AWS +28%, Oracle OCI accelerating every quarter from +52% to +93% — which is the opposite of the overbuild signature [**V**]. NVIDIA's days sales outstanding is **45.4 days today against 45.7 five quarters ago while revenue grew 85%**, which is the cleanest available refutation of the claim that NVIDIA is financing its own revenue [**C**]. Nine-year-old V100 silicon is still on Google's and Microsoft's *published* price lists, and Azure repriced it effective 1 January 2026 [**V**]. Only **32% to 45% of hyperscaler gross PP&E is the short-lived IT layer** at all [**C**]; the rest is shell and power, which independent

REITs depreciate over 39 and 60 years. And US nonfinancial corporate debt is **5.6x after-tax profits today against 19.7x at the 2000 peak [C]**.

And the bull case is fragile in exactly three places, which it should be judged on: capex has crossed operating cash flow at Amazon and Alphabet already; RPO is a promise, not cash, and Oracle's is 9.5x revenue while CoreWeave's is 12x; and corporate debt-to-GDP is **identical** to 2000 — the entire leverage improvement is an earnings phenomenon, and 2000-era earnings fell 47%.

1.4 The conclusion, stated plainly

Nothing in the record supports a forecast of collapse. A great deal in the record supports three narrower claims.

One. Reported earnings across this sector depend materially on an unobservable estimate that management chooses, that has moved in both directions within four years, and that two companies have now changed without quantifying. That is a measurable sensitivity, not an accusation, and Section 2 measures it.

Two. The obligations are increasingly located where the disclosure is weakest — in leases that have not commenced, in vehicles the sponsor does not name, in covenants that have already been amended once, and in contracts whose counterparty concentration is disclosed against revenue rather than backlog. Roughly $863 billion of committed lease obligations, on six balance sheets, sit outside all six of them.

Three. The distinction between the hyperscalers and the periphery is the entire analysis, and the credit market has only just begun to price it. Oracle's CDS at 203 basis points against an index at 79, and CoreWeave's secured spread doubling in two months on unchanged collateral, are that repricing beginning. Whether it stops there is the question this report cannot answer — and Section 10 states, in dates and numbers, exactly what would answer it.

2. Hyperscaler depreciation accounting

2.1 Why this is the right place to start

A server is an asset whose cost is spread over the period it renders service. ASC 360-10-35-4 puts it plainly, and a registrant quoted it verbatim to the SEC staff in 2021: "The cost of a productive facility is one of the costs of the services it renders during its useful economic life. Generally accepted accounting principles require that this cost be spread over the expected useful life of the facility in such a way as to allocate it as equitably as possible to the periods during which services are obtained from the use of the facility." **[V]**

Nothing in that sentence is objective. "Expected useful life" is an estimate made by management, reviewed by auditors, and disclosed in a footnote. Change it by twelve months on a hundred-billion dollar asset base and several billion dollars of pre-tax earnings appear or disappear, prospectively, with no cash consequence and no restatement of prior periods. It is the single largest discretionary lever in a hyperscaler's income statement, and in the last four years every one of these companies has pulled it.

This section establishes four things: exactly what each company assumes, exactly what each company disclosed when it changed, what the gap is between those assumptions and NVIDIA's actual product cadence, and what the earnings effect would be if the assumptions reverted.

2.2 Where every company stands today

Company	Server / compute life	Buildings	Separate server line in PP&E?
Meta Platforms	**5 to 5.5 years**	25–30 yrs	**Yes**
Microsoft	**6 years** (policy range 2–6 for "computer equipment")	5–15 yrs	No
Alphabet	**up to 6 years**	**7–40 yrs**	No
Amazon	**5 to 6 years** (subset at 5)	up to 40 yrs	**Yes, from FY2025**
Oracle	**6 years**	1–40 yrs	No
CoreWeave	**6 years** ("technology equipment")	n/a — no buildings	**Yes**
Nebius (comparator)	**4 years**	n/a	Yes

All **[V]**, from the most recent 10-K or 20-F policy note of each company. Two observations before the detail.

First, the range is not wide. Everyone except Nebius and Meta sits at six years; Meta sits at 5.5; Amazon has a subset at five. The dispersion in this industry is one and a half years, against a product cadence that has been annual since 2024.

Second, and more important for anyone trying to compare these companies: **only three of them disclose a server line you can actually isolate.** Meta breaks out "servers and network assets." Amazon began breaking out "servers and networking equipment" for the first time in its FY2025 10-K. CoreWeave discloses "technology equipment." Microsoft buries servers inside "computer equipment and software." Alphabet buries them inside "technical infrastructure," which also contains the data-centre shells that Alphabet depreciates over as much as forty years. Oracle buries them inside "computer, network, machinery and equipment." Any cross-company depreciation comparison that does not say this is comparing different things.

2.3 The register of disclosed changes

This is every useful-life change these companies have disclosed since 2020, with the effect **they reported**, not an effect this report has modelled.

Company	Effective	Change	Depreciation effect	Net income effect	Period
Amazon	1 Jan 2020	Servers 3 → 4 yrs	−$786m	+$602m	Q1 2020
Meta	Q2 + Q4 2022	Servers/network 4 → 5 yrs	−$860m	+$693m	FY2022
Amazon	1 Jan 2022	Servers 4→5; network 5→6	−$3,600m	+$2,800m	FY2022
Microsoft	1 Jul 2022	Server/network 4 → 6 yrs	n/d	+$859m	Q1 FY2023
Alphabet	1 Jan 2023	Servers 4→6; network 5→6	−$988m	+$770m	Q1 2023
Amazon	1 Jan 2024	Servers 5 → 6 yrs	−$3,200m	+$2,500m	FY2024
Oracle	Q1 FY2025	Servers/network 5 → 6 yrs	−$733m (opex)	+$573m	FY2025
Alphabet	FY2024 10-K	Buildings 7–25 → **7–40 yrs**	**not disclosed**	**not disclosed**	—
Meta	1 Jan 2025	Most servers/network 5 → 5.5 yrs	−$2,920m	+$2,590m	FY2025
Amazon	1 Jan 2025	**Subset of servers 6 → 5 yrs**	**+$1,400m**	**−$1,000m**	FY2025
Amazon	1 Jan 2025	Heavy equipment 10 → 13 yrs	n/d	+$900m (est.)	FY2025

Every row **[V]** from the filing named in the companion model's *Disclosed* tab. Four things fall out of this table.

Meta's FY2025 disclosure is the largest single quantified benefit in the set. Verbatim from the Q1 2025 10-Q: "In January 2025, we completed an assessment of the useful lives of property and equipment, which resulted in an increase in the estimated useful lives of most servers and network assets to 5.5 years, effective January 1, 2025… the financial impact of this change in estimate included a reduction in depreciation expense of $826 million and an increase in net income of $695 million, or $0.27 per diluted share, for the three months ended March 31, 2025." The FY2025 10-K's XBRL change-in-estimate tags give the full-year figures: depreciation **−$2,920 million**, net income **+$2,590 million**, diluted EPS **+$1.00** **[V]**.

Microsoft framed its benefit differently, and that matters. Its Q1 FY2023 10-Q reported "an increase in operating income of $1.1 billion and net income of $859 million, or $0.12 and $0.11 per basic and diluted share." Meta and Alphabet both reported a *reduction in depreciation expense*. Microsoft reported an *increase in operating income*. The economics are the same; the framing is not, and it means the two are not directly comparable without adjustment.

Microsoft and Alphabet each disclosed one quarter and never annualised. Neither company has ever restated the full-year effect of its change. The market's estimates of Microsoft's FY2023 benefit (commonly cited around $3.7 billion) are extrapolations, not disclosures **[G]** — the underlying narrative could not be retrieved from the FY2023 10-K, whose notes sit past the fetch limit of the tooling used here.

Amazon is the exception that damages the simple bear case, and it should be given full weight. Verbatim from its Q1 2025 10-Q: "Effective January 1, 2025 we changed our estimate of the useful lives of a subset of our servers and networking equipment from six years to five years. **The shorter useful lives are due to the increased pace of technology development, particularly in the area of artificial intelligence and machine learning.**" The FY2025 realised effect was **$1,400 million of additional depreciation and $1,000 million less net income**, or −$0.10 of diluted EPS **[V]**. Note also that the realised figure came in at double the $(700) million Amazon had forecast in its FY2024 10-K, because AWS added $96,496 million of gross property and equipment during 2025 **[V]**.

There is one important piece of context the bear framing usually omits. Amazon's shortening was a **partial reversal of an extension made twelve months earlier** — servers went 5 → 6 years effective 1 January 2024, worth $2,500 million of FY2024 net income, before going 6 → 5 for a subset effective 1 January 2025 **[V]**. And in the same footnote as the shortening, Amazon extended **heavy equipment from ten to thirteen years**, which its FY2024 10-K forecast would add **$900 million** to FY2025 net income [**V**]. The headline reversal was partly offset by a quieter extension.

2.4 Oracle quantified. Alphabet did not.

2.4.1 Oracle — and a correction to this report's own premise

This report's initial research pass concluded that Oracle disclosed no dollar effect for its five-to-six-year server extension. **That conclusion was wrong, and an adversarial re-check found it.** The quantification exists; it simply sits on a different page of the filing from the useful-life table, which is where the first pass looked.

Oracle's FY2025 10-K states the change and then quantifies it, verbatim [**V**]:

> "During the first quarter of fiscal 2025, we completed an assessment of the useful lives of our servers and networking equipment and increased the estimate of the useful lives from five years to six years, effective at the beginning of fiscal 2025... Based on the carrying value of our servers and networking equipment as of May 31, 2024, this change in accounting estimate **decreased our total operating expenses by $733 million and increased our net income by $573 million, or $0.21 per basic and $0.20 per diluted share, during fiscal 2025**."

That is a full ASC 250-10-50-4 disclosure and Oracle should be credited with it. It is recorded in the register above and in the companion model.

The reason this correction is left in the document rather than quietly absorbed is that it is the same failure mode the report is about: a conclusion drawn from where a number was not, rather than from where it was. The discipline that catches it is re-checking your own negatives.

The FY2026 10-K carries no further change-in-estimate language and restates the six-year life: "Comprised primarily of servers and networking equipment with estimated useful life of six years" [**V**]. For scale on what remains: Oracle's computer, network, machinery and equipment gross balance went from $30,345 million at 31 May 2025 to **$59,634 million** at 31 May 2026, and depreciation from $3,867 million to **$7,623 million —** up 97% in one year [**V**]. **The lever has been pulled, it was worth $573 million of net income in the year it was pulled, and on a base that has since doubled it cannot be pulled again without a further extension beyond six years.**

2.4.2 Alphabet

Alphabet's building lives moved from a stated 7-to-25-year range to a stated **7-to-40-year** range, and the category was relabelled from "Building" to "Data center and office buildings." EDGAR full-text search for "40 years" in Alphabet 10-Ks returns exactly two hits: the FY2024 10-K filed 5 February 2025 and the FY2025 10-K filed 5 February 2026 [**V**]. It appears in no 10-Q.

Search for the phrase "change in accounting estimate" across every Alphabet filing returns 21 hits, the most recent being the **FY2023 10-K filed 31 January 2024 —** nothing in any filing after that date through 25 July 2026 [**V**]. **Alphabet does not appear to have characterised the building-life extension as a change in accounting estimate anywhere, and therefore never quantified it.**

The premise this report started with — that Alphabet made a further change effective 1 January 2025 — is **wrong, and is corrected here.** Servers and network equipment are tagged at six years in both the FY2024 and FY2025 10-Ks, the Q1 2026 10-Q contains no change-in-estimate disclosure, and direct reads of the Q1 2024, Q3 2024, Q1 2025 and Q2 2025 10-Qs found nothing [**V, negative finding**]. The only change is the buildings range.

Whether that change is material depends on the shell portion of technical infrastructure, which Alphabet does not disclose [**G**]. Against $203,679 million of technical infrastructure at 31 December 2025, it is not obviously immaterial.

2.5 The regulator has not asked

A natural question is whether the SEC staff has pressed any of these companies on useful lives. On the record retrieved, it has not.

- **Meta:** SEC comment letters exist dated 2021 through June 2023. Three were read in full. They concern Reality Labs segment losses, advertising measurement, and advertising demand drivers. **None mentions PP&E, depreciation or useful lives.** No UPLOAD or CORRESP filings after 28 June 2023 [**V**].
- **Microsoft:** 35 UPLOAD filings back to 2006; the three most recent were read. They concern the January 2024 cybersecurity 8-K and an S-4 review. **No accounting content** [**V**].
- **Alphabet:** 18 UPLOAD/CORRESP filings, **all dated 2016 to February 2020**. The most recent is a review-completion letter with no numbered comments. **There are no Alphabet comment letters at all in the 2023–2026 window** [**V**].
- **CoreWeave:** ten UPLOAD letters with nine responses, January to September 2025. The September exchange was retrieved: it concerns qualified tax opinions in the Core Scientific S-4. **Not revenue recognition, not concentration, not the NVIDIA arrangement** [**V**].

The staff *is* active on long-lived assets generally — EDGAR full-text search returns 252 CORRESP filings citing ASC 360-10-35-21, and a 19 May 2026 comment letter to SpaceX expressly cites it regarding the long-lived assets of the former xAI business [**V**]. But no comment letter on hyperscaler server lives was located.

Critical audit matters are a genuine gap [**G**]. The auditor's report sits at the end of each 10-K, past the fetch limit of the tooling used, and CAMs are not XBRL-tagged. What can be said: the plural heading "Critical Audit Matters" appears in every Meta and Microsoft 10-K from FY2019 through FY2025; Alphabet's filings match only the singular, implying a single CAM; Oracle's and CoreWeave's FY2025/FY2026 10-Ks contain the phrase. **No CAM title was read, and the absence of a useful-life CAM cannot be asserted.** For CoreWeave specifically, EDGAR full-text search returns zero hits for every natural CAM heading involving useful lives or depreciation of technology equipment, which makes such a CAM unlikely but does not prove its absence [**V, indirect**].

2.6 The gap between assumption and cadence

2.6.1 What NVIDIA actually ships, and when

Platform	Announced	Availability	Interval since prior
Tesla P100 (Pascal)	5 Apr 2016	DGX-1 June 2016; OEM early 2017	—
Tesla V100 (Volta)	10 May 2017	"later in the year" [**G**] on exact date	~13 mo
A100 (Ampere)	14 May 2020	**Full production same day**	**~36 mo**
H100 (Hopper)	22 Mar 2022	Full production 20 Sep 2022	~22 mo
H200	13 Nov 2023	Shipping Q2 2024	~20 mo
B200 / GB200 (Blackwell)	18 Mar 2024	Ramp Q4 FY2025	~4 mo
GB300 (Blackwell Ultra)	18 Mar 2025	2H 2025; crossed over GB200 by Q3 FY2026	**12 mo**
Vera Rubin	GTC 16 Mar 2026	**Production racks running at CoreWeave, Google Cloud, Azure and OCI as of 21 July 2026**	**12 mo**
Rubin Ultra	roadmap	2H 2027	12 mo (planned)
Feynman	GTC 2026	2028	12 mo (planned)

All dates [**V**] from NVIDIA press releases and blog posts. Two corrections to widely used shorthand: **"B100" has no standalone launch release** and is not listed on NVIDIA's current Blackwell architecture page — treat it as a SKU, not a generation [**G**]. And **"VR200" appears in no NVIDIA primary source**; NVIDIA's own naming is "Vera Rubin NVL72" and "HGX Rubin NVL8" [**G**].

NVIDIA has codified the cadence in its own risk factors, which is the most useful disclosure in this entire section. From the FY2026 10-K: "We have introduced a new product and architecture cadence of our Data Center solutions where we seek to complete new computing solutions each year." And: "The increased frequency of these transitions and the larger number of products and product configurations may magnify the challenges associated with managing our supply and demand." And: "Customers may delay purchasing existing products as we increase the frequency of new products." [**V**]

Jensen Huang, Computex 2024: "Our company has a one-year rhythm." [**V**] GTC 2025: "NVIDIA will follow an annual rhythm for the buildout of AI infrastructure." [**V**] CES 2026: "With our annual cadence of delivering a new generation of AI supercomputers..." [**V**]

2.6.2 The honest version of the gap

The crude framing — "annual product cycle versus six-year depreciation, therefore the accounting is fictional" — does not survive the data, and this report will not make it. Three reasons.

First, the annual cadence is only two years old. Announce-to-announce intervals from 2016 to 2023 ran **13, 36, 22 and 20 months**. The twelve-month rhythm is empirically true only from GTC 2024 onward. The "one-year rhythm" statement of June 2024 postdated the first twelve-month interval by three months; it was a forward commitment that has now been delivered twice.

Second, and decisively: old GPUs are still commercially alive. Cloud on-demand list prices, fetched live on 25 July 2026 [**V**]:

GPU	Launched	AWS (per GPU-hr)	Azure	GCP	Lambda
V100 16GB	2017	$3.06	$3.365	—	$0.79
A100 40GB	2020	$2.745	$3.40	$3.673	$1.99
A100 80GB	2021	—	$4.096	$5.069	$2.79
H100	2022–23	$6.88	$12.29	$11.06–11.68	$3.99
B200	2024–25	$14.24	—	—	$6.69

Three details in that table matter more than the levels. **Azure's ND40rs_v2 (V100 32GB) carries an `effectiveStartDate` of 1 April 2020 — its list price has not moved in over six years. Azure's H100 list price is still exactly $98.32 per hour per eight-GPU VM, identical to AWS's July 2023 launch price, never cut**, while AWS cut its equivalent to $55.04. And on AWS, the **2017-vintage V100 is priced above the 2020-vintage A100 40GB** [**V**]. Hardware that is economically dead does not get repriced; it gets delisted. AWS still flags the P3 family as "current generation."

The counter-evidence is also in that data and is given equal weight: **Azure has fully delisted K80, P100 and V100-32GB SKUs**, and AWS has removed the P3/V100 family from its product pages [**V**]. Economic obsolescence is real. It simply arrives later than the crude framing implies.

NVIDIA's own CFO made the strongest statement on this, and the attribution matters — it is **Colette Kress, not Jensen Huang**, on the Q3 FY2026 call of 19 November 2025, verbatim:

> "The long useful life of NVIDIA's CUDA GPUs is a significant TCO advantage over accelerators... Thanks to CUDA, the A100 GPUs we shipped six years ago are still running at full utilization today, powered by vastly improved software stack." [**V**]

And: "The clouds are sold out and our GPU installed base, both new and previous generations, including Blackwell, Hopper, and Ampere, is fully utilized." She also noted the Hopper platform, in its thirteenth quarter, recorded approximately $2 billion of revenue in Q3 alone [**V**]. **No verbatim Huang statement to this effect exists in any NVIDIA primary source** — a point on which much secondary commentary is simply wrong [**G**].

Third, most of the asset is not the chip. Microsoft's gross PP&E at 30 June 2025 was $298,619 million, of which computer equipment and software was $132,836 million — **44.5%** — and land, buildings and leasehold improvements were $159,376 million — **53.4%** [**C**]. Amazon's servers and networking equipment were $172,492 million of $534,098 million gross — **32.3%** [**C**]. Independent data-centre REITs with no incentive to flatter hyperscaler earnings depreciate shells over **12–60 years** (Equinix) and **5–39 years** (Digital Realty) [**V**]. Alphabet's 7-to-40-year building range, whatever one thinks of the way it was disclosed, is corroborated by third parties.

The defensible statement of the gap is therefore this. Roughly one-third to 45% of the asset base turns over on a product cycle that has compressed from 20–36 months to 12 months in the last two years, while its assumed accounting life was simultaneously extended from four years to five and a half or six. The remaining 55–68% is long-lived and the extended lives are corroborated externally. There is no single "correct" number. There is a sensitivity, and it is large.

2.6.3 One decisive negative

No public company has ever disclosed a GPU-specific impairment. EDGAR full-text search returns **zero hits** for "impairment of GPUs," **zero** for "GPU impairment," and **zero** for "secondary market for GPUs." A search for GPUs plus impairment across 10-Ks from January 2025 to July 2026 returns 130 hits, all boilerplate risk-factor or accounting-policy language [**V**]. The only realised write-downs of GPU-class assets anywhere in EDGAR are the 2022 crypto-mining cohort. Amazon's $920 million "early retirement" accelerated depreciation — tagged in the FY2024 10-K but never described in narrative, and with its period unresolved [**G**] — is the closest hyperscaler analogue, and it is a depreciation acceleration, not an impairment.

Whether that is because the assets are genuinely unimpaired, or because ASC 360's recoverability test uses **undiscounted** cash flows and is therefore a low bar, is a question the filings do not answer.

2.7 The sensitivity, quantified

The companion workbook, `AI_Depreciation_Sensitivity_Model.xlsx`, computes this three ways from primary inputs. Every figure below is reproducible and every assumption is a labelled, editable cell.

2.7.1 Method 1: steady-state run rate

Annual depreciation on the disclosed server/IT asset base at alternative lives, straight-line, zero salvage. Delta = Gross × $(1/L_{alt} - 1/L_{current})$.

Company	Gross base ($m)	Stated life	vs 5-yr ($m)	vs 4-yr ($m)	vs 3-yr ($m)
Meta Platforms	105,987	5.5	(1,927)	(7,226)	(16,059)
Microsoft*	132,836	6.0	(4,428)	(11,070)	(22,139)
Alphabet*	217,886	6.0	(7,263)	(18,157)	(36,314)
Amazon	172,492	5.0	—	(8,625)	(22,999)
Oracle*	59,634	6.0	(1,988)	(4,970)	(9,939)
CoreWeave	26,627	6.0	(888)	(2,219)	(4,438)
Total pre-tax	**715,462**		**(16,493)**	**(52,266)**	**(111,888)**
Total after tax @21%			**(13,030)**	**(41,290)**	**(88,392)**
Clean-disclosure subset			*(2,815)*	*(18,070)*	*(43,495)*

*Asset line bundles non-server items — upper bound, not like-for-like. The "clean-disclosure subset" is Meta + Amazon + CoreWeave only, and is the defensible lower bound. All [**C**] from verified gross balances.

2.7.2 Method 2: ASC 250 prospective

A change in estimate applies to remaining carrying value over remaining revised life, so the first-year effect of an actual change is **larger** than the steady-state figure. Assuming a fleet age of 1.5 years and accumulated depreciation of 25% of gross on the server class — both stated assumptions, neither disclosed by any company — a move to a five-year life produces a pre-tax effect of **−$23.6 billion** [**C**]. The sensitivity to fleet age is severe:

Assumed fleet age	0.5 yr	1.0 yr	1.5 yr	2.0 yr	2.5 yr
Pre-tax effect ($m)	(15,008)	(18,595)	(23,648)	(31,097)	(42,755)

The gap between Method 1 and Method 2 is a measure of how much of the extensions' benefit has already been consumed.

2.7.3 Method 3: cumulative

Summing only the **disclosed full-year net income effects** — Meta FY2022 and FY2025, Amazon FY2022, FY2024, FY2025 and the heavy-equipment offset — gives **$8,483 million of cumulative disclosed benefit** [**C from V inputs**]. That figure deliberately excludes the quarter-only disclosures (Microsoft $859m, Alphabet $770m, Meta $695m), which no company ever annualised, and the two changes with no disclosed effect at all.

Projected forward on a 5-year reversion with the asset base held flat in year one and growing 25% thereafter, the cumulative three-year pre-tax effect is **−$62.9 billion** across all six, or **−$10.7 billion** on the clean-disclosure subset [**C**].

2.7.4 The validation, which is the part that matters

Any model of this kind is worthless unless it can reproduce a company's own audited arithmetic. Meta is the only available test, because it alone discloses a servers-only asset base, servers-only depreciation, and the dollar effect of its own change.

The naive calculation **fails**. Meta's average FY2025 servers/network gross base was $83,219 million; moving that from 5.0 to 5.5 years gives $1,513 million, against a disclosed $2,920 million. Roughly half.

Solving instead for the pre-change life that reproduces the disclosure gives **4.61 years** [**C**] — squarely inside the "**4 to 5 years**" range Meta stated in its FY2024 10-K. Re-running at 4.6 years gives **$2,960 million against a disclosed $2,920 million**, a difference of 1.4% [**C**].

The model ties. And the exercise produces a finding of its own: **Meta's real blended pre-2025 server life was approximately 4.6 years, not 5.0.** The January 2025 change was therefore roughly a 0.9-year extension, not a 0.5-year one — nearly twice what the headline range implies.

One further note from the validation. Meta's disclosed $2,920 million depreciation reduction became $2,590 million of net income — an **implied effective tax rate of about 11.3%**, not 21%. The model's 21% assumption is therefore conservative: it *understates* the after-tax benefit of extensions. The workbook lets the reader change it.

2.8 What this section concludes

Verified: every company in this set has changed server useful lives at least once since 2020. Meta's most recent change added $2.59 billion to FY2025 net income and $1.00 to diluted EPS on its own arithmetic. Amazon's most recent change subtracted $1.00 billion. Oracle's 2024 extension was worth $573 million and it disclosed that. Alphabet extended building lives to forty years and has never quantified it. NVIDIA's product cadence has compressed to twelve months and NVIDIA says so in its own risk factors. No public company has ever recorded a GPU impairment. Six-year-old A100s are running at full utilisation on NVIDIA's own account, and nine-year-old V100s are still on published price lists.

Analyst judgement, clearly labelled as such: the sensitivity is large enough to matter — a five-year reversion is worth roughly \$16.5 billion of annual pre-tax earnings across six companies, a four-year reversion roughly \$52 billion — but the case that current assumptions are *wrong* is not made by the evidence. It is made, at most, that they are *unverifiable from outside*, that two companies changed them without quantifying, and that the direction of travel has been overwhelmingly one-way while the product cycle moved the other way.

The thing to watch is not the assumption. It is **who changes it next, and whether they say by how much.** Section 10 dates that.

3. The circular-financing web

3.1 The right question

"Circular financing" is used loosely enough that it has stopped meaning anything. A supplier taking equity in a customer is ordinary industrial practice. What matters is narrower and testable:

Where is an entity simultaneously (a) an equity investor in, (b) a supplier to, and (c) a customer of the same counterparty — and is the money actually there?

That second clause is the one that does the work. This section separates every relationship in the complex into three categories: **funded and filed**, **binding but not yet funded**, and **announced and never filed**. The distribution turns out to be the finding.

3.2 The biggest number in the complex is not in any filing

On 22 September 2025, NVIDIA and OpenAI announced that "NVIDIA intends to invest **up to $100 billion** in OpenAI progressively as each gigawatt is deployed," under "a **letter of intent** for a landmark strategic partnership" covering "at least **10 gigawatts** of NVIDIA systems," with the first gigawatt "deployed in the second half of 2026 on the NVIDIA Vera Rubin platform," and the parties looking forward to "finalizing the details of this new phase of our partnership in the coming weeks" [**V, both companies' own releases**].

The phrase "non-binding" appears nowhere. Non-bindingness is inferable only from "letter of intent" and "intends to."

Now the filing trail, established by EDGAR full-text search against NVIDIA's CIK [**V, negative findings**]:

- **"up to $100 billion"** → exactly **one** hit in NVIDIA's entire filing history, and it is unrelated: an 8-K exhibit of 14 September 2020 describing the SoftBank Vision Fund. **No NVIDIA filing has ever described an up-to-$100-billion investment in OpenAI.**
- **"first tranche"** → **zero hits** in any NVIDIA filing, ever.
- **"10 gigawatts"** → one hit, in the Q3 FY2026 earnings press release filed as an 8-K exhibit. Never in a 10-K or 10-Q.

What the filings do say, in sequence:

- **Q3 FY2026 10-Q** (period 26 October 2025): contains "letter of intent" and an intention "to invest in OpenAI," **with no dollar amount and no gigawatt figure**.
- **FY2026 10-K** (filed 25 February 2026): replaces this with an unquantified "agreement with OpenAI," relocated into a **new risk factor titled "Commercial arrangements expose us to counterparty risks."**
- **Q1 FY2027 10-Q** (filed 20 May 2026): **does not mention OpenAI at all.**

The reported revision — to roughly $30 billion and five gigawatts, announced 27 February 2026 inside OpenAI's round that closed 31 March 2026 — has **no primary confirmation from either party** [**G**]. The only supporting statement located is Jensen Huang at the Morgan Stanley TMT conference on 4 March 2026, saying the original $100 billion was "probably not in the cards" [**R**].

The verdict: announced only. Not one dollar of the $100 billion is traceable to any filing. This is not an accusation of impropriety — a letter of intent creates no accounting obligation and NVIDIA was under no requirement to quantify it. It is a statement about what the public record supports, and the answer is: much less than the headline.

3.3 The control case: NVIDIA and Intel

For contrast, here is what a real transaction looks like in the documents.

Intel's 8-K of 18 September 2025 discloses a securities purchase agreement for **214,776,632 shares at $23.28 per share, an aggregate cash purchase price of $5.0 billion**, subject to HSR clearance. The exhibit describes Intel building "x86 system-on-chips that integrate NVIDIA RTX GPU chiplets" and "NVIDIA-custom x86 CPUs"

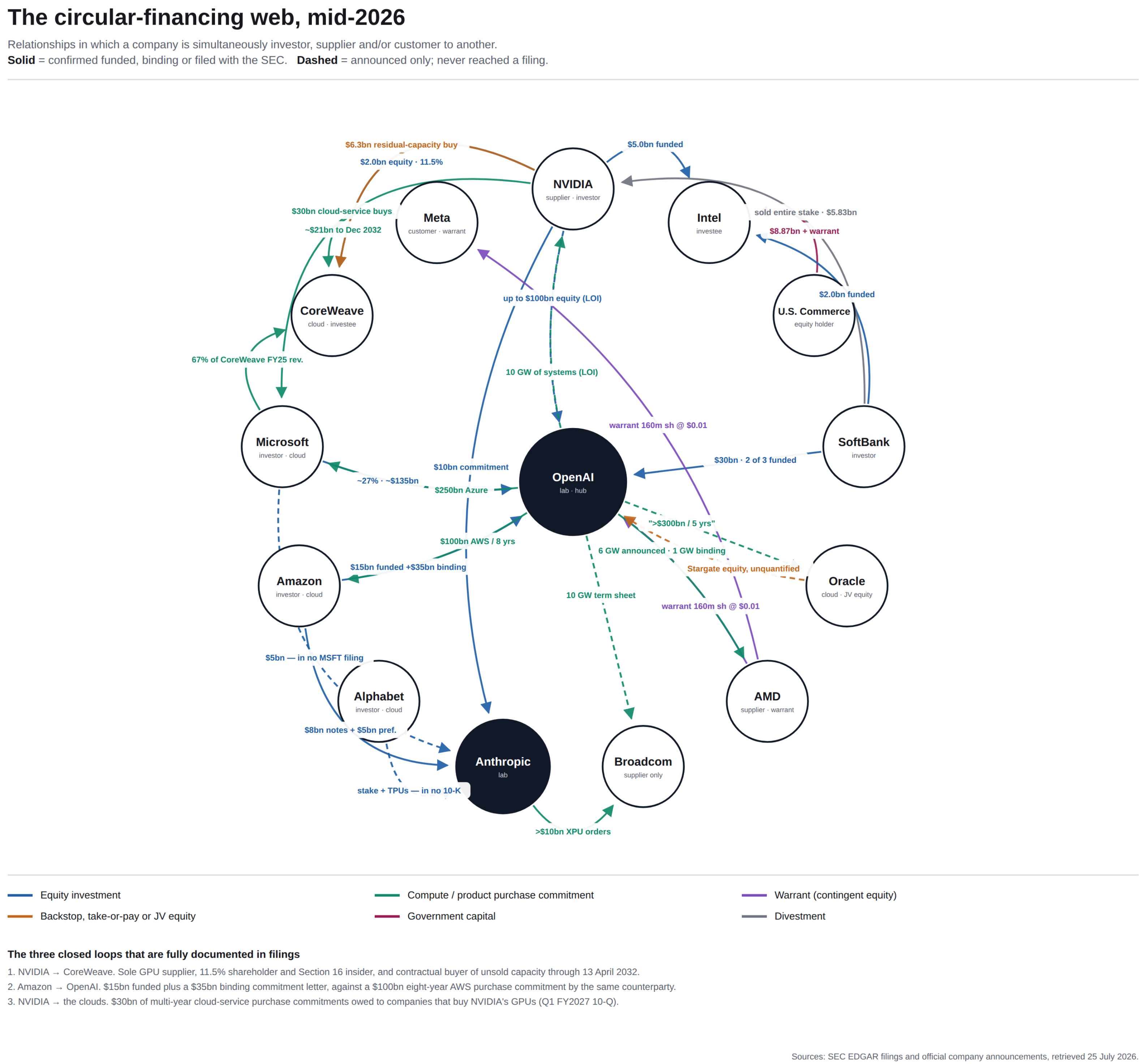

Figure 1: The circular-financing web at 25 July 2026. Solid edges are confirmed funded, binding or filed with the SEC; dashed edges were announced and never reached a filing. The complete edge register — 27 relationships with amounts, dates and sources — is in the companion artefact `circular_financing_map.html`.

for data centres, connected by NVLink [**V**]. A closing 8-K filed 29 December 2025 confirms: "On December 26, 2025, Intel Corporation completed the issuance and sale of 214,776,632 shares... for an aggregate purchase price in cash of $5.0 billion" [**V**].

The position then appears, independently, on NVIDIA's 13F-HR: 214,776,632 INTC shares worth $7.93 billion at 31 December 2025 and $9.48 billion at 31 March 2026 — **NVIDIA's largest disclosed 13F position** [**V**]. Against 5,044 million Intel shares outstanding, that is **4.26%**, below the 5% threshold, consistent with no Schedule 13D or 13G on file [**C**]. And it flows through NVIDIA's income statement: **equity gains of $5,491 million in Q4 FY2026 and $8,918 million for FY2026** [**V**].

Announcement, agreement, closing, ownership filing, and P&L effect — five independent documents. That is the standard against which the rest of this web should be judged.

One footnote worth recording: **no product from the NVIDIA–Intel co-development had shipped as of Intel's Q2 2026 earnings release on 23 July 2026** [**V**].

3.4 The loop that closes: NVIDIA and CoreWeave

CoreWeave acknowledged all three roles in a single sentence in its 8-K of 15 September 2025, verbatim:

> "...an **initial value of $6.3 billion**... in instances where the Company's datacenter capacity is not fully utilized by its own customers, **NVIDIA is obligated to purchase the residual unsold capacity through April 13, 2032**... The Company has determined that the MSA is a material agreement... because the MSA is **no longer immaterial** in amount or significance... **In addition to the MSA, NVIDIA supplies the Company with NVIDIA GPUs and is a stockholder of the Company.**" [**V**]

Then, on 23 January 2026, NVIDIA bought **22,935,780 Class A shares at $87.20 for $2.0 billion cash** in a Section 4(a)(2) private placement, and on 26 January filed a **Schedule 13G/A showing 47,213,353 shares, 11.5% of the class**, together with a **Form 3 as a 10% owner** [**V**]. CoreWeave's Q1 2026 equity statement corroborates: "Issuance of common stock in a private placement, net of issuance costs — 23 million shares, $1,985m net" [**V**]. NVIDIA's 13F history shows 24,182,460 shares from Q1 2025, held flat through Q4 2025, then 47,213,353 at Q1 2026. **NVIDIA has never sold a CoreWeave share** [**V**].

Set against CoreWeave's disclosed concentration — "approximately 67% of our revenue from our top customer, Microsoft, for the year ended December 31, 2025," falling to 45% for Customer A in Q1 2026 [**V**] — the picture is a company whose supplier is its largest shareholder and its buyer of last resort, and whose revenue comes overwhelmingly from a single hyperscaler.

Two things here are worth flagging as anomalies rather than accusations.

First, a disclosure anomaly. CoreWeave's Q1 2026 related-party note names **only Magnetar Financial and an unconsolidated joint venture. NVIDIA does not appear** — despite having become a greater-than-10% principal owner and Section 16 insider *inside the reporting period*. Under ASC 850, a principal owner is a related party [**V**].

Second, the backstop lives in exactly one document. The word "unsold" appears in **no CoreWeave 10-K or 10-Q, ever** [**V**]. The residual-capacity obligation is disclosed in a single 8-K and nowhere else.

And a widely circulated figure is wrong and is corrected here. There is no "$1.3 billion NVIDIA backstop through April 2032." The only 1.3 in CoreWeave's IPO prospectus is **"approximately 1.3 GW" of contracted power**; "April 2032" does not appear in the prospectus at all; and the word "unsold" appears there only as underwriting boilerplate [**V**]. The 8-K's own "no longer immaterial" language is affirmative evidence that no material NVIDIA purchase obligation had been disclosed before.

3.5 The node everyone under-weights: Amazon

Amazon's Q1 2026 10-Q is the most explicit circular-financing disclosure in the entire complex, and it is written in plain English [**V**]:

> "In Q1 2026, we invested **$15.0 billion in Series C Preferred Stock of OpenAI**, and we also entered into an **equity commitment letter agreement**... pursuant to which we agreed to purchase additional shares of Series C Preferred Stock... with an aggregate purchase price of **$35.0 billion**... we are **obligated to purchase all remaining Commitment Shares** upon the earlier to occur of (i) OpenAI meeting specified milestones, and (ii) OpenAI directly or indirectly consummating an **initial public offering** or direct listing..."

And on the other leg, verbatim: "AWS and OpenAI Group PBC announced an **expansion of the existing $38.0 billion multi-year commitment** and commercial arrangement with OpenAI **by $100.0 billion over 8.0 years**" — an aggregate of roughly $138 billion, not $100 billion, and a distinction most coverage gets wrong. The arrangement includes "the use of AWS chips" and "contractual obligations related to the performance of AWS chips" **[V]**.

Amazon has also invested **$8.0 billion in Anthropic convertible notes** from Q3 2023 to Q4 2025, carried at fair value as Level 3 assets, with portions converted to non-voting preferred in Q1 2026 producing a **$4.5 billion reclassification adjustment** out of accumulated other comprehensive income — plus a further **$5.0 billion of Anthropic non-voting preferred** in April 2026 **[V]**.

This is structurally identical to the NVIDIA–OpenAI arrangement — a supplier funding a customer's purchase of the supplier's own product — at a larger funded scale, and it is fully disclosed. The asymmetry in public attention between the two is not explained by the documents.

3.6 Oracle: the backlog with no named counterparty

Oracle has **never stated the $300 billion figure in any filing.** EDGAR full-text search for "300 billion" across all Oracle filings returns **zero hits [V]**. The number comes from OpenAI's own post of 23 September 2025: "In July, OpenAI and Oracle entered an agreement to develop up to 4.5 gigawatts of additional Stargate capacity. This represents a partnership that **exceeds $300 billion** between the two companies over the next five years" **[V, as an OpenAI statement]**.

What Oracle does disclose is RPO, and the trajectory is extraordinary **[V]**:

Quarter end	RPO	Change
31 May 2025	$137.8bn	+41% y/y
31 Aug 2025	**$455.3bn**	**+$317bn in one quarter**
30 Nov 2025	$523.3bn	+$68bn
28 Feb 2026	$552.6bn	+$29bn
31 May 2026	**$638bn**	+$85bn; +363% y/y

Oracle's own attribution for the Q1 FY2026 jump: "We signed four multi-billion-dollar contracts with three different customers in Q1" — **customers unnamed [V]**.

Oracle names OpenAI in exactly two SEC-filed documents: the 8-K of 10 March 2025 ("We have now signed cloud agreements with... OpenAI, xAI, Meta, NVIDIA and AMD") and a free-writing prospectus of 2 February 2026 ("contracted demand from our largest Oracle Cloud Infrastructure customers, including AMD, Meta, NVIDIA, **OpenAI**, TikTok, xAI and others") **[V]**. The second is bond marketing, not a periodic report.

The concentration gap is the structural point. Oracle's FY2026 10-K states that "No single customer accounted for 10% or more of our total revenues in fiscal 2026, 2025 or 2024" — and discloses **no customer's share of the $638 billion RPO anywhere [V]**. Because the largest contract begins around fiscal 2028, it contributes essentially nothing to FY2026 revenue while dominating backlog. The concentration is measured against the one denominator in which it cannot appear.

There is partial de-risking, and Oracle disclosed it in the Q4 FY2026 release: "Most of the RPO increase in both Q3 and Q4 were large scale AI contracts where the customer prepaid Oracle for the purchase of the GPUs, or the customer bought and supplied the GPUs to Oracle. **The prepaid and customer supplied hardware portions of our large AI contracts now total $75 billion**" **[V]**. This cuts both ways: bears read it as RPO inflation, but prepayment is the *opposite* of vendor financing — the customer is funding Oracle. The residual is that roughly **$563 billion of the $638 billion is not prepaid [C]**.

3.7 AMD: the warrant, and the accounting that almost nobody reports correctly

AMD's 8-K of 6 October 2025 discloses a warrant for "up to an aggregate of **160 million shares**" at "**$0.01 per share**," with the first tranche vesting on delivery of "the initial one (1) gigawatt," full vesting requiring purchase of "six (6) gigawatts," and vesting "subject to achievement of specified Company stock price targets that **escalate to $600 per share for the final tranche**," exercisable until 5 October 2030 **[V]**.

Two things are usually got wrong about this deal.

First, the commitment. OpenAI made "a **binding commitment to purchase**... the initial one (1) gigawatt." **The 6 GW headline overstates the contractual commitment by a factor of six [V]**.

Second, the accounting. From AMD's Q3 2025 10-Q, verbatim:

> "The Company expects to account for the warrant as a **liability-classified financial instrument**, unless the conditions for equity classification are satisfied." "**The grant date fair value of the warrant will be recognized as a reduction to revenue when revenue is recognized.**" **[V]**

That is consideration payable to a customer under ASC 606 — **contra-revenue, not a dilution charge, and liability-classified, not equity-classified.** Almost all secondary coverage of this deal describes it as equity dilution. It is not.

Nothing has been recognised yet. As of 28 March 2026: "none of the warrant shares had vested or become exercisable," and no warrant remeasurement line appears in FY2025 other income **[V]**. **The grant-date fair value has never been disclosed in dollars [G]** — and the operative economics are redacted: the filed warrant refers vesting to "Exhibit E" and exercise conditions to "Exhibit F," both marked [* * * *] **[V]**. The $600 ladder exists only in AMD's narrative summary, not in the filed legal text.

The structure has replicated. AMD issued a **parallel 160-million-share warrant to Meta** (announced 24 February 2026, 6 GW), and the Q1 2026 10-Q now uses the plural: "The **warrants** vest in tranches based on AMD Instinct GPU purchase milestones achieved by **OpenAI, Meta**, their affiliates, or indirectly through authorized third parties." Combined potential dilution: **up to 320 million shares at $0.01 [V]**.

One item that deserves more attention than it has received: AMD's FY2025 10-K subsequent events disclose that AMD "entered into an agreement to **guarantee a commercial partner's data center lease obligations** in the event of their default," with a **maximum gross exposure of $4.1 billion**. The counterparty is unnamed **[V/G]**.

3.8 Broadcom: the counter-example inside the complex

Broadcom is the node that proves the others are unusual. Its 13 October 2025 announcement with OpenAI covers "10 gigawatts of custom AI accelerators," deployment "to start in the second half of 2026, to complete by end of 2029" — and the parties "have signed a **term sheet**," not a definitive agreement. **No dollar figure. No equity. No warrant. No investment.** Broadcom is a pure fee-for-silicon vendor **[V]**.

Broadcom has never named OpenAI in an SEC filing. EDGAR full-text search for "OpenAI" across Broadcom's CIK returns **zero hits**, against a control test on AMD that returned 16 **[V]**.

And a major misattribution is corrected here. Broadcom's September 2025 disclosure of "over $10 billion of orders of AI racks based on our XPUs" from an unnamed "fourth and very significant customer" was widely attributed to OpenAI. **On 11 December 2025 Broadcom revealed the customer was Anthropic [V]**. Any model that treats that $10 billion as OpenAI demand double-counts.

On the Q2 FY2026 call of 3 June 2026 Broadcom abandoned the unnamed-customer convention altogether and named Google, Anthropic (1 GW in 2026, +5 GW from 2027), OpenAI (**1.3 GW in 2027** under the 10 GW/2029 commitment) and Meta (3 GW through 2028). Q2 FY2026 AI semiconductor revenue was **$10.8 billion, +143%**, against "bookings for AI semiconductors... over $30 billion" [**V**].

3.9 Microsoft, SoftBank, Anthropic

Microsoft–OpenAI. From OpenAI's own post of 28 October 2025: Microsoft's stake is "approximately $135 billion, representing roughly **27 percent on an as-converted diluted basis**"; OpenAI commits to "**$250B of incremental Azure services**"; IP rights run through 2032; AGI declaration is now "verified by an independent expert panel"; and critically, "**Microsoft will no longer have a right of first refusal to be OpenAI's compute provider**" [**V**].

Microsoft's accounting, verbatim from the Q3 FY2026 10-Q: other income "included $19 million of net losses and **$5.9 billion of net gains** for the three and nine months ended March 31, 2026... primarily net recognized gains (losses) on our **equity method investment**... The net gains recorded for the nine months ended March 31, 2026 **primarily relate to the dilution gain from the OpenAI Recapitalization**" [**V**]. Equity-method investments went from $6.0 billion at 30 June 2025 to $11.1 billion at 31 March 2026. **Microsoft's position is not fair-valued to the reported $135 billion**; it is equity-method with a dilution gain.

SoftBank is the reflexivity node. From its own IR materials [**V**]: a **$30.0 billion** follow-on OpenAI investment announced February 2026 in three tranches — tranche 1 closed 1 April 2026, **tranche 2 closed 1 July 2026 at "USD 10.0 billion (JPY 1,627.3 billion)" funded by drawing $10.0 billion under a bridge facility**, tranche 3 scheduled 1 October 2026, with closing "subject to acceleration in the event of a public listing of OpenAI Group PBC shares." Cumulative OpenAI investment reaches **$64.6 billion by October 2026** for roughly 13% ownership. The valuation marks used: $150bn (Sep 2024) → $260bn (Mar 2025) → $500bn (Oct 2025) → $730bn (Feb 2026). SoftBank sold its **entire NVIDIA stake for $5.83 billion** to help fund it, and reported FY2025 net income of **¥5,002.3 billion, +334% —** described as the highest ever by a Japanese corporation.

The circularity is explicit and SoftBank does not hide it: the record profit is largely the mark-up on OpenAI, whose valuation is set by rounds SoftBank itself leads and funds with borrowed money.

Anthropic is where the disclosure gaps cluster. NVIDIA's Q3 FY2026 10-Q XBRL names "Anthropic, PBC $10,000M" as an investment commitment [**V**] — but **"Anthropic" appears in neither NVIDIA's FY2026 10-K nor its Q1 FY2027 10-Q**, the 10-K reverting to an unnamed $11.4 billion aggregate [**V**]. Microsoft's announced up-to-$5 billion investment and Anthropic's ~$30 billion Azure commitment **appear in no Microsoft 10-Q or 10-K [V, negative]**. Alphabet's Anthropic stake and the October 2025 TPU agreement appear **only in DEF 14A proxies, never in a 10-K or 10-Q [V, negative]**. Amazon is the exception: fully filed and quantified.

3.10 How this is being accounted for, and who is looking

Arrangement	Disclosed treatment
AMD → OpenAI / Meta warrants	**Liability-classified**; grant-date FV as **reduction of revenue**
Microsoft → OpenAI	**Equity method**; $5.9bn dilution gain, 9M FY2026
Amazon → Anthropic	AFS convertible notes at fair value, **Level 3**; reclassified on conversion
Amazon → OpenAI	Series C preferred + commitment letter, mandatory on IPO
NVIDIA → Intel, CoreWeave	Marketable equity, FV through P&L ($5.5bn Q4 FY26 gain)
NVIDIA → private companies	**Non-marketable equity: $3.4bn → $42.3bn in five quarters**
Intel → U.S. Commerce	Escrowed shares as a **derivative liability**, marked to market

All [**V**]. Two of these deserve emphasis. NVIDIA's non-marketable equity book grew **$17,899 million in the single quarter to 26 April 2026 —** a near-doubling in thirteen weeks **—** with **$27 billion of further investment commitments** disclosed [**V**]. And Intel's derivative liability on the escrowed government shares grew from **$2,700 million to $15,600 million** between December 2025 and June 2026, producing **$12,873 million of fair-**

value losses in H1 2026 and a GAAP net loss of $(11.0) billion against non-GAAP net income of $2.2 billion in Q2 2026 alone [**V**]. The government's gain is Intel's reported loss.

On oversight, the finding is a null result and it is worth stating. No SEC comment letter, enforcement action, DOJ or FTC inquiry, or Congressional hearing addressing these circular arrangements was located for any company in this web [**G**]. The only verified regulator correspondence in the complex — ten UPLOAD letters to CoreWeave — concerns S-1/S-4 mechanics and tax-opinion qualification. The Federal Reserve's May 2026 Financial Stability Report registers AI as a top survey risk (50%, from 0%) and performs no analysis of the financing structures [**V**].

The most substantive public critique remains a short seller's: **Kerrisdale Capital, "CoreWeave Inc (CRWV): Artificial Returns," 15 September 2025** — published the same week as the $6.3 billion NVIDIA 8-K — which described "a debt fueled GPU rental business with no moat, dressed up as innovation" and explicitly identified NVIDIA as simultaneous investor and primary supplier [**R**].

3.11 What this section concludes

Verified: the fully-closed loop is NVIDIA–CoreWeave, and it closed in January 2026. The largest funded supplier-to-customer equity position in the complex is **Amazon's in OpenAI**, not NVIDIA's. AMD's warrant is contra-revenue and liability-classified, and the 6 GW headline covers a 1 GW binding commitment. Broadcom takes no equity and has never named OpenAI in a filing. Oracle has never stated the $300 billion figure and discloses no customer's share of a $638 billion backlog. NVIDIA's non-marketable equity book grew twelvefold in five quarters.

The structural finding, and it is the one to carry forward: **the deals that convert into filings and the deals that do not are not randomly distributed.** Intel, CoreWeave and Amazon converted — with purchase agreements, closing 8-Ks, ownership filings and P&L effects. NVIDIA's $100 billion for OpenAI, Broadcom's 10 GW term sheet, Microsoft's $5 billion for Anthropic and Alphabet's Anthropic stake did not. An investor reading press releases and an investor reading filings are looking at two different industries.

Analyst judgement: the circularity is real but smaller than the headlines and larger than the bulls concede. NVIDIA's equity book at $42.3 billion is roughly 13% of annualised revenue — a level that does not manufacture demand. The velocity — $17.9 billion added in one quarter, $27 billion more committed — is the number that would make the critique quantitatively correct if it persists for four more quarters. That is a monitorable, and it is in Section 10.

4. Off-balance-sheet debt

4.1 The finding that reframes the section

The expected story here is the special-purpose vehicle: Meta's Hyperion, Oracle's Abilene, xAI's Colossus. Those exist and they are documented below. But they are not the main event.

The dominant off-balance-sheet channel in AI infrastructure is ASC 842's commencement-date recognition rule. A lease creates no right-of-use asset and no lease liability until it *commences*. A sixteen-year data-centre lease signed in 2026 for 2029 delivery produces zero assets and zero liabilities for three years, and appears only in the disclosure sentence required by ASC 842-20-50-4(g).

Here is what that sentence contains, across six issuers, all figures [V] read from the filings:

Entity	Leases not yet commenced	As of	Recognised lease liabilities	Off : on
Oracle	**$261.0bn**	28 Feb 2026	$27.56bn ($21.31 op + $6.25 fin)	**9.5x**
Oracle	$260.0bn	31 May 2026	$30.19bn operating	8.6x
Microsoft	**$196.6bn**	31 Mar 2026	$85.17bn ($22.24 op + $62.93 fin)	2.3x
Meta	**$182.88bn**	31 Mar 2026	$28.02bn operating	6.5x
Amazon	**$106.3bn**	31 Mar 2026	$126.6bn undiscounted op+fin	0.8x
Alphabet	**$75.6bn**	31 Mar 2026	$18.38bn ($16.16 op + $2.21 fin)	4.1x
CoreWeave	**$40.7bn**	31 Mar 2026	$10.29bn	4.0x
Total	**~$863bn**	Q1 2026		

Roughly $863 billion of contractually committed lease obligations sits outside these six balance sheets. For comparison, total US data-centre ABS and CMBS issuance in 2025 — the best year on record, roughly triple the prior year — was about **$26–30 billion** [V].

The trajectories matter as much as the levels. Meta's uncommenced leases went **$58.14bn (30 Sep 2025) → $103.77bn (31 Dec 2025) → $182.88bn (31 Mar 2026)** — a 3.1x increase in six months [V]. Oracle's went **$43.4bn (31 May 2025) → $99.8bn (31 Aug 2025) → $248bn (30 Nov 2025) → $261bn (28 Feb 2026) — 6x in nine months** — while the stated lease terms on those uncommenced leases lengthened from 10–16 years to **15–19 years** [V].

Moody's put a number on the aggregate in a sector report published around 22 July 2026: capex of **$785 billion in 2026** approaching $1 trillion the following year; direct debt across the six of roughly **$460 billion**; and **lease commitments of $1.2 trillion, of which more than $820 billion has not yet commenced** — which Moody's called **"debt-equivalent liabilities"** [R].

Alphabet's disclosure is the clearest of the six and is worth quoting verbatim, because it shows how much information is in one sentence:

> "As of March 31, 2026, we have entered into leases primarily related to data centers that have not yet commenced with future lease payments of **$75.6 billion that are not yet recorded**... these leases"will commence between 2026 and 2031 with non-cancelable lease terms primarily between one and 25 years." [V]

Alphabet separately disclosed **$634 million of lease prepayments for leases not yet commenced**, expected to be finance leases, in Q1 2026 alone [V]. Cash is already moving on obligations that appear nowhere on the balance sheet.

Amazon is the outlier in the other direction, and deserves credit for it: it discloses uncommenced leases as a **column in its commitments table**, with a maturity profile. Of its $106.3 billion, **$69.8 billion falls after 2030**

[V]. Amazon's total contractual commitments at 31 March 2026 are **$569.3 billion** [V].

4.2 Meta's Hyperion: the template, in Meta's own words

Meta's Q1 2026 10-Q contains the single most informative paragraph written by any sponsor about any of these structures — and note where it sits: not in Commitments and Contingencies, but inside the **Non-Marketable Equity Investments** note. Verbatim, and at length, because the detail is the point:

> "Our non-marketable equity method investments include an arrangement, entered into in October 2025, to co-develop a data center campus in Louisiana (the Venture), in which we hold a **20% membership interest**... The parties have committed to fund their respective pro rata share of approximately **$27 billion** in total estimated development costs.
>
> Our lease agreements with the Venture... will **commence in 2029** and have an **aggregate initial lease commitment of approximately $12.31 billion**. Each leased property has an **initial four-year lease term** and options to renew for a total lease period of **up to 20 years**. In addition, we have provided **residual value guarantees (RVG) with an aggregate threshold of approximately $28 billion** that decreases over time... **RVG payments are not probable, and therefore no liability has been recorded to date.**
>
> Significant judgment is required to identify the activities that most significantly impact the Venture's economic performance... **decisions pertaining to remarketing the data center campus**, including but not limited to, negotiations with future lease tenants and individual property sales, were determined to have the most significant impact... **As we do not have the power to direct the activities that most significantly impact the Venture's economic performance, we are not the primary beneficiary and, therefore, do not consolidate the variable interest entity (VIE).** Our ongoing involvement with the VIE includes providing **construction management, administrative and property management services** to the Venture.
>
> As of March 31, 2026... the carrying value of our equity investment... was **$2.37 billion**... and our **maximum exposure to loss related to the Venture was $45.99 billion**, consisting of the carrying value of our equity investment, the lease commitments, our estimated future funding commitments, and the maximum RVG threshold." [V]

Read that consolidation argument again. It rests on a single hinge: that decisions about *remarketing* the campus — finding replacement tenants, selling individual properties — are the activities that most significantly affect the venture's economics, and Meta does not control them. Meta is the sole tenant. Meta is the construction manager. Meta is the property manager. Meta provides a $28 billion residual value guarantee. Meta carries $45.99 billion of maximum exposure against a $2.37 billion carrying value.

It is a defensible position under ASC 810 and Meta's auditors signed it. It is also a position that depends entirely on which activity you designate as "most significant," and Meta chose the one it does not control.

Meta never names its counterparty. EDGAR full-text search: **"Beignet" returns zero hits in any Meta filing, ever** (23 documents across eight companies contain the word; all are Popeyes and AFC Enterprises restaurant filings). **"Hyperion" returns two hits, both 2026 proxy documents — zero in any 10-K or 10-Q. "Residual value guarantee" (singular) appears in exactly one Meta filing before the Q1 2026 10-Q: the 10-Q for the quarter ended 30 September 2025 [V]**.

The financing terms themselves come from the trade press, because it is a 144A private placement and is not filed on EDGAR [**R throughout**]: issuer **Beignet Investor LLC**, a bankruptcy-remote SPV; **$27.294 billion**, priced 16 October 2025 at par; **6.581% fixed**, quarterly, 30/360; single tranche, fully amortising; expected final maturity **30 May 2049**; weighted average life ~23.6 years; **T+225bp**; 144A-for-life; rated **S&P A+**, one notch below Meta itself; Blue Owl-managed funds hold **80%** of the JV equity to Meta's 20%.

4.3 The dispute the rating agencies are having in public

This is the most important disagreement in the sector, and both sides are on the record.

S&P, approving non-consolidation: "the way that the Beignet deal was structured gives the hyperscaler **sufficient optionality** to warrant not consolidating." And on the guarantee: "We currently **do not make a debt adjustment for the RVG** due to significant headroom between the RVG and a third-party appraisal value." **[R, quoted in trade press]**

Moody's, dissenting: "The accounting at this time under the current rules is **not in line with the expected economics**." And: "We foresee a **material increase in adjusted debt and lease-related cash outflows** for these companies in the coming years." And: "These structures are unique and we plan to continue to evaluate potential adjustments on a **case-by-case basis**." **[R, same source]**

Moody's identified the mechanism precisely: **initial lease terms of roughly four years** against a historic 10–15, renewal options that escape ASC 842 recognition until "reasonably certain," and residual value guarantees that escape ASC 460 recognition until payment is "probable." Meta's own disclosure confirms every element of that description: four-year initial terms, renewals to 20 years, RVG not probable, no liability recorded.

Moody's published a Sector In-Depth on 23 February 2026 whose title says it outright: **"Hyperscalers' reported AI-related lease commitments may understate economic risk" [R]**.

Note also what S&P's **A+** rating on the Beignet notes implies. It is one notch below Meta's own AA−. S&P is effectively rating the vehicle on Meta's credit — treating the lease-plus-guarantee package as near-recourse to Meta — while simultaneously accepting Meta's conclusion that it does not control the vehicle. **The rating and the accounting reach opposite conclusions about who bears the risk.**

4.4 Oracle's structures

Oracle's off-balance-sheet position is the largest in the sector and its counterparties are the least documented.

The recognised position at 28 February 2026: operating leases undiscounted $28,726m less imputed interest $(7,416)m → **PV $21,310m**; finance leases undiscounted $9,191m less $(2,941)m → **PV $6,250m**. Total **$27.6 billion recognised against $261.0 billion committed [V]**.

The guarantee nobody discusses. Oracle's Q3 FY2026 10-Q tags a **"Lease guaranteed of lessor's borrowing" of $2.2 billion, maturing September 2026**; the FY2026 10-K restates it as **"a lease for which we have guaranteed up to $3.3 billion of the lessor's borrowing, which matures in September 2026" [V]**. This is the same economic species as Meta's RVG — explicit credit support to a lessor's debt — disclosed as a tagged fact rather than narrative, and it comes due within weeks. Oracle also discloses **$11.0 billion of unconditional purchase and other obligations**, plus a subsequent-event **$19 billion of unconditional purchase commitments for cloud infrastructure assets** commencing FY2027 on a five-year term **[V]**.

The sites. Ownership of the buildings sits with third parties in every case **[R throughout — none of these structures is filed on EDGAR]**:

- **Abilene, Texas (Stargate I):** developer/operator **Crusoe Energy**; JV equity from **Blue Owl Capital** and **Primary Digital Infrastructure**; a **~$9.6 billion** construction facility led by **JPMorgan**; second phase reported at **~$15 billion**. Oracle is the lessee; OpenAI is the compute offtaker. SPV legal entity names, tranching, coupons and lease terms are **[G]**.
- **Port Washington, Wisconsin and Shackelford County, Texas — Vantage:** a **$38 billion** bank debt package led by **JPMorgan and MUFG**, split **$23bn (Texas) / $15bn (Wisconsin)**; four-year maturities with two one-year extensions; preliminary pricing around **S+250**.
- **New Mexico:** a separate **~$18 billion** financing for a **STACK Infrastructure** site, led by SMBC, MUFG, BNP Paribas and Goldman Sachs. **This is commonly and wrongly attributed to Vantage; it is not a Vantage deal.**
- **Saline Township, Michigan — Related Digital**, confirmed 24 April 2026 by three concurrent press releases from Related, Blackstone and Bank of America **[V]**: **$16 billion** total, tenant **Oracle**, **>1 GW**;

equity from Related Digital and Blackstone-affiliated funds; debt described as "fixed-rate, long term debt financing **anchored by PIMCO-managed funds and accounts**"; Bank of America structuring agent.

Context worth recording: **Blue Owl withdrew from a $10 billion Oracle-linked Michigan financing in December 2025**, and Oracle's stock fell 5% on the news [**R**]. Whether PIMCO refinanced the same site is [**G**].

4.5 CoreWeave: the counterexample that proves the mechanism

CoreWeave uses the same bankruptcy-remote SPV architecture as everyone else. Its debt is **on** its balance sheet. The reason is instructive.

There are **five delayed-draw term loan facilities, not three**, each with its own ring-fenced borrower SPV [**V**]:

Facility	Size	Borrower SPV	Pricing	Maturity	Rating
DDTL 1.0	$2.3bn	—	~15% eff.	Mar 2028	unrated
DDTL 2.0	$7.6bn	—	~11% eff.	Aug 2030	unrated
DDTL 2.1	$3.0bn	[**G**]	~9% eff.	Mar 2031	[**G**]
DDTL 3.0	$2.6→3.0bn	CCAC V & CCAC VII	SOFR+400	Aug 2030	unrated
DDTL 4.0	**$8.5bn**	CCAC VIII	**SOFR+225** / ~5.9% fixed	Mar 2032	**A3 / A(low)**
DDTL 5.0	**$3.1bn**	CoreWeave Financing DDTL V, LLC	**SOFR+450**	Nov 2031	Ba2 / BB+

The collateral is not the GPUs as such. DDTL 4.0, verbatim: "Obligations outstanding under the DDTL 4.0 Facility are secured by perfected first priority pledges of and security interests in (i) **the equity interests of CCAC VIII** held by its direct parent and (ii) substantially all of the assets of CCAC VIII." Parent recourse is limited: the parent guarantees "on a **limited recourse basis for specified 'bad acts'**" — a bad-boy guarantee [**V**]. DDTL 5.0 reverses that and carries an **unconditional** parent guarantee, and is collateralised by contracts with "two large, **non-investment grade** customers" [**V**]. That single distinction is why one prices at SOFR+225 and the other at SOFR+450.

And yet none of it is off balance sheet. CoreWeave's FY2025 10-K sets out the ASC 810 VIE test in standard language, but there is **no separate VIE footnote**: the borrower SPVs are wholly owned and consolidated under the voting-interest model [**V**]. CoreWeave uses identical bankruptcy-remote structures to Meta and, lacking a third-party equity partner to take control rights, gets no deconsolidation.

That is the mechanism, isolated. **The SPV does not remove the debt. The third-party equity partner does.**

4.6 xAI, Vantage, and the ABS market

xAI — and a correction. There is **no closed "$12.5 billion SPV."** The $12.5 billion figure is the debt tranche of a *proposed* ~$20 billion structure floated in press reports in October 2025, and no primary source confirms it ever closed [**R/G**].

What exists [**V**]: **Valor Compute Infrastructure L.P.** (CIK 0002074951), a Delaware LP, whose Form D/A of 15 May 2026 discloses **$2,649,026,842 sold** to 1,141 accredited investors. And an Apollo press release of 7 January 2026 describing a **$5.4 billion** acquisition and lease of compute infrastructure, with Apollo-managed funds leading a **$3.5 billion capital solution**; structure is a **triple net lease**; lessee is "a subsidiary of xAI Corp"; assets are NVIDIA GB200 GPUs; and **NVIDIA invested as an anchor limited partner**, amount undisclosed. Apollo describes its $3.5 billion only as a "capital solution" and "flexible, asset-based capital" — **never as debt, notes or preferred**. The debt/equity characterisation in press coverage is inference, not sponsor language.

No rating agency rates xAI, X.AI Holdings, Valor Compute Infrastructure, or any Colossus issuer. No residual value guarantee, completion guarantee, backstop or bankruptcy-remoteness language exists in any primary source for these vehicles **[G, assessed as a true negative]**. SpaceX's S-1 of 20 May 2026 confirms the xAI merger effective 2 February 2026 and discloses **Q1 2026 AI-segment capex of $7,723 million**; EDGAR full-text search for "residual value guarantee" in SpaceX filings returns **zero hits** **[V]**.

The securitisation market, from two independent industry sources that agree to the decimal:

Year	ABS	CMBS	Total
2021	$6.2bn	$3.2bn	$9.4bn
2022	$1.0bn	$0.3bn	$1.3bn
2023	$5.9bn	$2.7bn	$8.6bn
2024	$8.4bn	$3.0bn	$11.4bn
2025	**~$15bn**	**~$11bn**	**~$26bn**

Sources: CREFC Data Center e-Primer (25 February 2026) and KBRA (4 March 2026), which independently report CMBS at **42% of 2025 structured-finance issuance against 25% over 2018–24** **[V]**. S&P puts global data-centre securitisation above **$30 billion in 2025, nearly tripling from just over $10 billion in 2024**, and notes that **tech-sector debt reached about 16.7% of global non-financial corporate bond issuance, from 11.6% twelve months earlier** **[R]**. Morgan Stanley projects **$30–40 billion a year in 2026–27** and roughly **$130 billion of net issuance across 2026–28** **[R]**.

Tenant concentration in that market, verbatim from CREFC: the top five companies were **73% of 2025 demand** — AWS 22%, Microsoft 18%, Google 15%, Meta 11%, Oracle 7% **[V]**.

One structural warning from a rating agency deserves quoting, because it is the only one that names the actual risk. **Scope Ratings**, on a Vantage German transaction, 8 May 2025: "the total 25 years until the final maturity may provide room for development of new technologies or tightening in artificial intelligence regulation, **resulting in declining demand for data centres**." And: "Note-to-value ratios based on stressed value of collateral properties... are **above 100% at inception** and significantly higher than comparable transactions" **[V]**.

4.7 Why the accounting works, and what it does not remove

ASC 810. A sponsor consolidates a VIE only if it is the primary beneficiary, which requires *both* the power to direct the activities that most significantly impact economic performance *and* the obligation to absorb losses or right to receive benefits that could be significant. The Hyperion design defeats the first prong: Blue Owl holds 80% of the equity and the control rights over construction, financing and disposition. S&P's phrase — "sufficient optionality" — describes optionality held by the *owner*, not the tenant.

ASC 842. Two exemptions do the heavy lifting. Recognition begins at commencement, not signature — that is the entire $863 billion. And only renewal options "reasonably certain" of exercise enter the lease term, so structuring initial terms at four years with renewals keeps the recognised liability a fraction of the economic obligation even after commencement.

ASC 460. A residual value guarantee is recognised at fair value at inception and thereafter only when payment becomes probable and estimable. With appraised value above the guaranteed floor, the RVG stays a footnote. Hence Meta's up-to-$28 billion guarantee with no liability recorded.

Bankruptcy remoteness is the lenders' protection: assets and liabilities ring-fenced in a separate legal entity, first-priority security over the property and the lease receivable, insulated from the sponsor's estate. The corresponding fragility, noted by Quinn Emanuel in a February 2026 client alert: "if it is later recharacterized as a secured loan, the assets **may be pulled back into the bankruptcy estate**" **[R]**.

What survives deconsolidation. Three things, and none of them is removed by any of the above:

1. **The lease itself** — a fixed, non-cancelable payment stream over 15 to 19 years, with priority in practice equivalent to senior unsecured debt.

2. **The residual value guarantee —** Meta's ~$28 billion, Oracle's $3.3 billion lessor-borrowing guarantee, AMD's $4.1 billion lease guarantee. These convert an equity-risk position into a debt-like one at precisely the moment asset values fall.
3. **Operational recourse —** a sponsor cannot walk away from a facility running its flagship model without stranding the compute that facility exists to provide.

Non-consolidation removes the liability from the balance sheet. It removes none of the cash flow.

4.8 The official sector has named it

The single most important public-institution statement on this subject:

> **BIS Quarterly Review, 16 March 2026 — Box A, "Financing the AI infrastructure boom: on- and off-balance sheet borrowing"** (Eren, Krohn & Todorov), pages 3–4:
>
> "These arrangements amount to **'shadow borrowing'**: obligations that are economically akin to debt but largely reside outside corporate balance sheets."
>
> "A common structure involves a dedicated vehicle — often a joint venture or special purpose entity — that acquires or develops data centre assets... The debt is serviced by lease cash flows and is held by **private credit funds and other institutional investors**."
>
> "Banks support the vehicles with funding lines, **potentially creating new shock transmission channels —** eg via refinancing pressures at the vehicle level, procyclical shifts in private credit appetite or **the activation of guarantees**." **[V]**

Supporting, all **[V]**:

- **BIS Annual Economic Report 2026**, 28 June 2026: "The most prominent is **circular financing**: chip makers and hyperscalers take equity stakes in AI labs or neocloud providers, who in turn commit to multi-year purchases of chips or computing power." And: "**The opacity of AI-sector financing compounds these vulnerabilities.**"
- **Bank of England FPC Record**, 7 July 2026: "AI-related companies' use of credit markets has accelerated rapidly, including in public markets, private credit, leveraged and structured finance, and is set to increase further." Risks turn on "the size of future compute demand, the timely availability of power to data centres, and **the depreciation rate of data centre facilities and AI chips**."
- **NY Fed Liberty Street Economics**, 20 May 2026: "Beneath those headline bond issues lies a more intricate layer — off-balance-sheet project finance vehicles funding data center construction, securitizations backed by lease cash flows, and **hundreds of billions in forward lease commitments that will not appear on balance sheets for years**."
- **IMF GFSR, April 2026**: firms have "increasingly relied on 'circular financing', **although the impact on financial stability appears modest currently**." The hedge is deliberate and should be read as written.

And the conspicuous absences, which are themselves findings [V]: the **Federal Reserve's** Financial Stability Report has no substantive treatment — its November 2025 AI box concerns algorithmic trading, and the May 2026 edition mentions AI once, in a market-contact survey. The **OFR's** 2025 annual report discusses AI only as internal productivity. And the **FSB's** May 2026 report on vulnerabilities in private credit **omits AI and data-centre lending entirely**, despite private credit being the principal funding channel and despite the FSB's own Box 4 elsewhere noting that **data centres were 34% of private credit deals in 2025 against a 17% five-year average**.

4.9 What this section concludes

Verified: approximately $863 billion of committed lease obligations sit off six balance sheets, growing 3x to 6x within nine months at Meta and Oracle. Meta discloses $45.99 billion of maximum exposure to a single unconsolidated venture against a $2.37 billion carrying value, on a non-consolidation argument that turns on remarketing rights, and never names the counterparty in any filing. Oracle has $261 billion of uncommenced leases against $27.6 billion recognised, plus a $3.3 billion guarantee of a lessor's borrowing maturing September 2026. CoreWeave uses identical SPVs and consolidates all of it, because it has no third-party equity partner. Two rating agencies publicly disagree about whether the accounting reflects the economics. The BIS calls it shadow borrowing.

Analyst judgement: the accounting is defensible and probably correct under current standards. That is precisely the problem the BIS is pointing at. The obligations are real, the cash flows are contractual, the guarantees activate exactly when values fall, and the standard-setters' commencement rule means the disclosure arrives years after the risk does. The number to watch is not the SPV count. It is Oracle's $260 billion line and Meta's $182.88 billion line, quarter by quarter, and in particular whether they ever fall without a matching increase in right-of-use assets — which would mean cancellation, not commencement.

5. Government and defence exposure

5.1 A warning about this section's evidence

Sections 5.2 and 5.3 rest on documents that were retrieved and read: SEC filings by accession number, Federal Register documents by document number, White House and Commerce pages, county government releases, and issuer investor-relations materials.

Sections 5.4 and 5.5 do not meet that standard. The web research budget for this session was exhausted before defence procurement and state-subsidy sources could be reached, and repeated retries against defense.gov, usaspending.gov, jlarc.virginia.gov and state PUC dockets failed. Those subsections are therefore written as **explicitly labelled verification roadmaps**, not findings. Contract identifiers that could not be confirmed are omitted rather than guessed — a wrong PIID is worse than a missing one. Every figure in them carries a **[U]** tag: *unverified recall, requires confirmation*.

This is stated plainly because the alternative — presenting recalled contract values with the same confidence as retrieved filings — is exactly the failure mode this report exists to avoid.

5.2 Stargate

5.2.1 What was announced

OpenAI's post of 21 January 2025, verbatim: "The Stargate Project is a new company which **intends to invest $500 billion over the next four years** building new AI infrastructure for OpenAI in the United States. **We will begin deploying $100 billion immediately.**" Governance: "The initial **equity funders** in Stargate are SoftBank, OpenAI, Oracle, and MGX," while "Arm, Microsoft, NVIDIA, Oracle, and OpenAI are the key initial **technology partners**." SoftBank has financial responsibility, OpenAI operational responsibility, Masayoshi Son is chairman **[V]**.

SoftBank issued a verbatim-identical release on 22 January 2025 and disclosed **no individual capital contribution figure [V]**.

The White House's own investment tracker lists the project as "Softbank, OpenAI, and Oracle | $500 Billion | Technology & AI | AI infrastructure (Project Stargate)" — **omitting MGX, one of the four named equity funders** — and references **no federal funding, guarantee or credit line** anywhere **[V]**.

5.2.2 The documentary hole at the centre

This is the most important negative finding in this section. No SEC filing, state corporate registration, operating agreement, ownership percentage table, or board roster for a capitalised "Stargate LLC" or "Stargate Project" joint venture could be located. Eighteen months after announcement, the ownership of the vehicle that purportedly intends to deploy $500 billion is undocumented in any public record **[G]**.

What *is* documented are the operating subsidiaries and site-level SPVs: the Abilene JV (Crusoe + Blue Owl Capital Real Assets + Primary Digital Infrastructure, described as "the second phase of a $15 billion joint-venture," ownership percentages undisclosed, and **neither OpenAI nor "Stargate" is a named party — the contractual tenant is Oracle**) **[V]**; SB OpenAI Japan (50/50, capitalisation undisclosed) **[V]**; and the SB Energy partnership (OpenAI and SoftBank Group **each investing $500 million**, plus **$800 million of redeemable preferred equity from Ares**) **[V]**.

5.2.3 Who has actually committed what

SoftBank [V]. Every disclosed SoftBank dollar is an investment in **OpenAI equity, not a capital contribution to the Stargate joint venture.** The 2025 commitment was $40.0 billion total, $30.0 billion effective to SBG after syndication; first closing April 2025 ($10.0bn), second December 2025 ($22.5bn), funded by asset monetisation including the **$5.8 billion NVIDIA stake sale**, $9.2 billion of T-Mobile and $2.4 billion of

Deutsche Telekom. The 2026 follow-on is $30.0 billion in three tranches, two closed. **SoftBank's only Stargate-specific IR disclosure is a presentation slide reading "Clear path to securing full $500B / 10GW commitment." It has never disclosed a Stargate equity contribution figure in any investor document [G].**

Oracle [V]. No disclosed equity contribution to Stargate at all. Its participation is contractual — lease and cloud offtake. EDGAR full-text search returns exactly **one** Oracle filing mentioning Stargate: an 8-K of 10 March 2025 quoting Safra Catz, "we look forward to signing our first Stargate contract."

MGX [G]. No primary disclosure of any dollar figure. **OpenAI [G].** No disclosure of its own equity contribution.

Of four announced equity funders, zero have disclosed a Stargate equity contribution figure in a primary document.

5.2.4 Announced versus built

Site	Partner	Planned	Operational (Apr 2026)
Abilene, TX	Crusoe / Blue Owl; Oracle tenant	1.2 GW	**0.3 GW; 4 of 8 buildings**
Shackelford Cty, TX ("Frontier")	Vantage	2.0 GW	0 — roofing
Doña Ana Cty, NM ("Jupiter")	STACK / Oracle	2.2 GW	0 — foundations
Milam Cty, TX	SB Energy	1.2 GW	0 — steel framing
Port Washington, WI	Vantage / Oracle	1.3 GW	0 — foundations
Saline Twp, MI	Related Digital / Oracle	1.4 GW	0 — foundations
Lordstown, OH	SoftBank / Foxconn	<0.3 GW	0 — mostly manufacturing

Site-by-site verification by Epoch AI (17 April 2026), using satellite imagery, regulatory filings and company announcements: **more than 9 GW planned, roughly 0.3 GW operational — a 30x gap [R]**. Epoch also documents OpenAI **withdrawing from a planned Abilene expansion and redirecting 900 MW elsewhere [R]**.

OpenAI stated on 29 April 2026 that its 10 GW goal was "surpassed" with "more than 3GW added in the last 90 days" **[V]** — but that aggregate covers OpenAI's entire compute portfolio, not Stargate-JV operational capacity.

The reconciliation of the headline is instructive: **$500 billion (Jan 2025) → ">$400 billion over three years" (Sept 2025) → ">$450 billion" (Oct 2025) [V]**. Documented cash actually deployed into Stargate-designated US assets is far smaller: Abilene $15 billion announced of which ~$11.6 billion raised ($7.1 billion of it a construction *loan*), SB Energy $1.8 billion, and Vantage's $25 billion and $15 billion announced but not drawn. **No issuer, filing or IR document anywhere reconciles announced to deployed capital for the Stargate entity [G]**.

5.2.5 Federal money: the precise answer

No federal money, loan guarantee, or tax expenditure has been identified as directly involved in Stargate.

The relevant instrument is **Executive Order 14318, "Accelerating Federal Permitting of Data Center Infrastructure,"** signed 23 July 2025, published 28 July 2025 (90 FR 35385) **[V]**. Section 3 directs Commerce to "launch an initiative to provide financial support for Qualifying Projects, which could include **loans and loan guarantees, grants, tax incentives, and offtake agreements**." A Qualifying Project is >100 MW new load and/or ≥$500 million capital commitment. Section 5(c) provides that "Federal financial assistance representing **less than 50 percent of total project costs shall be presumed not to constitute substantial Federal control**." The EO also provides FAST-41 designation, NEPA categorical exclusions, authority to lease federal lands, and EPA Brownfield streamlining. **No company is named. Stargate is not named.**

This EO is the single most important item for the endorsement-versus-money distinction: it creates the authority for federal loans and guarantees to AI data centres. No evidence was found that the Section 3 Commerce financing initiative was ever stood up, and no evidence any Stargate entity applied for or received anything under it **[G]**. That is the highest-value open question in this section.

Adjacent, and all **[V]**: DOE's federal-land AI data centre siting programme selected Idaho National Laboratory, Oak Ridge, Paducah and Savannah River **— no OpenAI, Oracle, SoftBank, Crusoe or Vantage involvement, and Stargate uses no federal land.** DOE's "Accelerating Speed to Power" is explicitly "**not** a funding opportunity, grant program, or regulatory proposal." The **Genesis Mission**, announced 22 July 2026, carries **">$5 billion in Federal commitments"** to DOE's American Science and Security Platform — the largest verified federal dollar commitment to AI infrastructure found anywhere in this research — with no private company named and no Stargate linkage **[V/G]**.

The actual public subsidy is state and local. Doña Ana County, New Mexico approved **"up to $165 billion in Industrial Revenue Bonds"** for Project Jupiter on **19 September 2025**, with a **thirty-year property-tax abatement offset by PILOT payments**, against 750 full-time jobs within three years **[V — county government release]**. This is conduit financing — the county is not the obligor — but the abatement is genuine multi-decade revenue forgone, and it is by an order of magnitude the largest public-finance instrument identified in the entire project.

5.2.6 International

Stargate UAE (22 May 2025): partners G42, Oracle, NVIDIA, Cisco, SoftBank; a 1 GW Abu Dhabi cluster with 200 MW live in 2026; "developed in close coordination with the U.S. government"; tied to the UAE's $1.4 trillion US investment pledge **[V]**. Commerce approved G42 and Humain each for up to the equivalent of **35,000 NVIDIA GB300 chips** on 19 November 2025 **[V] — an export authorisation, not funding. The US contribution is regulatory permission, which has real commercial value and zero fiscal cost.** A BIS final rule effective 10 July 2026 removed the UAE from Country Groups D:3/D:4 and added it to A:5, naming G42 and Core42 as licence-free for advanced computing items until 6 April 2027 **[V]**.

Also **[V]**: **Stargate Norway** (31 July 2025, Nscale/Aker 50/50, Narvik, 230 MW + 290 MW ambition); **Stargate UK** (16 September 2025, Nscale + NVIDIA, up to 8,000 GPUs scaling to 31,000); **Stargate Argentina** (10 October 2025, an LOI with Sur Energy, up to 500 MW, "$25 billion in investment at full scale," planning phase only, with no visible 2026 activity **[R]**). Stargate India is widely reported and has **no OpenAI primary announcement [G]. No EXIM Bank or DFC financing to any Stargate international project was identified [G]**.

5.3 CHIPS Act equity conversions

5.3.1 The Intel transaction, as actually contracted

Intel's Form 8-K of 25 August 2025 attaches the Stock and Warrant Purchase Agreement. **The widely reported "$20.47 per share" is an artefact — an average of two different mechanics [V]**:

- **274,583,000 shares issued at closing** against **$5,695,000,000** of accelerated CHIPS money — an implied **$20.741 per share** — for obligations under the November 2024 direct funding agreement other than Secure Enclave.
- **158,740,000 shares issued into escrow** against **$3,174,800,000** of **Secure Enclave** money at **exactly $20.00 per share**. Section 4.4(b) releases shares "equal to the quotient of (x) the amount of such Secure Enclave Disbursement, divided by (y) $20.00," and Section 1.2(b) claws back unissued shares on the same divisor.

Total: **433,323,000 shares for $8,869,800,000**. The press release cites "$11.1 billion total" including "$2.2 billion previously disbursed," and a **9.9%** stake **[V]**.

The warrant is a poison pill against a foundry spin-off, and it is priced. "Warrants exercisable to purchase up to **240,516,150 shares**" at **$20.00**, expiring five years after closing, exercisable **only "if, and from the date on which, the Company ceases to directly or indirectly own at least 51% of its foundry business" [V]**. That is roughly 5% of the company, contingent on a single corporate action.

Voting is contractually passive. SPA §4.6 requires Commerce to appear and "vote… all of the Voted Company Stock **in favor of each nomination and proposal recommended by the Board of Directors**," with carve-outs for agreement breaches and actions adversely affecting US government relationships. **No board seat [V]**.

Transfer is restricted to "on and after the first anniversary of the Closing Date" and only "in one or more **broadly-syndicated offerings** (including government sponsored auctions)." Escrowed shares carry "no ownership, voting, economic or other rights" until released **[V]**.

Two negative findings, both **[V]**:

- **EDGAR full-text search returns zero Schedule 13D filings by the Department of Commerce for August 2025 through July 2026. The US government has never filed a beneficial-ownership schedule on a 9.9% position.**
- **A Federal Register API search on "CHIPS Program Office" returns no rulemaking or notice implementing the equity conversion.** The only 2026 CPO items are routine Paperwork Reduction Act notices. **The conversion was executed through a private securities purchase agreement with no notice-and-comment and no Federal Register instrument.**

5.3.2 The private co-investors, and the pricing hierarchy

NVIDIA [V]: 214,776,632 shares at **$23.28**, $5.0 billion, agreement 15 September 2025, closed 26 December 2025, with the agreement expressly providing no governance or information rights beyond those of shareholders generally.

SoftBank [V]: 86,956,522 shares at **$23.00**, $2.0 billion, closed 26 September 2025.

The government paid $20.00–$20.74. SoftBank paid $23.00. NVIDIA paid $23.28. The taxpayer got the best price of the three.

5.3.3 The mark, and its mirror image

INTC closed at **$92.32 on 24 July 2026 [R — price from an aggregator, not an exchange feed]**, against 5,044 million shares outstanding per the 10-Q cover dated 17 July 2026 **[V]**.

Position	Shares	Cost basis	Value @ $92.32	Gain
Shares actually held (274.6m issued + ~16m released)	~290.3m	~$6.01bn	**$26.80bn**	**+$20.79bn**
Full package if all escrow releases	433.3m	$8.87bn	$40.00bn	+$31.13bn
Warrant intrinsic if the 51% trigger fires	240.5m @ $20	—	$22.20bn	+$17.39bn

All **[C]** from verified share counts and a **[R]** price. **The stake is diluting**: ~290.3 million held against 5,044 million outstanding is **5.76%**, not the 9.9% headline; SoftBank's and NVIDIA's issuances did the diluting **[C]**. Of 158,740,000 shares originally escrowed, roughly 3 million released in FY2025 and 13 million in H1 2026, leaving **143 million still escrowed at 27 June 2026 [V]**.

The mirror image is on Intel's books. The escrowed shares and warrant are a **derivative liability** marked against Intel's own rising stock. It grew from **$2,700 million (27 December 2025) to $15,600 million (27 June 2026)**, producing **$12,873 million of fair-value losses in H1 2026** — $12.5 billion in Q2 alone — and driving a **GAAP net loss of $(11.0) billion, $(2.16) per share, on $16.1 billion of revenue**, against non-GAAP net income of $2.2 billion and $0.42 per share **[V]**.

The taxpayer is up roughly $21 billion on an $8.87 billion outlay in eleven months, and the identical price move is what produced Intel's headline GAAP losses. The gain is unrealised and the position is illiquid until 27 August 2026 at the earliest.

5.3.4 The precedents, and where the real contingent liability is

MP Materials / Department of Defense [V, 8-K of 10 July 2025]: 400,000 shares of Series A convertible preferred at $1,000 stated value = **$400 million**, up to $350 million more, **7.0% PIK dividend**, conversion at $30.03, plus a warrant for 11,201,659 shares at $30.03 — preferred and warrant "collectively represent fifteen percent (15%)" of common at closing. And then the part that matters: a **$110/kg NdPr price floor for ten years**, a ten-year offtake at cost, and **quarterly payments equal to 25% of any shortfall against $140 million of threshold EBITDA**.

That is a far more aggressive structure than Intel's, and it contains the only genuinely open-ended federal exposure identified anywhere in this research. A price floor and an earnings guarantee are uncapped contingent liabilities. They receive almost no attention.

Lithium Americas / DOE [V, 8-K of 1 October 2025]: DOE receives **penny warrants** for a 5% equity stake plus a 5% economic stake in Thacker Pass at the JV level; DOE defers **$182 million** of debt service over five years; first draw $435 million of a **$2.26 billion** loan.

Trilogy Metals / Department of War [V, 8-K of 7 October 2025]: 8,215,570 units at $2.17 ≈ **$17.8 million**, each with three-quarters of a ten-year **$0.01 warrant**, aggregate 16,431,140 shares ≈ **10%**, plus a three-year right to appoint one independent director and a consent right over debt above $1 billion through 1 January 2029.

5.3.5 2026: the model was extended once, and not to a chipmaker

USA Rare Earth, Inc. [V]: an 8-K of 26 January 2026 discloses a non-binding LOI for **$277 million of CHIPS federal funding plus a $1.3 billion senior secured loan ≈ $1.6 billion**, with Commerce receiving roughly **16.1 million shares and 17.6 million warrants**. Definitive documents followed on 20 July 2026, referencing "extensive affirmative and negative covenants, domestic content and national security guardrail provisions."

Negative findings, all **[V]**: EDGAR full-text search for "Secure Enclave" in 8-Ks filed 1 January to 25 July 2026 returns **only Intel**. **There is no primary evidence that TSMC, Samsung, GlobalFoundries, Texas Instruments or Micron converted grants to equity.** Commerce is still writing conventional grants — TSMC incremental $100 billion (16 July 2026), Bosch $225 million, I-Pulse $250 million, SandboxAQ $500 million. **No sale or transfer of the Intel stake has occurred; the one-year lockup runs to approximately 27 August 2026 and has not lapsed.**

Lip-Bu Tan remains CEO [V]: Intel's DEF 14A of 23 March 2026 identifies him as CEO and the chair's letter states "Central to this reinvention was the appointment of Lip-Bu Tan as Chief Executive Officer." He published a CEO letter on 17 June 2026 and keynoted Computex on 2 June 2026. The August 2025 resignation episode could not be primary-sourced **[G]**.

Quantification gap [G]: the CHIPS appropriation splits into $39 billion of manufacturing incentives and $11 billion of R&D per NIST, but **no official obligated-versus-disbursed aggregate was retrieved**, and the Advanced Manufacturing Investment Credit's tax-expenditure score — plausibly the largest single federal fiscal item in the entire semiconductor programme — could not be sized.

5.4 Pentagon and defence AI contracts — verification roadmap, not findings

Everything in this subsection is [U]: unverified recall requiring confirmation. Zero primary documents were retrieved. Contract identifiers are omitted rather than guessed.

Palantir [U]. Project Maven Smart System, awarded via the Army's PEO IEW&S, reported at approximately **$480 million over five years** in May 2024, with a May 2025 modification reportedly raising the ceiling by roughly $795 million to **approximately $1.3 billion through 2029**. Army TITAN, March 2024, approximately

$178.4 million for ten prototypes. An **Army Enterprise Service Agreement** announced around July–August 2025, consolidating roughly 75 separate contracts into a single vehicle with a ceiling reported at up to **$10 billion over ten years**. NATO's Communications and Information Agency announced acquisition of Maven Smart System NATO around 25 March 2025, value undisclosed. FY2024 revenue $2.866 billion; FY2025 approximately $4.4 billion; US government revenue roughly $1.2 billion in FY2024, growing more than 50% in 2025.

The consolidation structure matters fiscally even if the numbers move: it converts many competitively-scoped awards into one sole-source-adjacent ceiling, which is the vehicle type GAO has historically flagged for competition risk. ***Palantir's own concentration disclosure is the single most decision-relevant number in this subsection and it must be read from the 10-K, not recalled.***

Anduril [U]. Microsoft and Anduril announced around 11 February 2025 that Anduril would assume the IVAS programme. The original 2021 Microsoft IVAS production contract carried a ceiling reported at up to **$21.88 billion over ten years — this is the source of the "$22 billion ceiling" figure, and it is a ceiling inherited from a predecessor contract, not a new award.** The Soldier Borne Mission Command award to Anduril was reported at approximately **$159 million initial** around September 2025. **The distinction between ceiling and obligation is the most commonly abused number in defence-AI reporting.** Arsenal-1, announced January 2025 near Rickenbacker International Airport, Ohio: approximately **$1 billion** and 4,000+ jobs by 2035 — and the Ohio state and local incentives attached to it are exactly the figure Section 5.5 needs and could not obtain [G]. Series G, June 2025, **$2.5 billion at a $30.5 billion valuation**.

Frontier labs [U]. An OpenAI CDAO contract announced around 16 June 2025 with a **$200 million ceiling**, one year, to "OpenAI Public Sector LLC." On approximately 14 July 2025, CDAO announced agreements with **Anthropic, Google, xAI and OpenAI, each with a ceiling of up to $200 million** — an aggregate ceiling of roughly $800 million, frequently misreported as $800 million *obligated*. GSA OneGov pricing: ChatGPT Enterprise at **$1 per agency for one year**, Anthropic Claude at **$1** across all three branches, Google Gemini at **$0.47**, xAI Grok at **$0.42**.

The fiscal analysis of those OneGov prices is counterintuitive and worth stating: they are below-cost offers that transfer value ***to*** *the government in year one, while creating switching costs and data gravity that shift bargaining power to vendors at renewal.* ***The taxpayer exposure is not the year-one price. It is the year-three renewal.***

The one verified item [V]: the **Genesis Mission**, announced 22 July 2026, carries **">$5 billion in Federal commitments"** to DOE's American Science and Security Platform. The appropriation source, public/private split and participating vendors are [G].

National-lab compute [U]: DOE–AMD announced two Oak Ridge systems (Lux ~2026, Discovery ~2028–29) and DOE–NVIDIA–Oracle announced Argonne's Solstice (~100,000 Blackwell GPUs) and Equinox, both in late October 2025. **The public/private cost split on each — the single most important fiscal question here — is unverified.** These were characterised as public-private partnerships, which typically means the appropriated share is well below headline system value.

5.5 Taxpayer risk channels

5.5.1 The conceptual frame

Four channels must be kept apart, because conflating them is the dominant error in this literature.

1. **Cash outlay —** appropriated money actually paid out. **Rare** in AI infrastructure.
2. **Tax expenditure —** revenue forgone through abatements and exemptions. Real money, appears in no budget line, rarely scored against project benefits.
3. **Contingent liability —** loan guarantees, price floors, equity backstops. The MP Materials NdPr floor and $140 million EBITDA guarantee are the clearest verified examples.
4. **Ratepayer cost-shifting — not taxpayer money at all.** Costs socialised across a utility's customer base by regulatory order. Larger in aggregate than the tax expenditures, and governed by state PUCs rather than legislatures, which means far less democratic scrutiny per dollar.

5.5.2 What is verified

Doña Ana County, New Mexico [V]: up to **$165 billion** in industrial revenue bonds with a thirty-year property-tax abatement offset by PILOT, approved 19 September 2025. Conduit financing; the county is not the obligor; the abatement is real.

Wisconsin [V]: Vantage's own release states WEC will apply a "dedicated electricity rate" designed to "protect other customers," with a minimum $175 million commitment to regional infrastructure.

Michigan [V]: OpenAI's release states grid "upgrades... funded by the project and **not local ratepayers**."

Both of the latter are developer self-characterisations, not PUC orders, and require verification against the actual tariff filings [G].

AEP, Q1 2026 10-Q, verbatim [V] — the most useful ratepayer-protection datapoint retrieved:

> "These new tariffs are designed to protect existing customers by strengthening and lengthening contract terms with large customers. These new protections include **contract lengths of up to 20 years and take-or-pay contractual minimums which can require a customer to pay for as much as 80-90% of their contracted demand**."

As of 31 March 2026, four such proposals were pending in Michigan, Oklahoma, Texas and Virginia **[V]**. **Utilities do not negotiate 80–90% take-or-pay against loads they believe are certain.**

5.5.3 What requires verification [U throughout]

- **Virginia JLARC, December 2024**: the data-centre sales-and-use tax exemption cost roughly **$130 million a year** and rising; the study found data centres likely raise electricity costs for other ratepayers and recommended cost allocation. **This is the best-documented state study in the field and should be the first retrieval when budget permits.**
- **Ohio PUCO Case No. 24-508-EL-ATA** (AEP Ohio data-centre tariff): PUCO approved a settlement around July 2025 requiring data centres above roughly 25 MW to pay for approximately **85% of contracted capacity for twelve years**, with exit fees. Moderate confidence in the docket number. **This is the most important single ratepayer-protection precedent in the country**, and it was appealed. Note that AEP's own Q1 2026 10-Q lists pending proposals in MI, OK, TX and VA **— Ohio is not among them**, which this research could not explain **[G]**.
- Also to verify: Louisiana LPSC approval of Entergy generation for Meta's Richland Parish campus and the ITEP exemption value; Georgia's sales-and-use tax exemption and the 2024 suspension attempt and veto; Michigan PA 144/145 of 2024 with House Fiscal Agency cost estimates; Wisconsin 2023 Act 19; Texas SB6 (2025) large-load interconnection and curtailment rules; Virginia SCC treatment creating a separate large-load class.
- **Contrary evidence, which is real and under-covered [U]:** multiple localities rejected data-centre projects in 2025–26, including Tucson, Arizona (Project Blue) and several Indiana and Georgia jurisdictions. Several states moved toward requiring data centres to bear their own grid costs. **This countervailing trend deserves the same rigour as the subsidies and did not receive it here.**

5.6 Federal money at risk versus federal endorsement without money

Federal money or balance-sheet risk — verified:

- **Intel equity: $8.87 billion committed, ~$6.01 billion disbursed against shares actually held.** Currently ~+$20.8 billion unrealised. Illiquid until ~27 August 2026, saleable only via broadly-syndicated offering. Real money, currently deeply in the money.
- **MP Materials: $400 million preferred plus up to $350 million more, plus a ten-year $110/kg NdPr price floor and a quarterly payment obligation against a $140 million EBITDA threshold.** The price floor and EBITDA guarantee are **uncapped contingent liabilities —** the most genuinely risky federal exposure identified here, and the least discussed.

- **Lithium Americas: $2.26 billion DOE loan with $182 million of debt service deferred.**
- **USA Rare Earth: $277 million of CHIPS funding plus a $1.3 billion senior secured loan**, definitive documents executed 19–20 July 2026.
- **Genesis Mission: >$5 billion in federal commitments** (22 July 2026); appropriation source unverified.

Federal endorsement without money — verified:

- **Stargate in its entirety.** A White House podium, a listing on the White House investments tracker, and permitting facilitation under EO 14318. **No federal dollars, no guarantee, no dedicated tax expenditure.** The $500 billion figure is a private-sector announcement that the federal government publicised. It is not a federal commitment and never was.
- **UAE and Saudi chip export authorisations.** Regulatory permission of substantial commercial value, zero fiscal cost.
- **EO 14318 permitting relief, NEPA categorical exclusions, FAST-41 designation.**
- **DOE federal-land siting** — real assets offered, but **Stargate is not a participant**.

The correction this section supports, stated plainly: the two largest AI-adjacent federal financial positions — Intel and MP Materials — have nothing to do with Stargate, and Stargate, the project that generated the most coverage of "government backing," has no identified federal money in it at all. The public subsidy attached to Stargate is county and state: property-tax abatements, industrial revenue bonds, and utility tariff design.

What taxpayer exposure would look like if the sector contracted. On the verified record: the Intel position is a mark-to-market equity exposure that would fall with the sector but is currently 3.5x in the money; the MP Materials price floor and EBITDA guarantee would activate precisely in a downturn and are uncapped; the DOE loans are principal at risk; and the largest exposure by far — thirty-year property-tax abatements and utility cost allocation — sits with counties and ratepayers, not the federal government, and is almost entirely unmeasured. **The AEP take-or-pay tariffs are the single most important development in the opposite direction: they push stranded-asset risk back onto the load, and their spread across states is a monitorable in Section 10.**

6. Meta: the case study

6.1 Why Meta

Meta is the cleanest test case in the sector for three reasons. It discloses a servers-only asset line, which nobody else except Amazon and CoreWeave does. It quantified its useful-life change, which Oracle and Alphabet did not. And in its Q1 2026 10-Q it wrote the most complete public description any sponsor has given of an unconsolidated data-centre venture — including a maximum-exposure figure that no other company discloses at all.

It is also the company where the tension between the two halves of this report is sharpest: a business with 48% Family of Apps operating margins and $115.8 billion of operating cash flow, running a capital programme that its own rating agency expects to produce **"limited to no free cash flow generation over the next two years."**

Timing note: Meta announced on 14 July 2026 that Q2 2026 results would be released after market close on **Wednesday 29 July 2026 [V]**. Everything below is therefore Q1 2026 or earlier.

6.2 The capex trajectory

Period	Purchases of P&E	Finance-lease principal	**Total capex**	y/y
FY2021	$18,567m	$677m	**$19,244m**	—
FY2022	$31,186m	$850m	**$32,036m**	+66%
FY2023	$27,045m	$1,058m	**$28,103m**	−12%
FY2024	$37,256m	$1,969m	**$39,225m**	+40%
FY2025	$69,691m	$2,524m	**$72,215m**	**+84%**
Q1 2026	$18,997m	$843m	**$19,840m**	+45%

All [V] from the cash-flow statements and 8-K exhibits. **Q1 2026 alone ($19.84 billion) exceeds all of FY2021 ($19.24 billion).**

The guidance ladder is the more revealing series, because it shows how fast the plan changed [**V, every figure from a filed CFO Outlook Commentary**]:

Date given	For	Guidance	Change
1 Feb 2023	FY2023	$30–33bn	*lowered* from $34–37bn
1 Feb 2024	FY2024	$30–37bn	—
29 Jan 2025	FY2025	$60–65bn	—
30 Apr 2025	FY2025	**$64–72bn**	raised
30 Jul 2025	FY2025	$66–72bn	raised
29 Oct 2025	FY2025	**$70–72bn**	raised
28 Jan 2026	FY2026	$115–135bn	—
29 Apr 2026	FY2026	**$125–145bn**	**raised within three months**

The April 2026 revision carries language that matters more than the number: "This reflects our expectations for **higher component pricing this year** and, to a lesser extent, additional data center costs to support future year capacity" [**V**].

Read that carefully. Part of the capex increase is price, not quantity. That is simultaneously a scarcity signal — consistent with everything in Section 8 — and a warning that headline capex overstates delivered compute.

On the forward path, Meta said on 29 October 2025: "our current expectation is that capital expenditures dollar growth will be **notably larger in 2026 than 2025**" [**V**]. On 2027 there is **no guidance at all**: on the Q1 2026 call Susan Li said Meta is "not providing a specific outlook for 2027 CapEx" and is "undergoing a very dynamic planning process ourselves as we are working through what our capacity needs will be over the coming years" [**R — transcript**]. That is a gap in Meta's own disclosure, not in this research.

Total expense guidance for FY2026 is **$162–169 billion**, unchanged in April, with the majority of growth attributed to "infrastructure costs, which includes third-party cloud spend, **higher depreciation**, and higher infrastructure operating expenses" [V]. Depreciation and amortisation ran **$11,178m (FY2023) → $15,498m (FY2024) → $18,616m (FY2025)** [V].

6.3 Where the free cash flow went

Period	Operating cash flow	Capex	**Free cash flow**	FCF margin
FY2021	$57,683m	$19,244m	**$38,439m**	32.6%
FY2022	$50,475m	$32,036m	**$18,439m**	15.8%
FY2023	$71,113m	$28,103m	**$43,010m**	31.9%
FY2024	$91,328m	$39,225m	**$52,103m**	31.7%
FY2025	$115,800m	$72,215m	**$43,585m**	**21.7%**
Q2 2025	$25,561m	$17,012m	$8,549m	18.0%
Q1 2026	$32,226m	$19,840m	**$12,386m**	22.0%

All [C] from [V] inputs. **The inflection is Q2 2025**, when quarterly free cash flow fell 18% year on year to $8.55 billion despite operating cash flow rising 21%. On a full-year basis, **FY2025 is the first year free cash flow declined in absolute dollars** — $52.10 billion to $43.59 billion, −16.3% — while revenue grew 22%.

The balance sheet tells the same story from the other side [V]:

Period end	Cash + marketable sec.	Long-term debt	Buybacks	Dividends
FY2021	$48,998m	$0	—	$0
FY2024	$77,818m	$28,826m	$30,125m	$5,072m
Q3 2025	**$44,448m**	$28,834m	—	—
FY2025	$81,589m	$58,744m	$26,248m	$5,324m
Q1 2026	$81,180m	$58,748m	**$0**	$1,346m

Two facts in that table. **Cash bottomed at $44.4 billion on 30 September 2025**, down $33 billion in nine months, which is precisely why a $30 billion bond deal arrived four weeks later. And **buybacks went to exactly $0 in Q1 2026**, against $12.75 billion in Q1 2025 [V]. The dividend continues; the buyback does not.

6.3.1 The debt

October/November 2025 — $30.0 billion, six tranches [V, 8-K of 3 November 2025]:

Maturity	Principal	Coupon	Maturity	Principal	Coupon
2030	$4,000m	4.200%	2045	$4,500m	5.500%
2032	$4,000m	4.600%	2055	$6,500m	5.625%
2035	$6,500m	4.875%	2065	$4,500m	5.750%

Net proceeds $29,887,775,000. Use of proceeds, verbatim: **"We intend to use the net proceeds from this offering for general corporate purposes."** That is the entire statement [V]. Bookrunners were **Citigroup and Morgan Stanley only** — two banks on a $30 billion trade. The order book peaked around **$125 billion**, described by bankers as the largest ever, with new-issue concessions of 10–15bp [R]. Rated **Aa3 / AA−** [R].

May 2026 — $25.0 billion, six tranches [V, 8-K of 4 May 2026]:

Maturity	Principal	Coupon	Maturity	Principal	Coupon
2031	$3,000m	4.550%	2046	$4,000m	**6.200%**
2033	$2,000m	4.875%	2056	$6,000m	**6.300%**
2036	$6,000m	5.250%	2066	$4,000m	**6.450%**

Compare the long end across the two deals. The 30-year went from 5.625% to 6.300% — 67.5 basis points wider in six months. The 40-year went from 5.750% to 6.450% — 70 basis points. That is Meta's own cost of capital deteriorating on unchanged credit ratings, and it is verifiable from two filed 8-Ks without reference to any third party.

Total face debt: **$29.0 billion (Dec 2024) → $59.0 billion (Dec 2025) → ~$84.0 billion (after May 2026) [V]**.

6.3.2 The commitments that are not on the balance sheet

At 31 March 2026 **[V]**:

- **$237.67 billion** of non-cancelable contractual commitments, "mostly related to third-party cloud capacity arrangements and continued investments in servers and network infrastructure, data centers" — up from **$131.05 billion** at 31 December 2025, an increase of **$106.6 billion in one quarter**.
- **$182.88 billion** of leases not yet commenced, "consisting of data centers, colocations, and certain network infrastructure, which will commence during the remainder of 2026 and 2036."
- On-balance-sheet operating lease liabilities: **$28.02 billion**.
- And subsequently: "In April 2026, we entered into additional multi-year infrastructure contracts, related to which our non-cancelable contractual commitments increased by **approximately $24 billion**."

On the Q1 2026 call Meta described "a **$107 billion step up** in our contractual commitments this quarter" driven by multi-year cloud deals **[R]**.

6.4 Scale AI: what Meta actually booked

Meta paid a reported **$14.3 billion for a 49% non-voting stake** in Scale AI in June 2025, and Alexandr Wang left to lead Meta Superintelligence Labs **[R]**. The non-voting structure is widely understood as antitrust avoidance — no change of control, no HSR filing.

The accounting is the interesting part, and it is verifiable. "Scale AI" appears in exactly four Meta filings: the Q2 2025 10-Q, the Q3 2025 10-Q, the FY2025 10-K, and the annual report **[V]**. Meta names it as a discrete line inside the **Non-Marketable Equity Investments** note, with an **initial cost of $13,790 million as of 30 June 2025 [V]**.

It is carried under the ASC 321 measurement alternative — cost less impairment, adjusted for observable price changes — not the equity method. That is provable arithmetically from the FY2025 10-K schedule **[V]**:

	31 Dec 2025	31 Dec 2024	Change
Initial cost	$20,271m	$6,342m	**+$13,929m**
Cumulative upward adjustments	$429m	$300m	+$129m
Cumulative impairment / downward	**$(624)m**	**$(624)m**	**$0**
Measurement alternative carrying value	$20,076m	$6,018m	+$14,058m
Equity method investments	$7,448m	$52m	+$7,396m
Total non-marketable equity	**$27,524m**	**$6,070m**	

The $13,929 million increase in measurement-alternative *initial cost* is almost entirely Scale AI's $13,790 million. And Scale AI **cannot** be equity-method: total equity-method investments at 31 December 2025 were $7,448 million — less than Scale AI alone.

Three forensic observations follow.

One. The booked cost is $13.79 billion, not $14.3 billion — a gap of roughly $510 million that Meta does not explain [G]. The most plausible reading is retention or compensation elements expensed rather than capitalised, but that is inference.

Two. No impairment has been recorded. Cumulative impairment and downward adjustments were **$(624) million at both 31 December 2024 and 31 December 2025 — unchanged** [V]. Not one dollar of the Scale AI position has been written down, despite documented customer flight: OpenAI began phasing out its Scale AI relationship within days of the announcement, and Google moved to discontinue work, both reportedly citing concerns about routing training data through a supplier part-owned by a competitor [R].

Three. There is no goodwill, no intangibles, no in-process R&D and no acquired-workforce intangible. Because this was structured as a minority non-voting stake rather than a business combination, none of ASC 805 applies. **The entire $13.79 billion sits as a single cost-basis financial asset with no amortisation and no recurring fair-value mark.**

That is a materially different earnings profile from an acquisition of the same economics. A business combination would have produced amortising intangibles and an annual goodwill impairment test. The non-voting minority structure produces neither.

For scale, Meta's 2020 Jio Platforms investment accounts for roughly $5,820 million of the same measurement-alternative bucket [V].

On the compensation packages reported for Meta Superintelligence Labs hires — the ~$200 million package, the $100 million signing bonus claim, the ~$1.5 billion offer — **this report declines to assert figures it could not verify** [G]. The verified proxy: share-based compensation expense ran **$14,027m (FY2023) → $16,690m (FY2024) → $20,427m (FY2025)**, +22.4% [V], and Meta named "employee compensation, driven by investments in technical talent" as the second-largest FY2026 expense driver [V].

6.5 Hyperion

Meta's own disclosure of the Louisiana venture is quoted in full in Section 4.2 and is not repeated here. The essential terms, all [V] from the Q1 2026 10-Q:

Item	Disclosed value
Meta's membership interest	**20%**
Total estimated development costs	**~$27 billion**, funded pro rata
Lease commencement	**2029**
Aggregate initial lease commitment	**$12.31 billion**
Initial lease term per property	**4 years**, renewable to **up to 20 years**
Residual value guarantee threshold	**~$28 billion**, decreasing over time
RVG liability recorded	**$0** — "not probable"
Carrying value of Meta's equity investment	**$2.37 billion** (31 Mar 26); $1.83bn (31 Dec 25)
Maximum exposure to loss	**$45.99 billion** (31 Mar 26); $45.95bn (31 Dec 25)
Meta's role	Construction manager, administrator, property manager, sole tenant
Consolidation	**Not consolidated** — Meta is not the primary beneficiary

Two further XBRL-tagged facts from the FY2025 10-K deserve notice: **assets contributed to the venture $4,300 million**, and **proceeds from venture distribution $2,550 million** [V]. Meta put $4.3 billion of assets in and took $2.55 billion of cash back out.

The financing terms — Beignet Investor LLC, $27.294 billion, 6.581%, priced 16 October 2025, expected final maturity 30 May 2049, S&P A+, Blue Owl 80% / Meta 20% — are [R] throughout, because it is a 144A private placement not filed on EDGAR. **Meta names neither Blue Owl nor Beignet in any filing** [V].

One reconciliation worth recording. Press accounts describe a "16-year" residual value guarantee. Meta's filing describes four-year initial terms renewable to twenty years with a ~$28 billion RVG threshold that decreases over time. **Meta's filing governs.**

And a second, unnamed unconsolidated VIE exists. Meta's Q1 2026 10-Q discloses total equity-method investments of $8,235 million at 31 March 2026, of which the Louisiana venture is only $2,370 million, plus a separate **VIE maximum exposure to loss of $5,790 million** distinct from the venture's $45,990 million. **Meta does not identify this entity** [**V/G**].

Campus scale: Hyperion in Richland Parish, Louisiana was originally ~2.1 GW and ~4 million square feet, targeted for completion June 2029 [**R**]. On **13 July 2026 Meta expanded Hyperion to 5 GW and lifted total Louisiana investment above $50 billion**, plus more than $1 billion of local infrastructure [**R**]. Prometheus in New Albany, Ohio is ~1.5 GW [**R**]. El Paso specifics are [**G**].

6.6 What Meta says the money is buying

This is the bull case in Meta's own words, and it should be given its full weight.

Q3 2025 prepared remarks [V]: Reels at "an annual run rate of over **$50 billion**"; end-to-end AI ad tools at a **$60 billion** annual run rate.

Q1 2026 call [R — transcript]: a **"6%+ increase in conversion rate"** on landing-page-view ads from the Lattice and GEM architectures; a **1.6% increase** in off-site conversion rates from adaptive ranking; **8 million+ advertisers** using generative-AI creative tools, with video generation showing "more than 3% boost in conversion rates in tests"; business AI conversations "**over 10 million... each week**, up from 1 million at the start of the year"; and the value optimisation suite at an annual revenue run rate "**now over $20 billion, more than doubling year over year**."

Zuckerberg's framing, Q3 2025: "We're building what we expect to be an industry-leading amount of compute," and "It's the right strategy to **aggressively frontload building capacity** so we're prepared for the most optimistic cases" [**R**]. Susan Li on the joint venture: "This gives us long-term optionality in supporting our future capacity needs given both the magnitude, but also **uncertainty** of what the capacity outlook in future years looks like" [**R**].

Note the disclosure asymmetry. Meta quantifies conversion-rate lifts to the tenth of a percent and has never disclosed AI-attributable revenue dollars, AI gross margin, or return on the incremental capex.

6.6.1 The precedent nobody applies

Year	Reality Labs revenue	Reality Labs operating loss
2021	$2,274m	$(10,193)m
2022	$2,159m	$(13,717)m
2023	$1,896m	$(16,120)m
2024	$2,146m	$(17,729)m
2025	$2,207m	$(19,193)m
Q1 2026	$402m	$(4,028)m

All [**V**]. **Cumulative 2021 through Q1 2026: $80,980 million of operating losses on $11,084 million of revenue —** a loss of roughly **$7.31 for every $1 of revenue**, sustained across five years [**C**]. Including 2020, cumulative losses are approximately $87.6 billion. Meta guided in January 2026 that Reality Labs operating losses would remain "similar to 2025 levels" in 2026 [**V**].

This is the governance precedent that matters, and it cuts both ways. **Meta has demonstrated that it will fund a segment losing $19 billion a year, indefinitely, without external pressure forcing a change.** Bulls should read that as evidence of conviction and staying power. Bears should read it as evidence that no market discipline operates on this management's capital allocation. Both readings are supported.

Meanwhile the engine keeps running [**V**]: Family of Apps revenue $198,759 million in FY2025 with operating income $102,469 million, a **51.6%** margin — though down from a **53.7%** peak in FY2024, and **48.1%** in Q1 2026. **That roughly 560-basis-point compression is the first visible margin cost of the infrastructure build.**

Consolidated FY2025: revenue $200,966m, operating income $83,276m, net income $60,460m, diluted EPS $23.49. Q1 2026: revenue $56,311m (+33%), operating income $22,872m (41% margin), net income $26,773m (+61%, flattered by an ~$8 billion tax benefit), diluted EPS $10.44. Headcount fell from 78,865 to 77,986 [**V**].

6.7 What the market and the agencies have said

Share price. Meta fell approximately **11%** on 30 October 2025 following Q3 results — its worst day in three years — and roughly **10%** on 30 April 2026 following the capex guidance raise, erasing around $170 billion of market capitalisation [**R**]. The October drop had two drivers: a one-time non-cash income tax charge of **$15.93 billion** that cut diluted EPS by $6.20 and reduced reported net income to $2,709 million on $51,242 million of revenue [**V**], and the "notably larger" 2026 capex language.

Credit. Moody's affirmed **Aa3, stable outlook**, on 27 February 2026 [**R**]. Its commentary is the most important credit datapoint in this section:

> "Moody's expects **limited to no free cash flow generation over the next two years**, driven by elevated capital expenditures that are expected to be **around 50% of revenue annually**."
>
> "For year-end 2026 and 2027, Moody's projects total **debt-to-EBITDA of less than 1x** and modest net debt balances." [**R**]

Note the tension inside that affirmation. Moody's models capex at roughly 50% of revenue and near-zero free cash flow, and holds Aa3 on the strength of the leverage ratio. **At the midpoint of Meta's own $125–145 billion guidance against roughly $240 billion of 2026 revenue, capex would be about 56% of revenue — above Moody's own assumption** [**C**].

And note the disagreement inside the ratings themselves. S&P rated **the Beignet notes A+**, one notch below Meta's AA−. S&P is rating the vehicle essentially on Meta's credit, treating the lease-plus-guarantee package as near-recourse, while Meta's accounting concludes it does not control the vehicle. **The rating agency and the accounting reach opposite conclusions about who bears the risk.**

6.8 What this section concludes

Verified: Meta's capex went from $19.2 billion to a guided $125–145 billion in five years, with the 2026 guide raised twice within six months and partly attributed to component prices rather than volume. Free cash flow declined in absolute dollars for the first time in FY2025. Buybacks stopped entirely in Q1 2026. Cash bottomed at $44.4 billion in September 2025 and $55 billion of bonds followed within seven months, at a long-end cost 70 basis points higher on the second deal. Scale AI is carried at $13.79 billion of cost with zero impairment and no goodwill. The Louisiana venture carries $45.99 billion of maximum exposure against a $2.37 billion carrying value, with a $28 billion residual value guarantee for which no liability is recorded. Reality Labs has lost $81 billion since 2021 and Meta has never blinked.

Analyst judgement: Meta is the strongest balance sheet running the most aggressive programme, and the two facts are related — it can afford to be wrong for longer than anyone can stay solvent betting against it. The genuine question is not solvency. It is whether a company that has funded $81 billion of Reality Labs losses without external discipline will apply discipline to a programme an order of magnitude larger. Meta's own answer, in Zuckerberg's words, is that it is "the right strategy to aggressively frontload building capacity so we're prepared for the most optimistic cases." That is a statement of strategy, not of expected return, and Meta has never published the expected return.

7. Credit market stress signals

7.1 The headline: the index says nothing is wrong

At the close on 23 July 2026, the ICE BofA US Corporate index option-adjusted spread was **79 basis points —** within six basis points of its post-1998 tights [**V, FRED `BAMLC0A0CM`**]. High yield was **277 basis points**. On the aggregate indices, this is one of the calmest credit markets in a quarter century.

Underneath, three things are happening simultaneously that the index does not show:

1. **Oracle's five-year CDS reached 203 basis points on 20 July 2026, the widest in the history of the series** [**R**], and S&P cut Oracle to **BBB−** on 9 July 2026 [**R**].
2. **CCC-rated spreads are at 991 basis points — 3.58x the high-yield index —** and rose every session from 16 to 23 July [**V**].
3. **Five issuers went from 3% of the outstanding investment-grade stock to more than 15% of year-to-date issuance in roughly four months** [**V, Bank of England**].

The story of 2025–26 AI credit is not index-level stress. It is **extreme dispersion, a migration of risk off balance sheets into vehicles and private credit, and a single issuer acting as the market's designated pressure valve.**

7.2 Issuance

Metric	Value	Source
Hyperscaler gross bond issuance, 2025 (AMZN, GOOGL, META, MSFT, ORCL)	**~$121bn**	LPL, 28 May 2026 [**R**]
Same cohort, 2020–24 annual average	**$28bn**	LPL [**R**]
BIS characterisation of 2025 hyperscaler issuance	**"topped $100 billion"**	BIS QR, 16 Mar 2026 [**V**]
Hyperscaler share of *outstanding* US IG debt stock, end-2025	**3%**	BoE FSR, Jul 2026 [**V**]
Hyperscaler share of US IG **YTD issuance**, early May 2026	**over 15%**	BoE FSR [**V**]
2026 projected AI-related IG supply	**~$300bn**	LPL [**R**]
Tech share of global non-financial corporate bond issuance	**16.7%**, from 11.6% a year earlier	S&P, 17 Feb 2026 [**R**]

The Bank of England figure is the single most important issuance statistic in this report. Going from 3% of the stock to more than 15% of the flow in four months is a duration and concentration shock to the investment-grade index. It is not, by itself, a credit event.

The deals, May–July 2026 — roughly $96 billion in ninety days, and none of it names AI capex as its use of proceeds [V]:

Issuer	Size	Priced	Long-end coupons
Meta	$25.0bn	1 May 2026	6.200% ’46 · **6.300% ’56** · 6.450% ’66
Alphabet	~$21bn equiv.	May 2026	EUR / CAD / JPY tranches; **no USD tranche**
NVIDIA	$25.0bn	17 Jun 2026	to 5.625% ’56; proceeds include refinancing
Amazon	$25.0bn	8 Jul 2026	6.000% ’46 · **6.100% ’56** · 6.250% ’66
Microsoft	**$0**	—	**No 424B2/424B5/FWP in the window**
Oracle	**$0 public**	—	**Despite an announced $45–50bn CY2026 programme**

Two of those rows are negative findings and both matter. **Microsoft, the largest capex spender, did not issue at all.** And **Oracle, which told investors it would raise $45–50 billion of gross cash proceeds during calendar 2026, priced no public debt between May and July 2026 —** while stating in its Q4 FY2026 release that it does "**not expect to issue additional debt in calendar year 2026**" [V]. Whatever Oracle is doing is private, bank-funded, or unpriced.

The earlier jumbo deals, for reference [V/R]: Meta's $30.0 billion on 30 October 2025 (order book ~$125 billion, sole bookrunners Citigroup and Morgan Stanley, concessions 10–15bp); Oracle's ~$18 billion in September 2025 (order book $90 billion — and the bonds then "**sold off materially, trading roughly 60–80bps wide of their issuance level**" [R]); and Oracle's $25 billion on 2 February 2026 (order book up to $155 billion, eight tranches plus a floater, 3-year at +95bp through 40-year at +195bp).

The Beignet Investor LLC template — $27.294 billion, 6.581%, 30 May 2049, S&P A+ preliminary, Blue Owl 80% / Meta 20%, 144A-for-life [R] — is precisely what the BIS calls shadow borrowing: minority equity, long-dated lease offtake, debt at the vehicle.

Data-centre ABS and CMBS, from two industry sources that agree independently [V]: **$9.4bn (2021) → $1.3bn (2022) → $8.6bn (2023) → $11.4bn (2024) → ~$26bn (2025)**, with CMBS rising to **42% of structured-finance issuance from 25% over 2018–24**. Cumulative market: **$48.69 billion across 88 US transactions since 2018 [R]**. Senior tranches are exclusively A-category; junior tranches run BBB to BB−. **2026 year-to-date issuance is a genuine gap [G] —** no citable public figure exists.

Tenant concentration in that market: the **top five companies were 73% of 2025 demand —** AWS 22%, Microsoft 18%, Google 15%, Meta 11%, Oracle 7% [V].

7.3 Spreads

Date	IG OAS	HY OAS	BBB OAS	CCC OAS
31 Dec 2024	0.82%	2.92%	1.02%	7.46%
30 Sep 2025	0.76%	2.80%	0.97%	8.07%
14 Nov 2025	0.83%	**3.07%**	1.06%	**9.02%**
31 Dec 2025	0.79%	2.81%	1.01%	8.85%
16–30 Mar 2026	**0.94%**	**3.46%**	**1.16%**	**10.20%**
30 Jun 2026	0.76%	2.75%	0.95%	9.70%
23 Jul 2026	**0.79%**	**2.77%**	**0.98%**	**9.91%**

All [V], FRED series `BAMLC0A0CM`, `BAMLH0A0HYM2`, `BAMLC0A4CBBB`, `BAMLH0A3HYC`.

Two AI-credit stress episodes are visible in the aggregate data:

- **Mid-November 2025.** Between 30 September and 14 November, IG widened 7bp, HY widened 27bp, and **CCC widened 95bp**.
- **Mid-to-late March 2026.** IG hit its 2026 wide of 0.94% on 16 March; **CCC hit 10.20% on 30 March, 135bp above year-end**.

Note the asymmetry. Investment grade's maximum drawdown across both episodes was roughly 20 basis points. CCC moved 135. **Index-level investment-grade calm is masking severe low-quality dispersion**, and the CCC-minus-HY spread of **7.14 percentage points** today — a ratio of 3.58x — is the single most informative number in this section.

Issuer-specific:

Item	Level	Date	Tag
Oracle 5-year CDS	**203bp — widest in series history**	20 Jul 2026	[R]
Oracle 5-year CDS	198bp	18 Jul 2026	[R]

Item	Level	Date	Tag
Oracle 2056 bond	+263bp (+8bp on the day)	~9–10 Jul 2026	**[R]**
Oracle Sept-2025 bonds	**60–80bp wide of issue**	by Feb 2026	**[R]**
Oracle Feb-2026 bonds	rallied ~30bp post-issue	Feb 2026	**[R]**
Hyperscaler new-issue concession	**~12bp** vs market ~2.5bp	2025–26	**[R]**
Post-issuance hyperscaler leverage	**0.4–0.7x** vs IG average ~3x	May 2026	**[R]**
Meta and Alphabet CDS	**began trading Nov 2025**	Nov 2025	**[R]**

That last row is worth pausing on. **Meta and Alphabet did not have traded CDS until November 2025.** A single-name credit default swap market appears when investors want to hedge a credit, and these credits did not previously require hedging.

The BIS noted, 16 March 2026: "**Credit default swap spreads rose, especially for hyperscalers with lower credit ratings**" **[V]**. The Fed's May 2026 FSR: "High-yield spreads for **speculative-grade technology firms** widened more notably" **[V]**.

Oracle at 203 basis points against an index at 79 means Oracle alone carries roughly 2.5x the index's default-risk premium — a BBB− name priced closer to crossover than to its rating cohort.

One important note on Oracle's spread and coupon evidence. The single most reliable indicator in this whole section requires no third party at all: Oracle's own two prospectuses. The 30-year coupon moved **5.950% → 6.700%** and the 40-year **6.100% → 6.850%** between September 2025 and February 2026 — **75 basis points in four months, on unchanged ratings, verifiable from the filed documents [V]**.

7.4 Ratings

Agency	Date	Issuer	Action	Rating
Moody's	28 Jul 2025	Oracle	Affirm; **outlook → negative**	Baa2 / negative
S&P	**9 Jul 2026**	**Oracle**	**Downgrade**	**BBB− / stable**
Moody's	Jul 2026	Oracle	maintained	Baa2 / negative
S&P	2025	CoreWeave	New issuer rating	**B+**; notes at B
S&P	9 Apr 2026	CoreWeave	Affirm; **outlook → positive**	B+ / positive
Moody's	2026	CoreWeave	corporate family	**Ba3**
Moody's / DBRS	31 Mar 2026	CoreWeave DDTL 4.0 SPV	New rating	**A3 / A(low)**
S&P	Oct 2025	Beignet Investor LLC	Preliminary	**A+ / stable**
Moody's	27 Feb 2026	Meta	Affirm	Aa3 / stable

Every rating action in this table is [R] — the agencies' own pages are JavaScript-rendered and could not be retrieved directly, so each is sourced to a press summary rather than read off the agency release **[G on the primary documents]**.

Moody's on Oracle, 28 July 2025, verbatim as reported:

> "While scaling up to be a major provider of AI infrastructure has the potential to drive dramatic growth and solid profitability, capital expenditure requirements will likely also drive **significant negative free cash flow and liability growth** as the AI business builds out."
>
> "The speed of the change, and potential scale of investment particularly as the AI industry is still evolving, introduce **an overhang to the ratings**."

S&P on Oracle, 9 July 2026 — the downgrade rationale as reported: FY2027 capex of **$90–95 billion**; FY2027 free operating cash flow deficit of **approximately −$42 billion**; leverage in the mid-4x area; **OpenAI**

approximately 50% of remaining performance obligations; and total S&P-adjusted debt of **$167 billion** against $129.5 billion reported — the gap being lease capitalisation. Verbatim: "If the AI industry's highly volatile competitive path or path to profitability stumbles, **Oracle is uniquely exposed**."

Note carefully: the "OpenAI ≈50% of RPO" figure is S&P's estimate. ***Oracle discloses no customer's share of RPO anywhere****, so this cannot be verified from any filing.*

Moody's sector report, around 22 July 2026 [R]: covering Microsoft, Amazon, Alphabet, Meta, Oracle and CoreWeave. Verbatim: "Previously, these companies relied on **asset-light structures** centered on software, intellectual property, and scalable cloud services that required modest capital investment... **The transition from asset-light to asset-heavy models requires unprecedented levels of investment and capital raising.**" Figures cited: capex **$785 billion in 2026** approaching $1 trillion the following year; direct debt across the six of roughly **$460 billion**; and lease commitments of **$1.2 trillion, of which over $820 billion has not commenced** — described as **"debt-equivalent liabilities."**

S&P on CoreWeave, 9 April 2026 [R]: 2025 revenue +168% to $5.13 billion; adjusted EBITDA margin 74.9%; **S&P-adjusted debt approximately $29.9 billion** from $10.6 billion; free cash flow deficit approximately −$7.2 billion; RPO $60.7 billion; backlog $66.8 billion from $15.1 billion; and **largest-customer concentration cut to ~35% from ~85%**. Upgrade triggers to BB−: remediate material weaknesses in internal controls by end-2026, FFO/debt above 12%, CFO/debt above 10%.

7.5 Coverage and leverage, with the arithmetic shown

7.5.1 Oracle

Inputs [**V**] from the FY2026 results and 10-K; ratios [**C**].

($m)	FY2025	FY2026
Revenue	57,399	67,357
Operating income (GAAP)	17,678	20,606
Depreciation	3,867	**7,623**
Amortisation of intangibles	2,307	1,671
EBITDA proxy	**23,852**	**29,900**
Interest expense	3,578	4,599
Total debt	92,568	**129,541**
Cash + marketable securities	11,203	31,894
Net debt	**81,365**	**97,647**
Operating cash flow	20,821	31,977
Capital expenditure	21,215	**55,663**

Ratios [C]:

- Gross debt / EBITDA: 92,568 ÷ 23,852 = **3.88x** → 129,541 ÷ 29,900 = **4.33x**
- Net debt / EBITDA: **3.41x** → **3.27x**
- EBIT / interest: 17,678 ÷ 3,578 = **4.94x** → 20,606 ÷ 4,599 = **4.48x**
- Free cash flow: 20,821 − 21,215 = **−$394m** → 31,977 − 55,663 = **−$23,686m**
- Capex / operating cash flow: **1.02x** → **1.74x**

Gross leverage crossed 4x — the level IFR identifies as S&P's downgrade trigger — in a single year. Debt rose $36.97 billion (+39.9%) while EBITDA rose $6.05 billion (+25.4%). Capex rose 162%. Free cash flow deteriorated by $23.3 billion. On S&P's adjusted basis, which capitalises leases, debt is $167 billion and leverage is materially higher than the reported-debt calculation above.

Note also what is *not* yet in the depreciation line: **$39,973 million of construction in progress — 32.6% of gross PP&E — is not depreciating at all [V]**. Depreciation nearly doubled in FY2026 and the base that drives it has not finished arriving.

7.5.2 CoreWeave

Inputs **[V]**; ratios **[C]**.

($m)	FY2024	FY2025	Q1 2026
Revenue	1,915	5,131	2,078
Operating income / (loss)	324	**(46)**	**(144)**
D&A	863	2,454	1,147
Interest expense, net	(361)	**(1,229)**	(536)
Operating cash flow	2,749	3,058	2,984
Capital expenditure	(8,702)	(10,309)	**(7,695)**
Total debt	7,926	21,373	**24,859**
Cash	1,361	3,127	2,244
Backlog	15,100	66,800	**99,400**

Ratios [C]:

- FY2025 EBITDA proxy (GAAP): −46 + 2,454 = **2,408**; gross debt/EBITDA = 21,373 ÷ 2,408 = **8.88x**; net = **7.58x**. On the company's adjusted EBITDA of 3,093: **6.91x** and **5.90x**.
- **FY2025 EBIT / interest = −46 ÷ 1,229 = −0.04x. Operating income does not cover interest at all.**
- Q1 2026 annualised: gross debt / annualised GAAP EBITDA = **6.20x**; EBIT/interest = **−0.27x**; adjusted EBITDA / annualised interest = **2.16x**.
- **Two-year cumulative free cash flow: −5,953 − 7,251 − 4,711 = −$17.9 billion**, funded almost entirely with debt (debt rose $16.9 billion over the same span).
- Cash to current debt at 31 March 2026: 2,244 ÷ 7,547 = **0.30x**.

CoreWeave's GAAP operating line has gone negative while D&A more than doubled. The depreciation schedule on GPUs is now consuming the entire operating margin. Interest expense grew 240% in 2025 and runs at roughly a **$2.1 billion annualised** rate against approximately $4.6 billion of annualised adjusted EBITDA.

7.5.3 The covenant amendment

This is the most important single document in this section, and it is an 8-K filed on **2 January 2026** amending the DDTL 3.0 credit agreement **[V]**. Verbatim:

> "reducing the **minimum liquidity amount** for the monthly payment dates ending on and after March 1, 2026 and prior to May 1, 2026 **to $100.0 million**"
>
> permits "**an unlimited number of equity cures** for any failure to satisfy the debt service coverage ratio and contract realization ratio financial covenants **prior to October 28, 2026**"

And the debt-service coverage ratio first test date was **postponed from April 2027 to 31 October 2027**, with the 1.40x level left intact. The stated purpose: to align the facility "for the timing of deliveries described by the Parent on its earnings call reporting financial results for the quarter ended September 30, 2025."

Unlimited equity cures, plus a thirty-month deferral of the first coverage test, plus a liquidity floor cut to $100 million, is material covenant loosening driven by delivery slippage. Lenders do not grant unlimited cures to borrowers who will not need them. The date the unlimited-cure window closes **— 28 October 2026 —** is the single hardest tripwire in this report.

The pledged customer contract under DDTL 3.0 is **OpenAI's** — the only counterparty affirmatively linked to a CoreWeave facility by a primary source, via the named "Phoenix 1 order form" in the amendment **[V]**.

And the cost of capital is bifurcating violently on the same collateral class [V]: DDTL 4.0 at **SOFR+225bp** in March 2026 (investment-grade rated, non-recourse but for a bad-boy guarantee, against an investment-grade customer) → DDTL 5.0 at **SOFR+450bp** two months later (unconditional parent guarantee, against "two large, **non-investment grade** customers") → **9.625% USD and 8.500% EUR unsecured** in June 2026.

7.5.4 The hyperscalers, for contrast

Company	Balance date	Cash + sec.	Total debt	Net position
Microsoft	31 Mar 2026	$78,272m	$40,262m	**$38.0bn net cash**
Alphabet	31 Mar 2026	$126,840m	$77,501m	**$49.3bn net cash**
Amazon (ex-leases)	31 Mar 2026	$143,089m	$119,074m	**$24.0bn net cash**
Amazon (incl. LT leases)	31 Mar 2026	$143,089m	$209,888m	$66.8bn net debt
Meta	31 Mar 2026	$81,180m	$58,748m	**$22.4bn net cash**
NVIDIA	26 Apr 2026	$50,335m	$8,470m	**$41.9bn net cash**
Oracle	31 May 2026	$31,894m	$129,541m	**$97.6bn net debt, 3.27x**
CoreWeave	~Q2 2026	—	$25,149m	**~$22.9bn net debt, ~6.3x**

[V] inputs, [C] net positions. **Four of the six largest buyers are in net cash. Oracle at 3.3x and CoreWeave at 6.3x are the leverage story, and they are roughly 10% of aggregate capex, not 90%.**

The direction of travel is uniform even where the level is conservative: **Alphabet quintupled gross debt from $22.6 billion to $112.8 billion in eighteen months and raised $84.75 billion of equity in June 2026 — the largest equity transaction ever by a listed corporate. Meta went from $28.8 billion of net cash to net debt [R for the aggregates; V for the components]**.

7.6 Private credit

Metric	Value	Source
Global private credit market	$1.5–2.0tn (end-2024)	FSB, 6 May 2026 [V]
US private credit including BDCs	>$1.6tn (YE2024)	OFR Brief 26-02 [V]
AI infra capex 2025–28	**$2.9tn**, of which $1.5tn external including **$800bn from private credit**	FSB Box 4 [V]
Data-centre share of private credit deals	**34% in 2025**, vs a 17% five-year average	FSB Box 4 [V]
Share of private credit financing AI investment	**9% (2024) → 34% (2025)**	OECD via BoE FSR [V]
Private credit borrower leverage	5–6x reported; **~7x on a "true" adjusted basis**	FSB [V]
Bank lending to private credit funds	~$220bn drawn/undrawn; $268–500bn on commercial data	FSB [V]
LP uncalled capital commitments	$300bn (~$100bn pensions)	OFR [V]
Software exposure of leveraged PC vehicles	**may exceed 50% of NAV**	IMF GFSR, Apr 2026 [V]
Life insurer allocation to private credit	~10%; PE-backed insurers control ~$900bn of liabilities	FSB [V]

The FSB's own framing, verbatim: "Private credit is playing a **critical role** in addressing the financing needs of data centre investments," and "**Asset-based finance is well-suited to the complexities of data centre financing**" [V].

Documented stress — and note carefully which direction it runs. The OFR records that "a selloff in **software loans** amid the threat of artificial intelligence disrupting business models has adversely impacted the broadly syndicated loan and private credit markets" [V]. The Fed's FSR notes "concerns about the potential for advances in AI to **disrupt industries, particularly software**. The software sector had become the largest sector in private credit portfolios" [V].

The verified private-credit damage so far is AI disrupting software borrowers — not data-centre loans going bad. That is an important distinction and it cuts against the simple bear framing.

And a genuine gap [G]: no specific mark-down, NAV decline, redemption gate or covenant breach at a named private credit fund with data-centre exposure was verified — including at Blue Owl, Apollo, Ares, Blackstone or PIMCO. **Any claim of a data-centre-driven private-credit markdown should be treated as unverified.** Nor were fund-level AUM allocations to digital infrastructure obtained; only Blue Owl's 80% of the Hyperion JV is verified.

On valuation opacity, the FSB notes that private credit valuations are updated "**typically quarterly**" and that "investors and other stakeholders may only have limited information" [V] — which is the mechanism by which a data-centre credit problem would surface slowly.

7.7 Vendor financing: what NVIDIA's balance sheet says

The classic vendor-financing tells are receivables and inventory. One confirms and one does not.

Quarter end	Revenue	AR net	Inventory	**DSO [C]**
27 Apr 2025	$44,062m	$22,132m	$11,333m	**45.7 days**
27 Jul 2025	$46,743m	$27,808m	$14,962m	**54.2 days**
26 Oct 2025	$57,000m	$33,391m	$19,800m	**53.3 days**
25 Jan 2026	$68,100m	$38,466m	$21,403m	**51.4 days**
26 Apr 2026	**$81,615m**	**$40,710m**	**$25,797m**	**45.4 days**

[V] inputs; DSO = AR ÷ quarterly revenue × 91 [C].

The receivables tell does NOT confirm. DSO is **45.4 days today against 45.7 days five quarters ago, while revenue grew 85%**. Accounts receivable grew 83.9% year on year against revenue growth of 85.2% — **receivables grew slightly slower than sales.** If NVIDIA were propping up customers on soft terms, receivables would balloon relative to revenue. They have not. **This is the cleanest available refutation of the claim that NVIDIA is financing its own revenue, and it should be given full weight.**

The inventory tell DOES confirm, partially. Inventory rose **127.6% year on year against revenue growth of 85.2%** — a 42-percentage-point divergence. But the composition argues against a demand problem: raw materials rose **74.6% quarter on quarter** to $6,647 million while finished goods rose only **4.9%** to $9,201 million [V]. The build is upstream, consistent with a forward production ramp rather than unsold product.

The supply commitments are the escalation that matters [V]:

Date	Commitment	Value
27 Apr 2025	Inventory purchase + long-term supply/capacity	$29.8bn
27 Jul 2025	Total purchase commitments	$45.8bn
26 Apr 2026	**Manufacturing, supply and capacity commitments**	**$119bn**
26 Apr 2026	**Multi-year cloud service agreements**	**$30bn**
26 Apr 2026	Other vendor commitments	$6bn

Verbatim from the Q1 FY2027 10-Q: "As of April 26, 2026, these commitments were **$119 billion for which $95 billion will be paid in the remainder of fiscal year 2027**." **Supply commitments grew 299% in twelve months, with $95 billion payable within nine months**, against $80.6 billion of total liquidity and only $8.47 billion of debt [V/C].

The $30 billion of cloud-service commitments is the purest circularity item in the entire complex: NVIDIA is contractually committed to purchase cloud capacity from companies to which it sells GPUs, and it took a $2 billion equity stake in one of them in the same quarter [V].

Customer concentration in receivables [V]:

Date	Disclosed concentrations	Top 3–4
27 Apr 2025	27%, 18%, 12%	57%
26 Oct 2025	22%, 17%, 14%, 12%	65%
25 Jan 2026	25%, 18%, 13%	56%
26 Apr 2026	**30%, 18%, 16%**	**64%**

The single largest direct customer rose from 27% to **30%** of receivables in a year; the top three from 57% to 64%. On $40,710 million of receivables, customer one is roughly **$12.2 billion, unsecured**, and **no allowance for credit losses is disclosed anywhere in the filing [V]**. NVIDIA also **dropped its quantified indirect-customer disclosure** this quarter — it defines the term and gives no percentage **[V]**.

7.8 What the official sector says

Already quoted at length in Section 4.8; the credit-specific additions:

- **BIS, 16 March 2026:** "Credit default swap spreads rose, especially for hyperscalers with lower credit ratings" **[V]**.
- **Bank of England, 7 July 2026:** AI companies are "around half" of the S&P 500, "up from around a quarter in 2022," and "**a reassessment of these prospects could trigger a fall in equity prices that might be amplified by high concentration, correlated momentum-driven positions**" **[V]**.
- **Federal Reserve, May 2026:** AI was cited by **50% of surveyed contacts as a financial-stability risk, up from 0% in Fall 2025 —** the sharpest single-period jump in the survey. And yet: "**Corporate bond spreads over comparable-maturity Treasury securities were roughly unchanged and remained low.**" **The Fed's formal four-pillar vulnerability framework does not name AI capex or data-centre lending at all —** a notable divergence from the BoE and BIS **[V]**.
- **Congressional:** Senator Elizabeth Warren and Senate Democrats requested an FSOC investigation on 22 January 2026 into "complex and opaque" data-centre financing **[R]**.

7.9 What this section concludes

Verified: the aggregate indices are at or near multi-decade tights while CCC spreads sit at 991 basis points, 3.58x the high-yield index. Five issuers went from 3% of the IG stock to over 15% of issuance in four months. Oracle's leverage crossed 4x, its free cash flow deteriorated by $23.3 billion in one year, and its own long-bond coupon widened 75 basis points in four months on unchanged ratings. CoreWeave's operating income does not cover its interest expense, its covenants were loosened with unlimited equity cures through 28 October 2026, and its secured spread doubled in two months. NVIDIA's DSO has not moved while revenue grew 85%, but its supply commitments grew 299% and its receivable concentration reached 64% with no disclosed allowance. Data centres went from 17% to 34% of private credit deals.

Analyst judgement: this is not a credit market in distress. It is a credit market that has correctly identified two names — Oracle and CoreWeave — and priced them apart from everything else, while leaving the aggregate indices at tights. The open question is whether that separation is the market working properly or the market pricing the visible portion of an exposure that is mostly invisible, because it sits in unrated private placements, unconsolidated vehicles and leases that have not commenced. The FSB's observation that private credit valuations update "typically quarterly" is the reason that question cannot be answered from market prices.

8. The counter-narrative: physical constraints

8.1 The claim this section makes

Everything in Sections 2 through 7 assumes, implicitly, that the risk is *too much capacity*. This section argues the opposite case as strongly as the evidence allows: **that the binding constraint on this buildout is power, grid interconnection, turbines, transformers and craft labour — not demand — and that if that is true, "overbuild" is the wrong frame entirely.**

This is not a rhetorical exercise. If capacity is genuinely scarce for the next three to five years, then the depreciation debate matters less (assets running at full utilisation are not impaired), the lease commitments are rational reservations of a scarce input rather than speculative overreach, and the credit risk is a *timing* problem rather than a *demand* problem.

The section also states, honestly, where the counter-narrative fails.

8.2 The core evidence: the people who make the physical objects have sold their output

8.2.1 GE Vernova

This is the single best datapoint available, and it is in an 8-K [**V, filed 22 July 2026**]:

- Total backlog **$176 billion**
- Gas Power equipment backlog **53 GW** (44 GW at Q1 2026)
- **Slot reservation agreements 63 GW** (56 GW at Q1 2026)
- Verbatim: "Gas Power equipment backlog and slot reservation agreements grew from **100 to 116 GW**; now anticipate reaching **at least 125 GW by year-end 2026**."
- CEO Scott Strazik, verbatim: "we remain on track to deliver **20 GW of annual gas turbine output in the third quarter of 2026, with 24 GW in 2028**, and we are implementing actions to **produce 30 GW in 2030**."
- "data center orders reaching **over $5 billion year-to-date, more than double our 2025 total**"
- Orders $24.2 billion, **+88% organic**; Power +134% organic; Electrification equipment backlog **$40.6 billion**
- **2026 free cash flow guidance raised from $6.5–7.5 billion to $11.5–12.5 billion**, while revenue guidance rose only about $1 billion

The arithmetic that matters: 116 GW under contract or reservation against 20 GW per year of output is roughly 5.8 years of sold-out capacity. Even against the aspirational 30 GW rate not reached until 2030, it is 3.9 years. GE Vernova has not used the words "sold out." This is what sold out looks like in numbers.

And the free-cash-flow guidance raise is the strongest single scarcity signal in this entire report. A $5 billion increase in free cash flow against a $1 billion increase in revenue is almost certainly customer cash arriving ahead of delivery. **Customers do not prepay for surplus.**

The FY2025 10-K adds the sentence the whole section turns on: "**Customer lead-times have increased as a result of demand outstripping supply**" [V].

Trajectory [**V**]: combined gas position **62 GW (FY2024) → 83 GW (FY2025) → 100 GW (Q1 2026) → 116 GW (Q2 2026)** — roughly doubled in six quarters.

8.2.2 The rest of the supply chain

Vertiv [V]: FY2025 backlog **$15.0 billion, +109%**; Q4 book-to-bill **~2.9x**; Q4 organic orders **+252% year on year**. CEO Giordano Albertazzi: "Robust market demand, particularly in AI infrastructure." **But note — and this is a real red flag, discussed in 8.6 — Vertiv disclosed no backlog, orders or book-to-bill figure in either its Q1 2026 press release or its Q1 2026 10-Q.**

Eaton [V, Q1 2026]: Electrical sector backlog **+48% year on year**; Electrical Americas backlog +44%; twelve-month rolling organic orders +42%; **book-to-bill 1.2x**. Acquired Boyd Thermal on 12 March 2026 for **$9.55 billion —** data-centre thermal.

Caterpillar [V, Q1 2026]: Energy & Transportation sales $7.031 billion (+22%); **Power Generation $2.817 billion, +41%**. Verbatim: sales increased "primarily **data center applications**." The CEO references "a record backlog" **— not quantified in the release [G]**. Note the counter-signal: segment margin **fell to 20.6% from 22.3%** despite +22% sales.

Cummins [V, Q1 2026]: Power Systems sales $1.956 billion (+19%); segment EBITDA margin **29.5% from 23.6%**; "record performance... driven by continued strong demand for **data center backup power**."

Installation labour — the constraint nobody can import [V]:

- **Quanta Services**, Q1 2026: total backlog **$48.5 billion, +37%**; twelve-month backlog **$28.2 billion, +45%**.
- **EMCOR**, Q1 2026: remaining performance obligations **$15.62 billion, a record, +32.9%**.

Electrical-construction backlog growing 33–45% is a direct read on installation capacity. **Craft labour cannot be expanded on a twelve-month cycle. It is the least elastic input in the chain.** Neither company quantified an explicit labour constraint in the retrieved documents [G].

8.3 Power

EIA Short-Term Energy Outlook, released 7 July 2026 [V]: US electricity consumption **+74 billion kWh in 2026** and **+130 billion kWh in 2027 —** roughly +1.8% and +3.1% against a ~4,100 TWh base. And a datapoint that cuts against the scarcity story: **summer wholesale prices are forecast to average $45/MWh, $4/MWh lower than last summer.** The STEO contains **no explicit data-centre attribution** anywhere in its electricity discussion.

Epoch AI [R]: global AI data-centre power capacity was approximately **30 GW** as of Q4 2025 — comparable to New York State's peak demand of ~31 GW — computed from quarterly AI chip sales × TDP × 2.5x infrastructure overhead. US AI data centres are approximately **1% of total US power consumption** today and could reach **100 GW by 2030**, roughly 5% of US generation capacity — growth rates "unseen since the 1980s."

And the local picture is where the scarcity actually is [R]: mid-Atlantic and Midwest wholesale prices rose **76% in Q1 2026**. Against that, Epoch also notes that Virginia, Oregon and Iowa — the *most* data-centre-concentrated states — saw **smaller** retail price increases than the national average from 2020 to 2025. **The constraint is local and seasonal, not national.**

PJM's 2028/2029 Base Residual Auction cleared at $325/MW-day, announced 14 July 2026 **[V, recovered from a Talen Energy 8-K exhibit since pjm.com was unreachable]**.

Major gaps [G]: the LBNL/DOE data-centre energy report, PJM and ERCOT interconnection queue data, FERC large-load interconnection reform documents, and EIA Electric Power Monthly generation tables were all unreachable. **No interconnection-queue length or time-to-energisation figure from any source is verified in this report.** That is a material hole in this section and it should be weighted accordingly.

8.4 The announced-versus-delivered gap

This is the strongest single argument in the counter-narrative, because it is measured in physical assets rather than dollars.

Stargate [R, Epoch AI, 17 April 2026, satellite imagery plus regulatory filings]: across seven sites, **more than 9 GW planned and approximately 0.3 GW operational — a 30x gap.** Only Abilene has any operational capacity, at four of eight buildings. Five of the seven sites are at foundations or steel framing.

CoreWeave's own disclosure shows the same shape [V]:

	Q3 2025	Q1 2026
Revenue backlog	$55.6bn	**$99.4bn**
Active power	~590 MW	**>1 GW**
Contracted power	~2.9 GW	**>3.5 GW**

Contracted 3.5 GW against active 1.0 GW. And CoreWeave's RPO recognition schedule is explicit: "36% is expected to be recognized over the initial 24 months... 39% between months 25 and 48, and the remaining balance... between months 49 and 84" **[V]**. That is a seven-year revenue tail on capacity that does not yet exist.

Oracle [V]: RPO $553 billion at Q3 FY2026 against nine-month capex of $39.17 billion. **RPO to annual capex is roughly 11x.** Either capex rises several-fold or the backlog stretches over a decade. Oracle's own words in the Q3 FY2026 release: "**The demand for cloud computing for AI training and inferencing continues to grow faster than supply.**"

Alphabet [V]: Q1 2026 capex $35.674 billion in a single quarter; Google Cloud revenue $20.028 billion (+63%); **cloud backlog "nearly doubling quarter on quarter to over $460 billion" —** roughly 23 quarters of current cloud revenue.

Meta [V]: the FY2026 capex raise was attributed partly to "**higher component pricing this year**." **Read that as a scarcity signal: the same money buys less compute.**

8.5 What management says about utilisation

The most useful evidence in this section, because it is management speaking against interest — admitting they cannot serve demand.

Microsoft, FY26 Q3 earnings call, 29 April 2026 [V]:

- **Amy Hood: "We expect to remain constrained at least through 2026."**
- Hood: "Broad and growing customer demand continues to exceed supply."
- Hood: "Strong customer demand across workloads, customer segments, and geographic regions continues to exceed available capacity."
- **Satya Nadella: "We added another gigawatt of capacity this quarter, and remain on track to double our overall footprint in just two years."**

Microsoft, FY26 Q2, 28 January 2026 [V]: Nadella — "we added nearly one gigawatt of total capacity this quarter alone." Hood — "Our customer demand continues to exceed our supply."

Two consecutive quarters of roughly 1 GW added, and still short.

Sundar Pichai, 22 July 2026 [V]: "**We continue to be supply constrained —** a sign of momentum and rapid adoption."

Jensen Huang, 19 November 2025 [V]: "Blackwell sales are off the charts, and **cloud GPUs are sold out**." And CFO Colette Kress on the same call: "The clouds are sold out and our GPU installed base, both new and previous generations, including Blackwell, Hopper, and Ampere, **is fully utilized**."

Oracle, Q3 FY2026 8-K exhibit [V]: "The demand for cloud computing for AI training and inferencing continues to grow faster than supply."

And the one that cuts the other way, which matters more than any of the above [V]: NVIDIA's Q1 FY2027 CFO commentary, 20 May 2026 — "**We have strategically secured inventory and capacity to meet demand beyond the next several quarters.**"

That is a statement that NVIDIA's *own* supply constraint has eased. It says nothing about power or grid, but it is the first crack in the "sold out" narrative from the company with the best view of the order book.

8.6 Where the counter-narrative fails — stated at full strength

This section would be worthless if it only presented the scarcity case. Eight items cut the other way, and several are serious.

1. GE Vernova's own risk factor undercuts its own backlog. From the FY2025 10-K, verbatim: "Some counterparties to slot reservation agreements **may not place orders equal to the value of their reservation amount or at all**," and "the volume of orders we expect under such agreements **may fail to materialize**" [**V**]. **63 of the 116 GW — 54% — are reservations, not orders.** This is the strongest single argument against the scarcity read, and it comes from the company itself. In Q2 2026, GE Vernova signed 18 GW of new *reservations* against only 2 GW of *firm orders* — a ratio worth tracking closely.

2. Gigawatt data centres can be built fast. Epoch AI's analysis finds "the time from starting construction to achieving 1 GW of total facility power ranges from **1 to 3.6 years**," with xAI's Colossus 2 on a roughly twelve-month timeline [**R**]. **If the build cycle is 1–3.6 years, a multi-year equipment backlog is a transient constraint, not a structural moat** — and a demand air pocket in 2027–28 would arrive before much of the backlog converts.

3. The binding constraint may be financial, not physical. Epoch AI, working from XBRL of the filings, finds that aggregate cash capex across Microsoft, Amazon, Alphabet, Meta and Oracle **crosses above operating cash flow around Q3 2026**, with operating cash flow growing ~23% a year against capex growing ~70% [**R**]. Oracle is already past the crossover; Amazon is crossing now; Alphabet around Q1 2027; Meta around Q3 2027; Microsoft around Q3 2028. **Capital markets, not turbine factories, may set the ceiling.**

4. Vacancy is not zero, and re-leasing spreads are ordinary. Digital Realty's Q1 2026 8-K discloses total portfolio occupancy of **90.1% — 2,725 MW occupied of 3,024 MW**, roughly 300 MW vacant — and renewal spreads of only **+5.0% cash and +6.3% GAAP** [**V**]. **In a genuine shortage, renewal spreads should be far higher than +5%.** This is the single best piece of contrary evidence in the section. (Digital Realty's backlog does carry a **19-month weighted-average lag to commencement**, which supports the constraint thesis, and ~6.3 GW is buildable under development, which supports supply response.)

5. Regulators are pricing the risk that the load never shows up. AEP's Q1 2026 10-Q, verbatim: "These new tariffs are designed to protect existing customers by strengthening and lengthening contract terms with large customers. These new protections include **contract lengths of up to 20 years and take-or-pay contractual minimums which can require a customer to pay for as much as 80-90% of their contracted demand**" [**V**]. **Utilities do not negotiate 80–90% take-or-pay against loads they believe are certain.**

6. Power prices are not uniformly signalling scarcity. The EIA has summer wholesale **falling** \$4/MWh. Epoch has mid-Atlantic wholesale **+76% in Q1 2026**. Both cannot be the whole story.

7. Announced capacity does get cancelled. OpenAI withdrew from an Abilene expansion and redirected **900 MW** elsewhere [**R**].

8. Backlogs are not uniformly scarce. GE Vernova's **Wind orders fell 40% organically** in Q2 2026. Caterpillar's E&T margin **compressed 170 basis points** despite +22% sales. Where demand is weak, it shows immediately.

And one that is neither confirming nor refuting but deserves flagging: Vertiv stopped disclosing. Its backlog, organic orders and book-to-bill — all disclosed in the Q4 2025 release — are absent from both the Q1 2026 press release and the Q1 2026 10-Q, confirmed by direct reads of both and by full-text search [**V**]. That is an omission, not a disclosed decline. But a company that stops publishing its best number is itself a datapoint, and it is in Section 10.

8.7 What would actually discriminate between the two frames

The overbuild thesis and the physical-constraint thesis **predict the same backlogs**. They diverge on four observables.

1. Prepayments. GE Vernova raised 2026 free-cash-flow guidance by \$5 billion against a \$1 billion revenue raise. Oracle disclosed that **$75 billion of its AI contracts are prepaid or customer-supplied hardware**. Customers do not prepay for surplus. **Strongest scarcity signal in the dataset.**

2. Equipment pricing. GE Vernova's equipment margin in backlog expanded about six points in FY2025; Eaton's book-to-bill is 1.2x; Vertiv's margin expanded 430 basis points. But Caterpillar's E&T margin *fell*. **Mixed, tilting scarce.**

3. Space pricing. Digital Realty's renewal spreads of +5.0% cash are **not** shortage pricing. **Tilts surplus.**

4. Conversion of reservations to firm orders. 54% of GE Vernova's gas position is reservations, and the quarterly conversion ratio deteriorated in Q2 2026 (18 GW of new reservations against 2 GW of firm orders). **This is the single most important number to track in the entire supply chain**, and it is disclosed quarterly.

8.8 What this section concludes

Verified: GE Vernova has roughly 5.8 years of gas turbine output sold or reserved and says customer lead times have increased because demand outstrips supply. Electrical construction backlog is growing 33–45%. Microsoft says it will remain capacity-constrained at least through 2026, having added roughly a gigawatt in each of two consecutive quarters. Alphabet and Oracle both say they are supply-constrained. More than 9 GW of Stargate is planned and roughly 0.3 GW runs. Meta's capex increase is partly component price rather than volume.

And equally verified: 54% of GE Vernova's position is cancellable reservations, on the company's own warning. Digital Realty is 90.1% occupied with +5% renewal spreads. Utilities are writing 80–90% take-or-pay tariffs. NVIDIA now says it has secured capacity "beyond the next several quarters." Gigawatt sites can be built in one to 3.6 years. And aggregate hyperscaler capex crosses above operating cash flow around Q3 2026.

The fairest reading is this. The constraint is real, physical and binding *today* — Microsoft's "constrained at least through 2026" is management's own admission against interest, and it is the single most credible statement in this section. But it is a **two-to-five-year** constraint held in place by turbine slots, transformers and craft labour, not a permanent moat. **The gap between 116 GW ordered and 20 GW a year produced is the surplus risk, deferred rather than avoided: it converts into overcapacity precisely when the equipment finally arrives, in 2029–2032, unless demand compounds at the rate the backlogs imply.**

That reframes the thesis rather than refuting it. The risk is not that too much capacity exists in 2026. It is that too much capacity has been *contracted for* delivery in 2029–2032, at prices set in 2025–2026, financed with four-year money, by counterparties whose ability to pay depends on revenue that does not yet exist.

9. The steelman: the strongest case against this entire thesis

9.1 The rules of this section

Everything above is the case for concern. This section is the case against it, argued at full strength and with the same sourcing discipline. It is not a list of objections. It is the argument a competent, honest bull would make if they had read every filing in this report and concluded the skeptics were wrong.

It wins several of the arguments outright. Where it does, this report says so.

9.2 The argument in one paragraph

The skeptic's case rests on three empirical claims: that AI revenue is vendor-recycled rather than end-customer-funded, that the assets are depreciated over lives far longer than their economic reality, and that this is a debt-financed mania in the mould of 1999–2001 telecom. **On the filed numbers, the first is weak, the second is largely wrong, and the third is right in direction but wrong by roughly 3.5x in magnitude.**

9.3 The revenue is real, contracted, and accelerating

This is the strongest single fact in the bull case, and it is decisive against the crude overbuild framing: cloud revenue growth is *accelerating* in the fourth year of the buildout. Decelerating growth against rising capex is the classic overbuild signature. It is not present anywhere.

Company	Latest quarter	Growth	Backlog / RPO
Microsoft Azure	Q3 FY2026	**+40%**	Commercial RPO **$627bn** (+99% y/y)
Microsoft Cloud	Q3 FY2026	+29% to $54.5bn	—
Microsoft AI business	Q3 FY2026	**+123%**, $37bn run rate	—
Google Cloud	Q1 2026	**+63%** to $20.0bn	Backlog **$467.6bn**
Google Cloud	Q2 2026	**+82%**	Backlog **$514bn**
AWS	Q1 2026	**+28.4%** to $37.6bn	RPO **$364.0bn**, 5yr 6mo avg life
Oracle OCI	FY2026	**+77%** to $18.1bn	RPO **$638bn**
NVIDIA Data Center	Q1 FY2027	**+92%** to $75.2bn	Q2 guide $91.0bn
Broadcom AI semis	Q2 FY2026	**+143%** to $10.8bn	RPO **$164.6bn**
CoreWeave	Q1 2026	**+112%** to $2.08bn	Backlog **$99.4bn**

All [V] from filings and earnings releases.

Oracle's OCI growth accelerated every single quarter from +52% to +93% while the revenue base tripled. That is not a bubble decay curve.

Google Cloud's operating income tripled year on year, from $2,177m to $6,598m — a 32.9% operating margin [C]. This matters enormously and is under-appreciated: **the incremental cloud dollar is now high-margin, not subsidised.**

And Oracle's $75 billion of prepaid and customer-supplied hardware cuts directly against the vendor-financing critique [V]. Bears read it as RPO inflation. But **prepayment is the inverse of vendor financing — the customer is funding the vendor.**

Honest caveats the bull case must carry: Microsoft's RPO jumped **$233 billion in a single quarter**, which is not broad-based demand but a small number of very large contracts; and Alphabet expanded its backlog methodology in Q1 2026 to include contracts of one year or less, adding roughly $7.3 billion — so the headline is about 1.6% inflated by a definitional change [V].

9.4 These are cash-rich buyers, not leveraged speculators

Company	Period	Capex	Operating cash flow	**Capex / OCF**
Microsoft	FY2025	$64,600m	$136,162m	**47.4%**
Microsoft	9M FY2026	$80,146m	$127,494m	**62.9%**
Alphabet	FY2025	$91,447m	$164,713m	**55.5%**
Alphabet	Q2 2026	$44,924m	$39,069m	**115.0%**
Amazon	FY2025	$131,819m	$139,514m	**94.5%**
Amazon	TTM to Mar 2026	$151,003m	$148,531m	**101.7%**
Meta	FY2025	$72,215m	$115,800m	**62.4%**
Four-company aggregate	**FY2025**	**$360,081m**	**$556,189m**	**64.7%**
Oracle	*FY2026*	*$55,663m*	*$31,977m*	***174.1%***
CoreWeave	*Q1 2026*	*$7,695m*	*$2,984m*	***257.9%***

All [**C**] from [**V**] inputs; arithmetic in the companion model.

The median hyperscaler still funds roughly two-thirds of capex from operations. And on net debt: **Microsoft −0.20x, Alphabet −0.31x, Amazon −0.15x (ex-leases), Meta and NVIDIA in net cash** — against **Oracle at 3.27x and CoreWeave at 6.29x** [**C**].

Four of the six largest buyers are in net cash. Oracle and CoreWeave are the leverage story, and they are roughly 10% of aggregate capex, not 90%. Aggregating them with the hyperscalers, as most bear commentary does, is the central analytical error on that side of the argument.

9.4.1 The 1999–2001 comparison, done honestly

The bull case usually asserts that this is nothing like telecom. The honest version is more interesting, and more damaging to the easy version of the bull argument.

Nonfinancial corporate debt to after-tax profits [C, from FRED `BCNSDODNS` and `NFCPATAX`]:

Quarter	Debt	NF profits after tax	**Ratio**
1999 Q4	$4,477.2bn	$319.6bn	14.0x
2000 Q4	$4,828.1bn	$245.4bn	**19.7x**
2001 Q4	$4,969.4bn	$128.9bn	**38.6x**
2026 Q1	**$14,453.8bn**	**$2,564.6bn**	**5.6x**

On an earnings-coverage basis, corporate America is roughly 3.5x less levered than at the 2000 peak. That is the strongest macro fact for the bulls, and it is real.

But here is the part the bull case must not hide. On a **debt-to-GDP** basis the two eras are essentially identical: **46.3% (2000 Q4) against 45.4% (2026 Q1) [C, using FRED `GDP`]'**. The entire improvement is an *earnings* story, not a debt-restraint story — and nonfinancial profits fell **47% from 1997 Q3 to 2001 Q4**. If AI-era earnings prove as cyclical as 2000-era earnings, 5.6x becomes double digits quickly.

A second uncomfortable fact: information-processing equipment investment is a **smaller** share of GDP today than in 2000 — **2.91% (2000 Q4) against 2.38% (2026 Q1) [C, FRED `Y034RC1Q027SBEA`]**. **The claim that "AI capex is unprecedented in scale" is not supported by the equipment data.** It is defensible only for structures and software.

Third, the actual funding mix in 2000 was far worse. The BIS documents the collapse: "Globally, gross issuance by telecoms operators **fell by 72% to $10 billion** between the fourth quarter of 2001 and the last quarter of 2002," and "In the fourth quarter of 2001, **AT&T Corporation alone had been responsible for $10.1 billion in gross issuance** of bonds and notes, **a quarter of total gross issuance by US non-financial corporations**" [**V**]. **The 2000-era telcos were issuing bonds to build. The 2026-era hyperscalers are still mostly spending cash flow.** That is a genuine structural difference — for now.

9.5 The depreciation skepticism is largely wrong

Section 2 already concedes most of this, and it is worth restating as the bull would.

The decisive evidence is that old GPUs are still on published price lists. Nine-year-old V100 silicon is sold today by AWS, Azure and Lambda. Five-and-a-half-year-old A100s are sold by all three hyperscalers plus specialists. **Azure repriced its V100 SKU effective 1 January 2026** and its ND40rs_v2 carries an `effectiveStartDate` of **1 April 2020 — unchanged in six years [V]**. Hardware that is economically dead gets delisted, not repriced. AWS still flags the P3/V100 family as "current generation."

NVIDIA's CFO says the same thing on the record: "the A100 GPUs we shipped **six years ago** are still running at **full utilization** today," and the installed base "both new and previous generations, including Blackwell, Hopper, and Ampere, **is fully utilized**" **[V]**. The Hopper platform recorded roughly $2 billion of revenue in its *thirteenth* quarter **[V]**.

Most of the asset is not the chip. Microsoft: computer equipment and software is **44.5%** of gross PP&E; land, buildings and leasehold improvements are **53.4% [C]**. Amazon: servers and networking equipment is **32.3%** of gross P&E **[C]**. And the long lives are corroborated by parties with no incentive to flatter hyperscaler earnings: **Equinix depreciates buildings over 12–60 years; Digital Realty over 5–39 years [V]**. **Alphabet's 7-to-40-year range, however it was disclosed, is not an outlier.**

And what does the standard actually require? ASC 360's impairment trigger uses **undiscounted** cash flows: an asset is impaired only if carrying value exceeds the sum of *undiscounted* expected cash flows **[V]**. **That is a low bar — and that is exactly the point.** A five-to-six-year server life is defensible under ASC 360 if the asset renders service for five to six years, which the price lists demonstrate it does. Nor is the staff asleep: 252 CORRESP filings cite ASC 360-10-35-21, and the SEC issued a comment letter on 19 May 2026 to SpaceX expressly citing it regarding the former xAI business's long-lived assets **[V]**.

The bull's honest concession: Amazon *shortened* server lives citing "the increased pace of technology development, particularly in the area of artificial intelligence," and Fed Governor Barr noted on 17 February 2026 that "computer chips have historically been depreciated over three years, some firms have stretched the depreciation of AI chips to five years or more" **[V]**. **The depreciation debate is genuinely live, and the operator with the most fleet data moved toward the bears.**

9.6 The circularity critique is overstated

Scale. NVIDIA's non-marketable securities were **$42,336 million** at 26 April 2026, plus $30,237 million of marketable equity. Against annualised Q1 FY2027 revenue of $326,400 million, that is **13.3%**; against FY2026 revenue of $215,900 million, **20.1%**; against total assets, **16.7% [C]**. **NVIDIA generated more operating cash flow in one quarter ($50.3 billion) than the entire carrying value of its non-marketable book ($43.4 billion).** A supplier holding equity worth ~13% of one year's revenue is not manufacturing its own demand.

The receivables evidence is decisive and it is worth restating. NVIDIA's DSO is **45.4 days today against 45.7 days five quarters ago, while revenue grew 85% [C]**. Accounts receivable grew **83.9%** against revenue growth of **85.2%** — receivables grew *slower* than sales. **If NVIDIA were propping up customers on soft terms, receivables would balloon relative to revenue. They have not. This is the cleanest available refutation of the "NVIDIA is financing its own revenue" thesis, and this report finds it persuasive.**

Supplier equity in customers is ordinary industrial practice across semiconductors, aerospace, automotive and pharmaceuticals. And the direction of capital in the largest single arrangement runs the *other* way: **Oracle's customers prepaid $75 billion for GPUs.**

The bull's honest concession, and it is a real one: NVIDIA's non-marketable securities went from **$22.251 billion to $43.364 billion in a single quarter — $17,899 million of net additions in thirteen weeks [V]**. The *level* is defensible. **The *velocity* is the bear case's best fact, and if it continues for four more quarters the critique becomes quantitatively correct rather than merely rhetorically available.**

9.7 Adoption and productivity

Census Bureau BTOS, current AI use [V]:

Reference period	Current AI use
23 Oct – 5 Nov 2023	**3.8%**
2-week period ending 3 May 2026	**19.8%**

A 5.2x increase in roughly 2.5 years [C]. By size in May 2026: 250+ employees **37%**; 100–249 **32%**; four or fewer **<20%**. By sector: Information **39.7%**; Finance and Insurance **33.9%**. Census notes use "increased among firms with at least 20 employees but didn't change significantly among firms with fewer than 20 employees."

Token volume — the cleanest usage series available [V, Alphabet CEO letters]: 10 billion tokens per minute (Q4 2025) → 16 billion (Q1 2026) → 22 billion (Q2 2026). +120% in two quarters.

Enterprise depth [V]: Alphabet reports nearly **500 Cloud customers each processing over 1 trillion tokens annually**, more than **2,000 enterprises consuming over 100 billion tokens**, nearly **90% of the Fortune 100** on Google Cloud, and **over 50% of existing customers exceeding their commitments**. Microsoft's AI business is at a **$37 billion run rate, +123% [V]**. OpenAI reports "more than 5 million people use Codex every week," of whom "more than 1 million... for work outside software development," and Samsung deployed ChatGPT Enterprise and Codex to all Korean employees and the whole DX division worldwide [V].

The productivity evidence, presented with its tensions intact:

- **Brynjolfsson, Li & Raymond (NBER w31161)**, N=5,179 support agents: generative AI "increases productivity... by **14% on average**, including a **34% improvement for novice and low-skilled workers**" [V].
- **Bick, Blandin & Deming (NBER w32966)**: ~40% of US adults use generative AI; 23% of the employed use it weekly; but "**between 1 and 5 percent of all work hours** are currently assisted by generative AI," with time savings of roughly **1.4% of total work hours** [V].
- **Acemoglu (NBER w32487)** — the strongest counterweight, included deliberately: "**no more than a 0.66% increase in total factor productivity**" over ten years, and under 0.53% on stricter assumptions [V].
- **Fed Governor Barr, 17 February 2026**: research estimates AI could add "between **0.3 and 0.9 of a percentage point** to annual total factor productivity growth over the next decade," with a "**J-shape** after technology adoption: adjustment costs lead to short-run losses before firms realize larger, longer-run gains" [V].

The honest synthesis: the micro RCT evidence is strong (+14%, +34% for novices). The economy-wide evidence is weak-to-contested (0.5–0.9pp of TFP over a decade at best; Acemoglu says ≤0.66%). Anyone claiming the macro productivity case is settled is overreaching — in either direction.

9.8 The historical base rates, honestly

Railroads. Hickman's NBER study of $71.5 billion par of straight corporate bonds offered 1900–1943 found **$14.6 billion (20.4%) defaulted**, with **railroads the worst sector at 28.0% of par —** "the railroads turned in by far the poorest performance" [V].

And yet railroad bondholders earned a realised yield of 4.4%, with promised and realised yields converging at 5.3–5.4% on large issues; Hickman found investors "actually received a **higher rate of yield than was promised**... in spite of the heavy default losses" [V]. **Default is not total loss.** Reorganisation converted bonds into new senior claims on assets that kept moving freight.

Equity holders were not so lucky — routinely wiped out or massively diluted in reorganisation. A primary quantification of railroad equity-holder losses could not be obtained [G].

And the funding mix is the sharpest contrast available. Ulmer's NBER study finds railroads 1880–1890 funded **54.9% from bonds, 43.0% from stock, and 90.3% from external financing overall [V]**. **Compare 2026: roughly 65% of hyperscaler capex funded from internal operating cash flow.** That is the bull case's best historical argument.

Dotcom. Robert Gordon (NBER w7833, August 2000) decomposed the 1995–99 productivity rebound into 0.54pp cyclical and 0.81pp trend, and found that **productivity growth outside computer-producing durables actually decelerated [V]**. That is the strongest primary-source caution against extrapolating economy-wide gains from a technology investment boom, and it transfers directly.

What the official sector says now, and it is not friendly to the bulls. The **BIS Annual Economic Report 2026** (28 June 2026): "The five largest hyperscalers are set to spend **over a trillion US dollars** on AI-related capital expenditure from 2025 through 2026. **These commitments are outpacing earnings and the free cash flow of these firms.**" And on the parallel: "The canal mania of the 1830s, the British railway mania in the 1840s, the electrification exuberance of the late 1920s… and the dotcom boom of the late 90s all shared one common trait: **a genuine technological breakthrough that attracted capital in excess of what commercial returns could ultimately justify**" [V].

The counterbalance, from **Fed Governor Cook, 27 May 2026**: "companies have announced more than **$1.5 trillion in data-center plans, only a small portion of which have been realized**," and "the increasing use of leverage to finance investments in an emerging technology carries risk" — but also, "**even under very ambitious investment and debt-issuance projections, we would be unlikely to return to peak leverage levels observed before the Global Financial Crisis**" [V].

9.9 What would have to be true

The bull case holds if — and only if — all five of these hold:

1. **Cloud revenue growth stays above roughly 25%** at Microsoft, Google and Amazon for another six to eight quarters. Current: Azure +40%, Google Cloud +82%, AWS +28%. There is margin of safety, but Azure decelerating below 30% would break the RPO conversion arithmetic.
2. **Capex to operating cash flow for the big four stays below roughly 110%.** At 64.7% aggregate in FY2025, with Amazon already at 101.7% TTM and Alphabet at 115% in Q2 2026, there is perhaps 18–24 months of headroom before the sector becomes structurally debt-funded — at which point the 2000 analogy strengthens materially.
3. **RPO converts at historical rates.** Microsoft discloses only ~30% of RPO recognising within twelve months; Amazon's has a 5.5-year weighted life; Oracle's is 9.5x annual revenue and converts 12% in the next year. **These are promises, not cash.** Conversion is the load-bearing assumption of the entire bull case.
4. **The shell, power and cooling half of capex genuinely lasts 20–40 years.** This is the best-evidenced plank, corroborated by two independent REITs.
5. **NVIDIA's equity-stake growth decelerates.** $17.9 billion of net additions in one quarter cannot continue for long without the circularity critique becoming quantitatively correct.

9.10 Where the bull case is fragile — stated without hedging

- **Concentration.** Microsoft's RPO jumped $233 billion in one quarter. CoreWeave's largest customer was 72% of revenue in Q1 2025 and is still 45% in Q1 2026. **A single counterparty's distress reprices multiple balance sheets simultaneously.**
- **Oracle should not be aggregated with the hyperscalers.** FY2026 free cash flow −$23.7 billion; capex 174% of operating cash flow; net debt/EBITDA 3.27x; $46.1 billion of debt issued in FY2026 with roughly $40 billion of further financing planned for FY2027. **If you are bullish, you are bullish on the hyperscalers *despite* Oracle, not because of it.**
- **CoreWeave at 6.3x net debt to LTM EBITDA and 258% capex to operating cash flow**, with a backlog 12x trailing revenue, is a levered pure-play. The equity may work. **It is not a cash-rich buyer by any**

definition.

- **Amazon shortened server lives**, and the Fed has publicly flagged depreciation-schedule extension as an earnings-flattering practice. **The bulls' depreciation argument is strong on the shell and weaker on the silicon.**
- **The macro productivity case is unproven.** Acemoglu's ≤0.66% over a decade and Gordon's finding that the 1990s boom did not lift productivity outside tech manufacturing are serious objections that no amount of token-growth data answers.
- **Corporate debt to GDP is identical to 2000.** The entire leverage improvement is an earnings phenomenon, and 2000-era earnings fell 47%.

9.11 What this section concludes

The strongest version of the bull case is narrower than the popular version, and this report finds that narrow version persuasive.

The *hyperscalers* are funding a genuinely long-lived asset base out of genuinely compounding, genuinely contracted, genuinely end-customer-paid cash flows, with net-cash balance sheets and residual GPU values demonstrably intact six to nine years after launch. The *periphery* — Oracle, CoreWeave, private-credit-funded developers, and labs whose finances are undisclosed — is doing something much closer to what the skeptics describe.

Both statements are true simultaneously, and conflating them is the central analytical error on both sides of this argument. Section 10 is designed to test which of them starts to fail first.

10. What to monitor

10.1 The rule this section follows

A thesis you cannot lose is not a thesis. Every tripwire below carries a **refutation** condition of equal weight to its confirmation condition, and the confirming and refuting thresholds are stated in numbers, not adjectives.

Estimated dates are labelled **[EST]** with the pattern they were inferred from. Confirmed dates are labelled **[V]**. No date in this section was guessed silently.

The companion artefact `monitoring_dashboard.html` renders this section as a filterable, checkable tracker.

10.2 The four dates that carry most of the information

Everything else in this section is texture. These four are load-bearing.

Date	Event	Why it dominates
26 Aug 2026 [V]	NVIDIA Q2 FY2027	Inventory, DSO, $119bn of supply commitments, 64% receivable concentration and $42.3bn of non-marketable equity all print at once. Confirmed on NVIDIA's own IR calendar.
28 Oct 2026 [V]	CoreWeave's unlimited-equity-cure window expires	From 29 October, cures are capped at three of any four consecutive months, on **both** the debt-service coverage ratio and the contract realisation ratio. Binary and unavoidable.
Jan–Feb 2027 [EST]	FY2026 10-Ks for Meta, Alphabet, Amazon	**Useful lives.** The last remaining earnings lever, and the one disclosure that cannot be spun. Dates inferred from the 2026 pattern: Meta 29 Jan, Alphabet 5 Feb, Amazon 6 Feb.
31 Oct 2027 [V]	CoreWeave DDTL 3.0 first debt-service coverage test	Minimum 1.40x on a trailing three-month basis, tested monthly. Already deferred once, from April 2027, with the level left intact.

10.3 The dated calendar, August 2026 to mid-2028

Date	Event	The one metric that matters
28–29 Jul 2026 [V]	FOMC	Policy rate
29 Jul 2026 [V]	**Meta Q2 2026**	**Non-cancelable contractual commitments** — $237.67bn at Q1, after a $106.6bn single-quarter increase
29 Jul 2026 [V]	Microsoft FY26 Q4	Capex *including finance leases* (the release gives cash PP&E only)
~29–31 Jul 2026 [EST]	**Microsoft FY2026 10-K**	Any change to "computer equipment 2 to 6 years". FY2025 10-K was filed 30 Jul 2025
~30 Jul 2026 [EST]	Amazon Q2 2026	Capex to operating cash flow — already 101.7% TTM
~late Jul 2026 [EST]	Vertiv Q2 2026	**Whether backlog, organic orders and book-to-bill return to the release**
4 Aug 2026 [V]	AMD Q2 2026	Data Center revenue and MI450/Helios ramp language
~4–6 Aug 2026 [EST]	Palantir Q2 2026	US commercial growth against the $4.5bn RPO

Date	Event	The one metric that matters
~5–12 Aug 2026 [**EST**]	CoreWeave Q2 2026	Backlog ($99.4bn) and cash against current debt (0.30x)
14 Aug 2026 [**V**]	13F-HR deadline, Q2 2026	**NVIDIA's positions in CoreWeave and Intel**
26 Aug 2026 [**V**]	**NVIDIA Q2 FY2027**	Inventory, DSO, supply commitments, concentration, equity book
27 Aug 2026 [**V**]	Intel — U.S. Commerce lockup first anniversary	Transfers permitted, and only in broadly-syndicated offerings. Note: **27**, not 26 August — the 8-K states the closing occurred 27 Aug 2025
~3–10 Sep 2026 [**EST**]	Broadcom Q3 FY2026	AI semiconductor revenue against the $16.0bn guide; RPO against $164.6bn
Sep 2026 [**V**]	Oracle's $3.3bn guarantee of a lessor's borrowing matures	Whether it is refinanced, called, or extended
~8–15 Sep 2026 [**EST**]	Oracle Q1 FY2027	RPO ($638bn) and 12-month conversion (12%)
15–16 Sep 2026 [**V**]	FOMC (with SEP)	
30 Sep 2026 [**V**]	CoreWeave DDTL 5.0 DSCR ≥1.35x begins	First test of the newest facility
1 Oct 2026 [**V**]	SoftBank third $10bn OpenAI tranche	Subject to acceleration on an OpenAI listing
~20–29 Oct 2026 [**EST**]	Alphabet Q3, Intel Q3, GE Vernova Q3	Alphabet capex/OCF (115% in Q2); **GEV slot reservations against the 125 GW target**
28 Oct 2026 [**V**]	**CoreWeave unlimited-equity-cure window expires**	Any cure exercised, or any ratio shortfall, thereafter
Late Oct 2026 [**EST**]	Microsoft FY27 Q1, Meta Q3, Amazon Q3	As above
3 Nov 2026 [**C**]	US midterm elections	Policy risk to CHIPS, export controls, federal permitting
Nov 2026 [**EST**]	NVIDIA Q3 FY2027	As August
31 Dec 2026 [**V**]	GE Vernova year-end gas position	**≥125 GW target**, and the reservation-to-firm-order conversion rate
1 Jan 2027 [**V**]	ASU 2024-03 (DISE) effective for calendar-year filers	Depreciation disaggregated within each income-statement caption
Jan–Feb 2027 [**EST**]	**FY2026 10-Ks**	**USEFUL LIVES**
Apr 2027 [**V**]	Oracle $2,250m 2.87% notes mature	First refinancing step at the new cost of capital
Jun 2027 [**V**]	CoreWeave DDTL 4.0 commitment termination; DSCR ≥1.15x begins	
Aug 2027 [**V**]	**Oracle term loan matures — ~$4,645m balloon**	Of a $5,137m facility at 5.29%. First hard refinancing test
Jun 2027 – May 2028 [**C**]	Oracle–OpenAI revenue ramp	Consistent with Oracle's own RPO schedule: 12% in FY2027, then 34% across FY2028–29
31 Oct 2027 [**V**]	**CoreWeave DDTL 3.0 DSCR first test, ≥1.40x**	
Nov 2027 [**V**]	Oracle $2,750m 3.29% notes mature	
1 Jan 2028 [**V**]	ASU 2024-03 interim effective; ASU 2025-06 annual effective	
Jan–Feb 2028 [**C**]	**First DISE-compliant 10-Ks**	Depreciation disaggregated by expense caption for the first time
Mar 2028 [**V**]	CoreWeave DDTL 1.0 matures	$1,438m at ~15% effective — the most expensive money comes due first
1 Jun 2028 [**V**]	PJM 2028/2029 delivery year begins at $325/MW-day	The price the data-centre load must absorb

Technology milestones [V]: NVIDIA Vera Rubin is **already shipping —** "Vera Rubin NVL72 production is ramping up with racks running at partners CoreWeave, Google Cloud, Microsoft Azure and Oracle Cloud Infrastructure," 21 July 2026. *This corrects a common assumption that it is a forward event.* AMD's first 1 GW MI450 deployment for OpenAI is set for H2 2026, and **as of 28 March 2026 no warrant tranche had vested**. GE Vernova gas turbine output: 20 GW/yr in Q3 2026, 24 GW in 2028, 30 GW in 2030. NVIDIA GTC Berlin

20–22 October 2026; GTC San Jose 15–18 March 2027.

10.4 The tripwires

Every threshold below is a number. Current values are as at 25 July 2026.

10.4.1 Credit

Metric	Now	**Confirms**	**Refutes**
HY OAS (`BAMLH0A0HYM2`)	2.77% (23 Jul)	>4.50% for 10 sessions	<2.50% sustained
CCC OAS (`BAMLH0A3HYC`)	**9.91%**, rising daily 16–23 Jul	>13.00%	<8.50%
CCC minus HY dispersion	**7.14pp; ratio 3.58x**	>9.00pp	<6.00pp
BBB OAS (`BAMLC0A4CBBB`)	0.98%	>1.60%	<0.90%
IG OAS (`BAMLC0A0CM`)	0.79%	>1.30%	<0.75%
10-year Treasury (`DGS10`)	**4.71%** (23 Jul, Fed H.15)	>5.50%	<4.00%
Fed funds effective	3.63% (Jun 2026), from 4.33% a year earlier	Cuts stop while spreads widen	Easing continues, spreads stable

The dispersion line is the most informative single number in this report. The investment-grade index is within six basis points of its post-1998 tights while CCC prices at 991. Index calm is masking the tail.

10.4.2 NVIDIA — the supplier's balance sheet

Metric	Now	**Confirms**	**Refutes**
Inventory	$25,797m; raw materials +74.6% q/q; ~115 days	>$35bn while sequential revenue growth <10%	Inventory grows slower than revenue, 2 quarters
DSO	**45.4 days**, vs 45.7 five quarters earlier	>65 days	≤45 days
Supply commitments	**$119bn**, $95bn due within FY2027; +299% y/y	Sequential decline >15%	Grows in line with revenue
Receivable concentration	Top 3 = **64%**; **no allowance disclosed**	Top 3 >70% with rising DSO, or a first-ever allowance	<55%
Non-marketable equity	**$42,336m**, +$17,899m in one quarter; $27bn more committed	Impairments >$3bn in a quarter, or additions stop abruptly	Additions continue, net gains positive
13F positions in CRWV / INTC	11.5% of CRWV; 214.8m INTC	Either position trimmed	Either increased

10.4.3 Oracle — the levered node

Metric	Now	**Confirms**	**Refutes**
RPO and 12-month conversion	$638bn; **~12% ($76.6bn)** within a year	Conversion <10%, or RPO declines sequentially	RPO grows *and* conversion ≥14%
Leases not yet commenced	**$260bn** vs $30.19bn recognised (8.6x)	Falls >10% q/q **without** a matching ROU-asset increase	Converts to ROU assets on schedule
Capex / operating cash flow	**1.74x**; FCF −$23.7bn	>2.0x with FY2027 revenue below the $90bn guide	<1.30x
New-issue long-bond yield	6.74% (30y) / 6.89% (40y), Feb 2026	Next 30y >7.50%, or the CY2026 programme is not completed	Next 30y <6.25%

10.4.4 CoreWeave — the levered pure-play

Metric	Now	**Confirms**	**Refutes**
Equity-cure usage after 28 Oct 2026	Unlimited until then; capped thereafter; CRR ≥0.85x tested monthly since 28 Feb 2026	**Any cure exercised, or any shortfall, after 28 Oct 2026**	No cure used through Q1 2027
Backlog to revenue	\$99.4bn vs \$8.3bn annualised = **~12x**; only 36% of RPO converts in 24 months	Backlog declines sequentially, or 24-month conversion <30%	Ratio falls below 8x **via revenue growth**
Cash to current debt	\$2,244m vs \$7,547m = **0.30x**	<0.25x in a quarter with no completed raise	>0.50x
Marginal cost of capital	SOFR+225 (Mar) → SOFR+450 (May) → 9.625% unsecured (Jun)	Next unsecured >11.00%, or secured wider than SOFR+500	Next unsecured <8.50%

10.4.5 The hyperscalers

Metric	Now	**Confirms**	**Refutes**
Meta non-cancelable commitments	**\$237.67bn**, +\$106.6bn in one quarter; \$182.88bn of uncommenced leases	Sequential decline (cancellation), or growth while revenue growth <15%	Grows alongside revenue growth >20%
Capex / operating cash flow	Alphabet **115.0%** (Q2 26); Amazon **101.7%** TTM; Microsoft 62.9%; Meta 61.6%	Four-company aggregate >110% for two consecutive quarters	Aggregate <85%
Useful lives	Alphabet 6y · Amazon 5–6y · Meta 5.5y · Oracle 6y · Microsoft 2–6y · CoreWeave 6y	**Any extension beyond six years**	**Any shortening**, as Amazon did in 2025

10.4.6 The supply chain

Metric	Now	**Confirms**	**Refutes**
GE Vernova gas backlog + reservations	**116 GW** vs 20 GW/yr output; guided ≥125 GW by YE2026; **54% is reservations**	<120 GW at YE2026, or any disclosed slot cancellation	≥125 GW at YE2026
Vertiv orders / backlog / book-to-bill	**NOT DISCLOSED in Q1 2026.** Last: \$15.0bn, +252% orders, 2.9x	Continued non-disclosure, or backlog <\$15.0bn	Disclosure resumes with backlog >\$18bn
Broadcom RPO	\$164.6bn, ~30% within 12 months	Declines sequentially, or 12-month share <25%	Grows; Q3 AI semis at or above \$16.0bn
Data-centre renewal spreads	Digital Realty **+5.0% cash** at 90.1% occupancy	Turns negative	Above +15%, which would be genuine shortage pricing
AEP-style large-load take-or-pay tariffs	Pending in MI, OK, TX, VA; 80–90% minimums	Spread to more states, or minimums raised	Rejected or diluted by PUCs

10.5 Watch the disclosure itself, not only the numbers

This is the pattern that emerged most consistently across nine research passes, and it deserves its own heading. **Companies stop disclosing numbers before those numbers go bad, and the omission is observable in real time.**

All **[V]**, all within the last eighteen months:

- **Vertiv** disclosed backlog, organic orders and book-to-bill every quarter, then disclosed none of them in Q1 2026 — in neither the press release nor the 10-Q.
- **NVIDIA** dropped its quantified indirect-customer percentage in Q1 FY2027; it defines the term and gives no number. It also stopped naming Anthropic and OpenAI in periodic filings after naming both in the Q3 FY2026 quarter.
- **Oracle** discloses no customer's share of a $638 billion backlog while disclosing customer concentration against revenue, where it cannot appear; and its Q4 FY2026 earnings release contains **no name-attributed executive quotations —** a departure from every prior quarter.
- **Alphabet** extended building lives to forty years and never characterised it as a change in accounting estimate.
- **Meta** does not name Blue Owl or Beignet Investor LLC anywhere, and does not identify the second unconsolidated VIE to which it discloses $5.79 billion of exposure.
- **CoreWeave** identifies customers only as A, B, C and D, with an explicit warning that the letters may map to different customers in different periods.

None of this is unlawful and none of it is even unusual. But the direction is uniform, and the cumulative effect is that the sector is becoming *less* legible from outside precisely as its obligations grow.

10.6 The largest threat to this framework is losing the data

SEC Release 33-11414, File S7-2026-15, proposed 5 May 2026, comment period closed 6 July 2026, would permit semiannual reporting: an optional Form 10-S replacing three 10-Qs, elected by checkbox on the 10-K cover page [V]. No effective date has been proposed.

If it is adopted and elected, roughly **half the observation points in this section disappear** for any issuer that opts in — and because the election binds the *following* fiscal year and is made on a cover page, the loss of visibility would arrive with essentially no warning.

A second item points the other way. **ASU 2024-03** (**DISE**), effective for calendar-year filers from 1 January 2027 with interim application a year later, requires **depreciation to be disaggregated within each income-statement caption [V]**. That is the first genuinely new disclosure in this area in a decade, and it will make the questions in Section 2 materially easier to answer — from the FY2027 10-Ks filed in early 2028.

10.7 How to read all of this

1. **Read the refutation column first.** If a refuting condition is met and you find yourself reaching for a reason it does not count, the thesis has become a belief.
2. **Weight the hard dates above the soft metrics.** A covenant test either happens or it does not. A spread level is a matter of judgement.
3. **Separate the two trades in every observation.** A datapoint that damages Oracle or CoreWeave is not automatically a datapoint against Microsoft or Alphabet, and the reverse. Section 9 is right about that, and it is the discipline most commentary lacks.
4. **The single most informative disclosure in the next twelve months is the useful-life line in the FY2026 10-Ks.** Not because six years is wrong — Section 2 does not claim it is — but because it is the last lever, Oracle has already pulled it, and whoever pulls it next will have to say so.

11. The companion artefacts

Three working artefacts accompany this report. They are not illustrations; each contains material that could not be compressed into prose without losing its usefulness.

11.1 `AI_Depreciation_Sensitivity_Model.xlsx`

A fully formula-driven workbook behind every figure in Section 2. Seven tabs.

- **README —** colour convention, method, and an explicit statement of what the model is *not*.
- **Inputs —** every hardcoded figure, with the filing, the date and the EDGAR URL it came from, and a note on each row explaining what the asset line does and does not include. Every modelling assumption is a yellow-filled, editable cell. Nothing else should be edited.
- **Disclosed —** the register of every useful-life change these six companies have actually disclosed and the dollar effect *they* reported. No modelling on this tab at all.
- **Run_Rate —** Method 1. Steady-state depreciation at 3, 4, 5, 6 and 7-year lives; pre-tax and after-tax effects; and a panel expressing each as a share of reported depreciation. The clean-disclosure subset (Meta, Amazon, CoreWeave) is broken out separately, because the other three companies' asset lines bundle non-server items.
- **Prospective —** Method 2. ASC 250 prospective remeasurement, with a sensitivity grid across assumed fleet ages from 0.5 to 2.5 years.
- **Cumulative —** Method 3. Cumulative disclosed benefit already taken, plus a three-year forward projection with editable growth assumptions, plus a scenario table across all five reversion cases.
- **Validation —** the tab to read first. It attempts to reproduce Meta's own disclosed FY2025 effect from the model's inputs. The naive calculation fails by half; solving for the implied pre-change life gives **4.61 years**, inside Meta's own stated "4 to 5 years" range, and re-running at that life ties to within **1.4%**. **If this tab does not tie, nothing else in the workbook should be trusted.**

Every formula was recalculated in LibreOffice before delivery: **282 formulas, zero errors.**

11.2 `circular_financing_map.html` and `circular_financing_map.svg`

The visual map reproduced in Section 3, plus the complete edge register: **27 relationships**, each with the parties, the flow type, the amount, whether it is funded / binding / announced-only, the date, and the primary source. Rows shaded amber are announced-only — that is, they never reached a filing.

The HTML version is self-contained, adapts to light and dark, and is the one to open. The SVG is the map alone, for embedding elsewhere.

11.3 `monitoring_dashboard.html`

Section 10 as a working tracker: 27 tripwire cards filterable by theme, each with its current value, its confirmation threshold and its refutation threshold; the dated calendar from August 2026 to mid-2028 with severity marking; and the four "how to read this" rules. Self-contained and offline.

12. The gap register: everything this report could not verify

This section exists because a research document that only reports what it found is not honest about its own reliability. What follows is every material gap, grouped by cause. Nothing here was quietly dropped.

12.1 The structural constraint on this research

The web-search budget available to this session was exhausted partway through — 200 of 200 calls — after which all retrieval was by direct fetch against URLs constructed from known patterns. That works extremely well for SEC EDGAR, FRED, and named institutional domains, and not at all for discovery. Additionally, several classes of document are simply unreachable by this tooling:

- **Large filings truncate.** Oracle's FY2026 10-K is 6.87 MB; CoreWeave's FY2025 10-K is roughly 18 MB. Both truncate on fetch well before the financial-statement notes. The workaround used throughout was EDGAR's XBRL-rendered `R##.htm` pages, which are the filing's own tagged data and are equally primary — but they carry numbers reliably and narrative unreliably. **This is why several figures in this report are verified as tagged facts while their accompanying sentences are not.**
- **Rating agency sites are JavaScript-rendered.** S&P Global Ratings, Moody's and Fitch all returned empty shells. **Every rating action in this report is therefore sourced to a press summary, not read off the agency release.**
- **Some domains require per-URL permission** that could not be granted in an unattended session: CBRE, LBNL, PJM, ERCOT, FERC, JLARC Virginia, Ohio PUCO, FASB, PCAOB, and several IR sites.

12.2 Gaps by section

12.2.1 Section 2 — depreciation

- **The Oracle quantification was initially missed.** See Section 12.3, item 7. The lesson is recorded here because it is the most instructive error in this document: a negative finding ("company X discloses no…") is only as good as the completeness of the search behind it, and large filings truncate. **Every negative finding in this report should be read with that caveat.**
- **Critical audit matters.** No CAM title was read for any company. The auditor's report sits past the fetch limit and CAMs are not XBRL-tagged. **The absence of a useful-life CAM cannot be asserted.** Indirect evidence: Meta and Microsoft 10-Ks match the plural heading; Alphabet matches only the singular; CoreWeave's natural CAM headings return zero full-text hits.
- **Microsoft's FY2023 dollar benefit** from the four-to-six-year change beyond the disclosed first quarter. Commonly cited around $3.7 billion; **not disclosed, not verified**.
- **Microsoft's FY2021 3→4-year change.** Full-text search shows the phrase "server and network equipment" first appearing in the Q1 FY2021 10-Q of 27 October 2020, consistent with a change effective 1 July 2020, but the text was not read.
- **Alphabet's building-life extension** has no explanatory or quantifying sentence anywhere. This is a gap in Alphabet's disclosure, not in the research.
- **Amazon's $920 million "early retirement" accelerated depreciation —** the value is XBRL-tagged in the FY2024 10-K, but the rendered column mapping is demonstrably unreliable and Amazon uses the phrase "early retirement" nowhere in any filing. **Period attribution unresolved; most consistent reading is Q4 2024.**
- **NVIDIA's V100 volume-shipment date**; **"B100"** has no standalone launch release; **"VR200"** appears in no NVIDIA primary source.
- **Original NVIDIA list prices** for P100/V100/A100/H100 were never published, so any "percent of original list" residual figure in circulation rests on an unverified baseline.
- **Secondary-market GPU transaction data.** Only a single dealer's asking-price list was obtained. eBay sold listings, auction results and independent resale indices were unreachable.
- **Michael Burry's reported ~$176 billion industry depreciation-understatement estimate** could not be

retrieved and is not relied on anywhere in this report.

12.2.2 Section 3 — circular financing

- **The NVIDIA–OpenAI $30 billion revision** has no primary confirmation from either party.
- **Whether any NVIDIA tranche to OpenAI ever closed.** "First tranche" returns zero hits across all NVIDIA filings, ever.
- **AMD's warrant grant-date fair value** — never disclosed in dollars. **Exhibits E and F of the warrant, containing the actual vesting and exercise ladders, are filed redacted as `[****]`.**
- **AMD's $4.1 billion data-centre lease guarantee counterparty** — unnamed.
- **Oracle's Stargate equity commitment** — never quantified in any filing.
- **Microsoft's cumulative funded amount in OpenAI** — could not be tied to a filing.
- **NVIDIA's Nscale, Lambda, Crusoe, Firmus, Mistral, Wayve and Figure AI positions**, and the reported ~$2 billion NVIDIA slice of the xAI Colossus 2 vehicle — no primary document found. (Confirmed from 13F: CoreWeave, Intel, Coherent, Synopsys, Nokia, Nebius, Generate Biomedicines.)
- **No SEC comment letter, enforcement action, DOJ or FTC inquiry, or Congressional hearing addressing the circular arrangements themselves** was located for any company.

12.2.3 Section 4 — off-balance-sheet

- **Meta's residual value guarantee sentence in the Q3 2025 10-Q.** Full-text search confirms the phrase appears in exactly that one filing; the passage sits in MD&A past the truncation boundary. **Location verified, wording not.**
- **The Beignet offering memorandum.** A 144A private placement, not on EDGAR. **All Beignet terms are press-sourced**, and reported figures for Blue Owl's equity contribution conflict across sources (~$7bn vs ~$2.2bn) and could not be reconciled.
- **The S&P and Moody's primary reports** on the Meta vehicle — their sites returned JavaScript shells; both agencies' quoted positions come from trade press.
- **Crusoe/Abilene:** no borrower entity, tranching, or Oracle lease term located. **Vantage/Frontier:** no SPV names, lease term, debt/equity split or completion guarantees are public. "Vantage Data Centers Jupiter Issuer LLC" appears **not to exist** and should not be cited.
- **xAI/Valor:** no tranche names, sizes, coupons, tenors, lease term, ratings, RVG or bankruptcy-remoteness language exists in any primary source. **No agency rates xAI or any xAI SPV.** The reported $12.5 billion debt tranche has no primary confirmation and this report does not treat it as having closed.
- **CoreWeave DDTL 2.1** has no 8-K; lenders, borrower SPV, collateral and covenants are unverified. DDTL 4.0's and 5.0's customer counterparties are deliberately unnamed by the company.
- **2026 year-to-date data-centre ABS issuance** — no citable public figure exists.

12.2.4 Section 5 — government

- **Sections 5.4 and 5.5 rest on no retrieved primary sources at all** and are labelled [U] throughout. Contract identifiers were deliberately omitted rather than guessed.
- **No Stargate entity documentation exists in any public record** — no operating agreement, ownership table, or board roster, eighteen months after announcement.
- **No Stargate equity contribution figure has been disclosed by any of the four announced equity funders**, and **no announced-to-deployed reconciliation exists** from any issuer.
- **CHIPS obligated-versus-disbursed aggregate** not retrieved. GAO and CRS are robots-disallowed to this tooling.
- **The Advanced Manufacturing Investment Credit's tax-expenditure score** — plausibly the largest single federal fiscal item in the semiconductor programme — could not be sized.
- **Whether the EO 14318 §3 Commerce financing initiative was ever stood up.** This is the highest-value single open question in Section 5.

- **Intel's closing price of $92.32** is from an aggregator, not an exchange or SEC source. All mark-to-market arithmetic inherits that dependency.
- **The Lip-Bu Tan August 2025 episode** could not be primary-sourced.
- **The Virginia JLARC December 2024 study and Ohio PUCO Case 24-508-EL-ATA** — the two most important state-level documents in this field — were both unreachable.
- **No verified data-centre moratorium, rejected permit or referendum** with jurisdiction and date was obtained. This is the most significant hole in the counter-evidence.

12.2.5 Section 6 — Meta

- **Q2 2026 results were not yet released** (29 July 2026). All 2026 data is Q1 only.
- **Meta Superintelligence Labs compensation packages** — deliberately not asserted.
- **The $510 million gap** between the reported $14.3 billion Scale AI consideration and the $13,790 million initial cost booked is unexplained by Meta.
- **The second unconsolidated VIE** with $5.79 billion of maximum exposure is disclosed but unnamed.
- **The Q4/FY2025 share-price reaction**, exact S&P and Fitch action dates, and the body of the 22 July 2026 Moody's sector report.
- **El Paso campus specifics.**

12.2.6 Section 7 — credit

- **All rating actions are press-summarised**, not read off agency releases. Specifically unverified: the full S&P Oracle downgrade rationale, the S&P CoreWeave positive-outlook rationale, and the Moody's A3 rating on the CoreWeave SPV.
- **No Fitch action on any named issuer** was verified.
- **Oracle's 203bp CDS level** rests on one aggregator citing ICE Data Services; the "widest in series history" characterisation is that source's.
- **No private credit fund-level stress was verified** — no mark-down, NAV decline, redemption gate or covenant breach at any named fund with data-centre exposure. **Treat any such claim as unverified.** Nor were fund-level digital-infrastructure AUM allocations obtained.
- **NVIDIA's FY2026 10-K financial statements** could not be retrieved; Q4 FY2026 DSO could not be computed. Balance-sheet figures at 25 January 2026 come from the comparative columns of the Q1 FY2027 10-Q, which is reliable.
- **Oracle's FY2026 10-K lease obligations** were obtained from XBRL tags, not narrative; the ~$37 billion gap between reported and S&P-adjusted debt is inferred.
- **CoreWeave secondary bond levels**, Meta's 2065 post-issue performance and Hyperion secondary trading levels were all unobtainable.
- **The exact date and title of the July 2026 Moody's hyperscaler report** are unconfirmed; its figures are single-sourced through a mirror.
- Two Fed FSR publication dates are ambiguous (8 vs 28 May 2026) and were not resolved.

12.2.7 Section 8 — physical constraints

- **The LBNL/DOE data-centre energy report** (TWh and share of national load) — unreachable.
- **PJM, ERCOT and FERC interconnection-queue data** — unreachable. **No interconnection-queue length or time-to-energisation figure from any source is verified in this report.** This is the largest single hole in Section 8.
- **Vertiv's current backlog** — not disclosed by the company.
- **Caterpillar's backlog figure** — referenced by the CEO as a record, not quantified.
- **Siemens Energy, Schneider Electric, MYR Group and Bloom Energy** backlog data — not retrieved.
- **Amazon's, Alphabet's and Oracle's earnings-call capacity quotes** — release-only; no transcripts. Microsoft's quotes come from its own IR event pages via a summarising fetch and should be treated as

very likely but not character-exact.

12.2.8 Section 9 — steelman

- **FCC telecom capex data 1996–2002** — every FCC host returned 403. BEA and FRED series were substituted, which is arguably better evidence but is not the same comparison.
- **Fiber "lit versus dark" utilisation percentages — found in no primary source. The widely circulated "2–5% lit" figures are unsourced and are not used in this report.**
- **Railroad equity-holder losses** and mileage in receivership — Census historical statistics are robots-disallowed.
- **David and Devine's original electrification-lag quantifications —** outside reachable domains.
- **OpenAI's and Anthropic's paying-customer counts, seat counts or revenue run-rates — not disclosed anywhere primary.** The "end-customer money is real" claim is therefore strongly supported for audited Microsoft, Google and Amazon cloud revenue and only inferentially supported for the labs themselves. Bears are entitled to that gap.
- **Census BTOS series beyond 3 May 2026 —** the data-download endpoints are JavaScript-only.

12.3 Claims actively refuted, not merely unverified

Six widely circulated claims were checked and found to be **wrong**. They are corrected in the body and collected here:

1. **There is no "$1.3 billion NVIDIA–CoreWeave backstop."** The only 1.3 in the IPO prospectus is **1.3 gigawatts** of contracted power; "April 2032" does not appear in the prospectus at all.
2. **Broadcom's $10 billion XPU order was Anthropic, not OpenAI.** Broadcom disclosed this itself on 11 December 2025.
3. **Alphabet made no useful-life change effective 1 January 2025.** Servers remain at six years since 2023; the only change is the buildings range.
4. **The ~$18 billion Oracle-linked New Mexico financing is a STACK Infrastructure deal, not Vantage.** The $25 billion Frontier project cost and the $38 billion two-campus bank package are different quantities and should not be conflated.
5. **CoreWeave has five DDTL facilities, not three, and its SPV debt is consolidated**, not off-balance-sheet.
6. **NVIDIA's Vera Rubin is already shipping —** production racks were confirmed running at CoreWeave, Google Cloud, Azure and OCI on 21 July 2026 — not a forward event.

And four corrections to this report's own working premises, found by an adversarial re-check of its own findings and left visible rather than absorbed:

7. **Oracle DID quantify its useful-life extension.** The first research pass concluded it had not, because the quantification sits on a different page of the FY2025 10-K from the useful-life table. Oracle disclosed a $733 million reduction in total operating expenses, a $573 million increase in net income, and $0.21 basic / $0.20 diluted EPS for fiscal 2025. Section 2.4 records the correction in full. **Alphabet's building-life extension remains the only unquantified change.**
8. **The Amazon–OpenAI AWS figure is an expansion, not a total.** The 10-Q describes "an expansion of the existing $38.0 billion multi-year commitment… by $100.0 billion over 8.0 years" — roughly $138 billion aggregate.
9. **"Up to $100 billion" is not absent from NVIDIA's filings —** it appears once, in a 2020 8-K exhibit describing the SoftBank Vision Fund. The correct and still-striking statement is that **no NVIDIA filing has ever described an up-to-$100-billion investment in OpenAI.**
10. **Meta's $30 billion bond is labelled the "November 2025 Notes"** in Meta's own filing, and **Meta's blended pre-2025 server life was approximately 4.6 years, not 5.0 —** which the Section 2 validation established by solving from Meta's own disclosed figures.

13. Principal sources

Every claim in this report carries an inline source. This is the register of the document families relied on, so that a reader can reproduce the work.

13.1 SEC EDGAR — the spine of the report

Filings were retrieved by accession number from www.sec.gov/Archives/edgar/data/<CIK>/<accession>/ with financial-statement notes read from the XBRL-rendered R##.htm pages reached via each filing's FilingSummary.xml. Full-text presence and absence tests used efts.sec.gov/LATEST/search-index.

Company	CIK	Principal filings used
Meta Platforms	0001326801	FY2022–FY2025 10-K; Q1 2025–Q1 2026 10-Q; 8-Ks of 3 Nov 2025 and 4 May 2026
Microsoft	0000789019	FY2023–FY2025 10-K; Q1 FY2023, Q1–Q3 FY2026 10-Q
Alphabet	0001652044	FY2022–FY2025 10-K; Q1 2023, Q1 2026 10-Q; Q2 2026 8-K
Amazon	0001018724	FY2022, FY2024, FY2025 10-K; Q1 2020–Q1 2026 10-Q
Oracle	0001341439	FY2024–FY2026 10-K; Q1–Q3 FY2026 10-Q; 424B2 of Sep 2025 and Feb 2026; FWP of 2 Feb 2026
NVIDIA	0001045810	FY2025–FY2026 10-K; Q1 FY2026–Q1 FY2027 10-Q; 13F-HR; 424B5 of 17 Jun 2026
CoreWeave	0001769628	S-1/424B4; FY2025 10-K; Q1 2026 10-Q; 8-Ks of 9 Sep 2025, 2 Jan 2026, 23 Jan 2026, 15 May 2026
AMD	0000002488	FY2025 10-K; Q3 2025, Q1 2026 10-Q; 8-K and warrant exhibit of 6 Oct 2025
Broadcom	0001730168	Q2 FY2026 10-Q; quarterly 8-K exhibits
Intel	0000050863	8-Ks of 21 and 25 Aug, 18 Sep, 29 Dec 2025; FY2025 10-K; Q2 2026 10-Q; DEF 14A 2026
GE Vernova	0001996810	FY2025 10-K; 8-Ks of 28 Jan, 22 Apr, 22 Jul 2026
Vertiv	0001674101	8-Ks of 11 Feb and 22 Apr 2026; Q1 2026 10-Q
Others	—	Eaton, Caterpillar, Cummins, Quanta, EMCOR, Digital Realty, Equinix, AEP, SpaceX, MP Materials, Lithium Americas, Trilogy Metals, USA Rare Earth

13.2 Official-sector publications

- **BIS**, *Quarterly Review*, 16 March 2026 — Box A, "Financing the AI infrastructure boom: on- and off-balance sheet borrowing" (Eren, Krohn & Todorov). bis.org/publ/qtrpdf/r_qt2603.pdf
- **BIS**, *Annual Economic Report 2026*, 28 June 2026, Chapter I. bis.org/publ/arpdf/ar2026e.pdf
- **BIS**, *Quarterly Review*, March 2003 — telecom issuance collapse data.
- **Bank of England**, *Financial Stability Report* and FPC Record, July 2026.
- **IMF**, *Global Financial Stability Report*, April 2026, Chapter 1.
- **Federal Reserve**, *Financial Stability Report*, November 2025 and May 2026; Governor Barr, 17 February 2026; Governor Cook, 27 May 2026.
- **Financial Stability Board**, *Report on Vulnerabilities in Private Credit*, 6 May 2026.
- **Office of Financial Research**, Brief 26-02, 12 March 2026.
- **Federal Reserve Bank of New York**, Liberty Street Economics, 20 May 2026.

- **FRED** series: `BAMLC0A0CM`, `BAMLH0A0HYM2`, `BAMLC0A4CBBB`, `BAMLH0A3HYC`, `DGS10`, `FEDFUNDS`, `BCNSDODNS`, `NFCPATAX`, `GDP`, `Y034RC1Q027SBEA`.
- **US Census Bureau**, Business Trends and Outlook Survey, AI-use series.
- **EIA**, *Short-Term Energy Outlook*, 7 July 2026.
- **Federal Register**: EO 14318 (90 FR 35385); BIS UAE final rule (FR doc 2026-14132); DOE RFI (FR doc 2025-18058); Treasury/IRS §48D final rule (FR doc 2024-23857).
- **White House**: Stargate announcement coverage, investments tracker, Genesis Mission release (22 July 2026).
- **Doña Ana County, New Mexico**, industrial revenue bond approval, 19 September 2025.

13.3 Company primary material

OpenAI (`openai.com/index/...`), NVIDIA newsroom and IR transcripts, SoftBank Group (`group.softbank/en/news/press/...`), Apollo, Blackstone, Related, Vantage, Crusoe, Oracle IR, Meta IR, Microsoft IR, Alphabet CEO letters, Broadcom IR, AWS / Azure / GCP / CoreWeave / Lambda published pricing.

13.4 Industry and third-party research

CREFC *Data Center e-Primer* (25 February 2026); KBRA commentary (4 March 2026); CRA International (December 2025); RBC Capital Markets (13 January 2026); Scope Ratings (8 May 2025); Epoch AI (10 November 2025, 16 January 2026, 17 April 2026, 16 June 2026, 14 July 2026); IFR; Quinn Emanuel client alert (February 2026); Kerrisdale Capital (15 September 2025); LPL Research (28 May 2026); FactSet Insight (23 July 2026); NBER working papers w7833, w31161, w32487, w32966, and Hickman and Ulmer on corporate bond experience.

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